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PHYS491: INTRODUCTION TO RESEARCH IN PHYSICS I / PROJECT REPORT

Project: Skyrmion-Based Physical Reservoir Computing

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Abstract

This study investigates a skyrmion-based physical reservoir computing system in a Pt/Co bilayer using COMSOL Multiphysics and OOMMF. The spatial distributions of the electric current density and the current-induced magnetic field were calculated in COMSOL and transferred to OOMMF for micromagnetic simulations. The skyrmion was driven by spin–orbit torque (SOT) generated by the applied current.

The inputs were encoded using the 16 possible four-bit binary sequences. In this encoding scheme, a binary value of 1 was represented by a positive current pulse, while a binary value of 0 was represented by a negative current pulse. Different pulse sequences could produce input-dependent skyrmion trajectories and final physical states.

The final horizontal position (x), vertical position (y), and effective radius (r) of the skyrmion were extracted as reservoir-state features. These features were used individually and in different combinations to train a readout classifier for the MNIST handwritten-digit classification task. The combined (x, y, r) representation achieved the highest classification accuracy of 90.27%.

The results demonstrate that SOT-driven skyrmion dynamics can transform bipolar current-pulse sequences into distinguishable physical states, highlighting the potential of Pt/Co skyrmion devices for physical reservoir computing applications.

Keywords: Physical reservoir computing, magnetic skyrmions, spin–orbit torque, Pt/Co bilayer, OOMMF, COMSOL Multiphysics, MNIST classification.

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1 Introduction

1.1 Background and Motivation

Machine-learning methods are increasingly used for classification, prediction, and the analysis of complex dynamical data. Tasks involving temporal or sequential information are commonly addressed using recurrent neural networks, since recurrent connections allow the internal state of a network to depend on both the current input and previously received information. However, conventional recurrent neural networks are generally more difficult and computationally demanding to train than feed-forward networks. In particular, training recurrent connections commonly requires iterative optimization methods such as backpropagation through time, in which the network is unfolded over many time steps and its internal parameters are repeatedly updated.

Reservoir computing provides an alternative framework for processing sequential information. Instead of training all recurrent connections, it uses a fixed dynamical system to transform an input sequence into a collection of internal states. Only the final readout layer is trained, usually using a relatively simple regression or classification algorithm. This separation between a fixed reservoir and a trainable readout substantially simplifies the learning procedure and reduces the computational cost associated with training recurrent systems [5, 6].

An additional advantage of reservoir computing is that the reservoir does not need to be designed specifically for each individual task. The same dynamical system can be stimulated by different inputs, while task-specific computation is achieved by changing only the readout weights. This characteristic makes reservoir computing attractive for applications in which rapid training, limited training data, or repeated adaptation to different tasks is required.

Although reservoir computing was originally developed for temporal and sequential signals, it can also be applied to static data such as images. For this purpose, a two-dimensional image can be converted into a sequence of smaller input groups and presented to the reservoir in temporal order. The reservoir then transforms each input group into a physical or numerical state, and the complete collection of these states can be used for image classification [5].

1.2 Reservoir Computing

A typical reservoir computing system contains three main components: an input layer, a reservoir layer, and a readout layer. The input layer encodes and couples the original data to the reservoir. The reservoir is a nonlinear dynamical system whose internal parameters remain fixed during training. Finally, the readout layer maps the reservoir states to the required output, such as a class label or a predicted numerical value.

The primary role of the reservoir is to transform the original input into a richer state representation. Inputs that may be difficult to distinguish in their original form can become more easily separable after being mapped into a higher-dimensional state space. The readout layer can then classify or reconstruct the information using a comparatively simple mapping.

An effective reservoir is generally expected to possess several important properties. First, the response of the reservoir should be nonlinear, since a purely linear transformation cannot separate input patterns that are not already linearly separable. Second, the reservoir should exhibit a form of fading memory, meaning that its current state is influenced by recent inputs while the influence of inputs from the distant past gradually disappears. Third, the reservoir should generate sufficiently diverse and distinguishable states for different input sequences. These properties are commonly described in terms of nonlinearity, fading memory, state richness, and separation capability [5].

Only the readout parameters are trained; the reservoir dynamics provide a fixed transformation. This feature is particularly important for the physical implementation of reservoir computing, because the internal parameters of a physical material or device do not need to be individually trained or modified.

1.3 Physical Reservoir Computing

The reservoir in a reservoir computing system does not have to be implemented as a software-based recurrent neural network. Any physical system that produces distinguishable dynamical responses to external inputs may, in principle, be used as a reservoir. Physical reservoir computing exploits the natural evolution of a material, device, or physical structure as a computational resource [6].

In a physical reservoir, the input is represented by a controllable physical stimulus, such as an electrical voltage, electric current, magnetic field, optical signal, mechanical force, or strain. The response of the physical system is then measured through suitable state variables. These measured variables form the reservoir-state representation supplied to the readout layer.

The main motivation for physical reservoir computing is that complex transformations can be generated directly by the intrinsic dynamics of the physical system. Instead of numerically calculating every recurrent interaction on a conventional processor, the physical substrate naturally performs the state transformation as it evolves. A wide range of systems has therefore been studied as potential physical reservoirs, including optical, electronic, mechanical, memristive, and magnetic devices [5, 6].

The suitability of a physical system depends not only on the complexity of its dynamics but also on how effectively the input can be applied and the output states can be measured. A useful physical reservoir must respond reliably to the input, produce repro-

ducible yet sufficiently diverse states, and retain relevant information over an appropriate time interval. The physical output must also be accessible through numerical simulation or experimental measurement.

1.4 Skyrmion-Based Physical Reservoirs

Magnetic systems are promising candidates for physical reservoir computing because magnetization dynamics can exhibit nonlinear, history-dependent, and high-dimensional behavior. Among the magnetic structures investigated for this purpose, magnetic skyrmions are particularly attractive. A magnetic skyrmion is a localized spin texture with a nontrivial topological structure. Its position, size, shape, and velocity can change in response to external stimuli while the texture remains relatively stable.

In heavy-metal/ferromagnet structures, magnetic skyrmions can be stabilized by interfacial Dzyaloshinskii–Moriya interaction and manipulated through current-induced spin–orbit torque. The application of a current pulse can therefore move or deform a skyrmion. Since the resulting response depends on the pulse amplitude, pulse duration, device geometry, and previous state of the skyrmion, different input sequences can produce distinguishable final physical states.

Skyrmion displacement and deformation provide several possible reservoir variables. The horizontal and vertical coordinates of a skyrmion describe its position in the device, while its effective radius provides information about its size and deformation. These variables may contain complementary information about the applied input history. Recent study has shown that the nonlinear response, memory of previous manipulations, displacement, and deformation of skyrmions can be utilized in physical reservoir computing systems for classification and time-series processing tasks [7].

A numerical skyrmion-based reservoir was previously demonstrated using a dumbbell-shaped Pt/Co bilayer [1]. In that framework, the spatial distributions of the electric current density and the current-induced Oersted field were calculated using COMSOL Multiphysics. These distributions were then incorporated into OOMMF micromagnetic simulations, where a skyrmion was driven by current-induced spin–orbit torque.

For the handwritten-digit classification task, each binarized 28×28 MNIST image was divided into 196 groups containing four pixels each. Every four-bit group was encoded as a sequence of current pulses and applied to the skyrmion reservoir. The final horizontal position of the skyrmion after each four-pulse sequence was recorded as the reservoir output. The resulting set of physical states was then supplied to a trained readout classifier. The original study reported a maximum MNIST classification accuracy of 87.6% using the final horizontal skyrmion position [1].

Figure 1 illustrates the general processing workflow of the skyrmion-based physical reservoir. The figure connects the binary image encoding, the application of four-bit pulse

sequences, the physical evolution of the skyrmion, and the final readout classification.

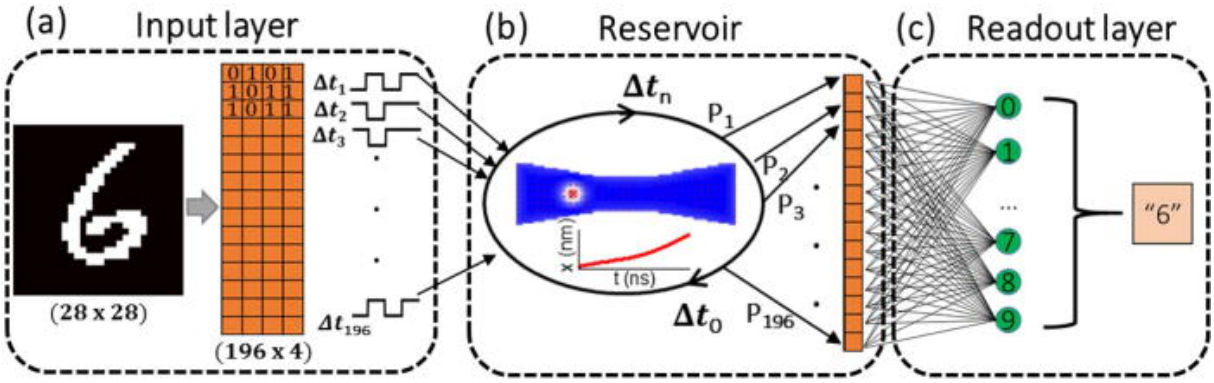


Figure 1: General processing workflow for MNIST handwritten-digit classification using a magnetic-skyrmion physical reservoir. A binarized image is divided into four-bit groups, encoded as current-pulse sequences, transformed into physical skyrmion states, and classified by the readout layer. Reproduced from Jiang et al. [1].

1.5 Aim and Scope of the Study

The present study investigates a skyrmion-based physical reservoir computing system in a Pt/Co bilayer using COMSOL Multiphysics, OOMMF, and a software-based readout classifier. The general device and input-processing framework follows the skyrmion-memristor reservoir model introduced by Jiang et al. [1].

The current-density distribution and the corresponding current-induced magnetic field are obtained from COMSOL and incorporated into OOMMF. Within the micromagnetic simulations, the applied current generates spin–orbit torque and drives the skyrmion through the magnetic device. The spatially nonuniform current distribution and the confining geometry produce nonlinear skyrmion motion, allowing different pulse sequences to generate distinguishable final states.

The binary inputs are represented using all 16 possible four-bit sequences. In the encoding scheme used in this work, a binary value of 1 corresponds to a positive current pulse, whereas a binary value of 0 corresponds to a negative current pulse. Each four-bit sequence therefore applies a specific combination of positive and negative current pulses to the device.

Following each pulse sequence, three physical features are extracted from the final skyrmion state: the horizontal position x, the vertical position y, and the effective radius r. These features are investigated individually and in different combinations to determine how much information each physical variable contributes to the classification task.

The principal objective of the study is to determine whether the use of multiple skyrmion features provides a richer reservoir-state representation than the horizontal position alone. The study therefore compares the classification performance of x, y, and r, as well as combined representations such as $x+y$, $x+r$, $y+r$, and $x+y+r$. The extracted

states are used to construct the physical representation of the MNIST input images and to train the readout classifier.

The main research question addressed in this report is whether spin–orbit-torque-driven skyrmion dynamics can transform bipolar four-bit current-pulse sequences into sufficiently distinguishable physical states for handwritten-digit classification. A related question is whether position and size information contain complementary computational information. The results show that the combined $x + y + r$ representation provides the highest classification performance among the investigated physical feature sets, reaching an accuracy of 90.27%.

The remainder of this report is organized as follows. The relevant micromagnetic theory, skyrmion dynamics, and spin–orbit torque are presented in the following section. The COMSOL and OOMMF simulation procedures are then described, followed by the extraction of the physical reservoir states. The classification results are presented and compared for different feature combinations.

2 Theoretical Background

This section summarizes the physical principles required to describe the skyrmion reservoir considered in this work. The discussion focuses on the micromagnetic energy contributions, magnetization dynamics, magnetic skyrmions, spin–orbit torque, the skyrmion Hall effect, and geometrical confinement relevant to the simulated Pt/Co device.

2.1 Micromagnetic Energy Contributions

The magnetization is written as

$$\mathbf{M}(\mathbf{r}, t) = M_s \mathbf{m}(\mathbf{r}, t),$$

where M_s is the saturation magnetization and $\mathbf{m}$ is the normalized magnetization vector, with $|\mathbf{m}| = 1$.

The total micromagnetic energy of the magnetic Co layer in the Pt/Co system may be expressed schematically as

$$E_{\text{tot}} = E_{\text{ex}} + E_{\text{ani}} + E_{\text{demag}} + E_{\text{Z}} + E_{\text{DMI}}, \tag{1}$$

where the terms represent exchange, anisotropy, demagnetizing, Zeeman, and Dzyaloshinskii–Moriya energies, respectively.

2.1.1 Exchange Energy

The exchange interaction originates from the quantum-mechanical interaction between neighboring spins. In a ferromagnetic material, it energetically favors parallel alignment of adjacent magnetic moments. Within the continuous micromagnetic approximation, the exchange energy for an isotropic material can be written as

$$E_{\text{ex}} = \int_V A \left|\nabla \mathbf{m}\right|^2 \, dV, \tag{2}$$

where A is the exchange-stiffness constant.

The exchange energy penalizes rapid spatial variations in the magnetization direction. Therefore, it tends to maintain a smooth magnetization texture and resists strong compression or deformation of the skyrmion.

2.1.2 Magnetic Anisotropy Energy

Magnetic anisotropy describes the preference of magnetization to align along particular crystallographic or interfacial directions. In an ultrathin Co layer with perpendicular magnetic anisotropy, the preferred easy axis is normal to the film plane.

For a uniaxial easy axis $\mathbf{e}_u$, the anisotropy energy may be written as

$$E_{\text{ani}} = \int_V K_u \left[1 - (\mathbf{m} \cdot \mathbf{e}_u)^2\right] \, dV, \tag{3}$$

where K_u is the uniaxial anisotropy constant. In the present system,

$$\mathbf{e}_u = \hat{\mathbf{z}}, \tag{4}$$

and therefore

$$E_{\text{ani}} = \int_V K_u \left(1 - m_z^2\right) \, dV. \tag{5}$$

A positive value of K_u favors magnetization parallel or antiparallel to the z direction. The anisotropy contributes to the stability and equilibrium radius of the skyrmion. For a fixed DMI strength, increasing the perpendicular anisotropy generally favors a more compact magnetization texture and may reduce the diameter of an isolated skyrmion [8].

Spatial variations in K_u also produce a nonuniform magnetic-energy landscape. A skyrmion approaching a region with a different anisotropy experiences a change in its total energy, which can act as an effective confining or repulsive interaction.

2.1.3 Demagnetizing Energy

The demagnetizing field originates from the long-range dipole–dipole interaction associated with the magnetization distribution. It depends strongly on the geometry, thickness,

surfaces, and boundaries of the magnetic structure.

The demagnetizing energy can be written as

$$E_{\text{demag}} = -\frac{\mu_0 M_s}{2} \int_V \mathbf{m} \cdot \mathbf{H}_{\text{demag}} \, dV, \tag{6}$$

where $\mathbf{H}_{\text{demag}}$ is the demagnetizing field and μ_0 is the vacuum permeability.

The factor of $1/2$ appears because the demagnetizing field is generated by the magnetization itself. This energy generally penalizes magnetic surface charges and competes with perpendicular magnetic anisotropy. It therefore influences the shape, size, confinement, and deformation of the skyrmion, particularly near device boundaries and constrictions [9].

2.1.4 Zeeman Energy and the Oersted Field

The Zeeman energy describes the interaction between magnetization and an externally imposed magnetic field. It is given by

$$E_{\text{Z}} = -\mu_0 M_s \int_V \mathbf{m} \cdot \mathbf{H}_{\text{ext}} \, dV, \tag{7}$$

where $\mathbf{H}_{\text{ext}}$ is the externally applied magnetic field. The Zeeman energy is minimized when the magnetization aligns with the applied field.

An electrical current also generates an Oersted magnetic field. Therefore, the current-induced Oersted field contributes through the same field–magnetization interaction. For the present system, the external field contribution may be written as

$$\mathbf{H}_{\text{ext}} = \mathbf{H}_{\text{bias}} + \mathbf{H}_{\text{Oe}}, \tag{8}$$

where $\mathbf{H}_{\text{bias}}$ represents any externally imposed bias field, if present, and $\mathbf{H}_{\text{Oe}}$ is the Oersted field generated by the electrical current. No separate bias magnetic field was applied in the present simulations. Therefore, during the current-driven stages, $\mathbf{H}_{\text{ext}} = \mathbf{H}_{\text{Oe}}$.

Because the current density is nonuniform in the dumbbell-shaped device, the Oersted field is also spatially dependent. In this work, its spatial distribution is obtained from COMSOL and supplied to the micromagnetic simulation.

2.1.5 Dzyaloshinskii–Moriya Interaction

The Dzyaloshinskii–Moriya interaction is an antisymmetric exchange interaction that favors a chiral alignment of neighboring magnetic moments. It arises in systems combining strong spin–orbit coupling and broken inversion symmetry. These conditions occur at the interface between a heavy-metal layer such as Pt and an ultrathin ferromagnetic layer such as Co.

For an ultrathin film with interfacial DMI, Rohart and Thiaville express the DMI energy as

$$E_{\mathrm{DMI}} = t \int_S D \Big[m_x \frac{\partial m_z}{\partial x} - m_z \frac{\partial m_x}{\partial x} + m_y \frac{\partial m_z}{\partial y} - m_z \frac{\partial m_y}{\partial y} \Big] d^2 r, \tag{9}$$

where D is the interfacial DMI constant and t is the thickness of the magnetic layer [10].

The conventional exchange interaction favors parallel magnetization, while DMI favors a fixed sense of rotation. Their competition allows a localized chiral texture to exist. Together with perpendicular anisotropy and demagnetizing energy, DMI determines the chirality, equilibrium radius, and stability of the skyrmion.

DMI also modifies the micromagnetic boundary conditions. As a result, the magnetization near the edge of a finite structure generally tilts away from a perfectly uniform configuration. The interaction between this edge magnetization and the skyrmion creates an effective boundary-confinement potential [10].

The DMI affects magnetization dynamics through its contribution to the effective field,

$$\mathbf{H}_{\mathrm{DMI}} = -\frac{1}{\mu_0 M_s} \frac{\delta E_{\mathrm{DMI}}}{\delta \mathbf{m}}. \tag{10}$$

DMI modifies the local effective field, affecting skyrmion precession, deformation, chirality, and stability through the magnetization dynamics.

2.2 Landau–Lifshitz–Gilbert Dynamics

The effective magnetic field is obtained from the functional derivative of the total magnetic energy,

$$\mathbf{H}_{\mathrm{eff}} = -\frac{1}{\mu_0 M_s} \frac{\delta E_{\mathrm{tot}}}{\delta \mathbf{m}}. \tag{11}$$

For the present system, it can be decomposed as

$$\mathbf{H}_{\mathrm{eff}} = \mathbf{H}_{\mathrm{ex}} + \mathbf{H}_{\mathrm{ani}} + \mathbf{H}_{\mathrm{demag}} + \mathbf{H}_{\mathrm{DMI}} + \mathbf{H}_{\mathrm{ext}}. \tag{12}$$

The magnetization dynamics are described by the Landau–Lifshitz–Gilbert equation. Including current-induced spin–orbit torques, it can be written as

$$\frac{\partial \mathbf{m}}{\partial t} = -\gamma_T \mu_0 \mathbf{m} \times \mathbf{H}_{\mathrm{eff}} + \alpha \mathbf{m} \times \frac{\partial \mathbf{m}}{\partial t} + \boldsymbol{\tau}_{\mathrm{SOT}}, \tag{13}$$

where $\gamma_T > 0$ is the magnitude of the Gilbert gyromagnetic ratio in $\mathrm{s}^{-1}\,\mathrm{T}^{-1}$, $\mathbf{H}_{\mathrm{eff}}$ is

in A/m, and α is the Gilbert damping constant. The equivalent field-based coefficient is $\gamma_H = \mu_0 \gamma_T$ in $\mathrm{m\,A^{-1}\,s^{-1}}$. The term $\boldsymbol{\tau}_{\mathrm{SOT}}$ has units of $\mathrm{s^{-1}}$ [9, 11].

The first term produces precession of the magnetization around the effective field. Without damping, the magnetization would continue to precess while approximately conserving magnetic energy.

The Gilbert damping term represents energy dissipation. It gradually aligns the magnetization toward a lower-energy state and stabilizes the system after an applied pulse. The damping constant α therefore affects both the transient response and the velocity of current-driven magnetic textures.

2.3 Magnetic Skyrmions

A magnetic skyrmion is a localized, particle-like spin texture in which the magnetization rotates continuously between the skyrmion core and the surrounding ferromagnetic background. The core magnetization is oriented opposite to the magnetization far from the center.

In ultrathin heavy-metal/ferromagnet structures, interfacial DMI typically stabilizes Néel-type skyrmions. In a Néel-type skyrmion, the in-plane components of the magnetization point radially inward or outward. In contrast, the in-plane magnetization of a Bloch-type skyrmion rotates tangentially around the center.

The topological charge of a two-dimensional magnetic texture is

$$Q = \frac{1}{4\pi} \int \mathbf{m} \cdot \left(\frac{\partial \mathbf{m}}{\partial x} \times \frac{\partial \mathbf{m}}{\partial y} \right) dx\, dy. \tag{14}$$

An isolated skyrmion normally has $Q = \pm 1$, whereas a uniform ferromagnetic state has $Q = 0$. The sign of Q depends on the relative orientation of the skyrmion core and the surrounding magnetization.

Figure 2 shows Néel-type skyrmions with opposite topological charges. The magnetization rotates radially in both configurations, but the core and background directions are reversed.

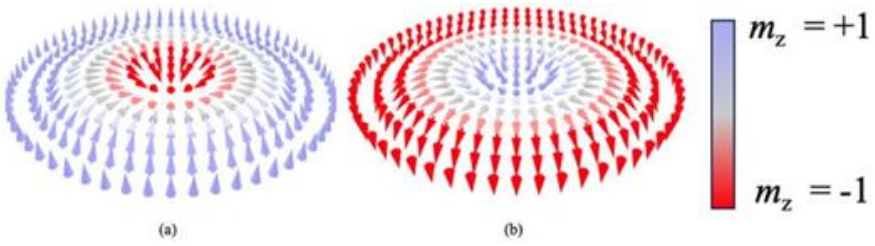


Figure 2: Spin configurations of Néel-type magnetic skyrmions with opposite topological charges. Panels (a) and (b) show skyrmions with $Q = +1$ and $Q = -1$, respectively. The color scale represents the out-of-plane magnetization component m_z. Reproduced from Yang et al. [2].

In addition to its topological charge, a skyrmion has continuous physical properties such as its horizontal position, vertical position, effective radius, and deformation. These quantities can change under electrical driving and can therefore serve as physical reservoir states.

2.4 Spin Hall Effect and Spin–Orbit Torque

Pt is a heavy metal with strong spin–orbit coupling, whereas Co is the ferromagnetic layer in which the skyrmion is hosted. In the commonly used spin-Hall picture, an in-plane charge current in Pt generates a transverse spin current. The transfer of spin angular momentum to the adjacent Co layer produces a spin–orbit torque on the magnetization [11].

The most general lowest-order spin–orbit torque can be separated into field-like and damping-like components,

$$\boldsymbol{\tau}_{\mathrm{SOT}} = \boldsymbol{\tau}_{\mathrm{FL}} + \boldsymbol{\tau}_{\mathrm{DL}}. \tag{15}$$

2.4.1 Field-Like Torque

The field-like torque has the form

$$\boldsymbol{\tau}_{\mathrm{FL}} = \tau_{\mathrm{FL}} \mathbf{m} \times \boldsymbol{\sigma}, \tag{16}$$

where $\boldsymbol{\sigma}$ is the direction of the current-induced spin polarization and τ_{FL} represents the strength of the field-like torque.

This term acts similarly to the torque produced by an effective magnetic field directed along $\boldsymbol{\sigma}$. It therefore primarily modifies the precessional motion of the magnetization. Depending on the material stack and interface properties, it can also contribute to skyrmion deformation, breathing dynamics, and changes in the Skyrmion Hall angle.

2.4.2 Damping-Like Torque

The damping-like torque is written as

$$\boldsymbol{\tau}_{\mathrm{DL}} = \tau_{\mathrm{DL}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \,. \tag{17}$$

Here τ_{DL} is a signed coefficient, and $\boldsymbol{\sigma}$ denotes the polarization direction defined for the positive-current reference geometry. For the OOMMF implementation in Sec. 3.3.6, $\mathbf{m}_p = \boldsymbol{\sigma} = (0, -1, 0)$ and $\tau_{\mathrm{DL}}(J) = -|\gamma_H|\beta(J)\epsilon$. Thus, a positive J and positive P give a negative coefficient in Eq. (17).

This term acts in a manner analogous to a modification of magnetic damping. In heavy-metal/ferromagnet bilayers, the damping-like torque is commonly the principal

contribution responsible for current-driven translation of chiral Néel magnetic textures [11].

The coefficients τ_{FL} and τ_{DL} depend on the current density, material composition, spin–orbit coupling, layer thicknesses, and interface properties. The exact torque components included in the numerical model are specified in the Simulation Methodology section.

The signed current density represents pulse reversal while the reference polarization vector is held fixed in the numerical implementation. Changing the sign of J reverses the current-induced torque; the polarization vector is not reversed a second time.

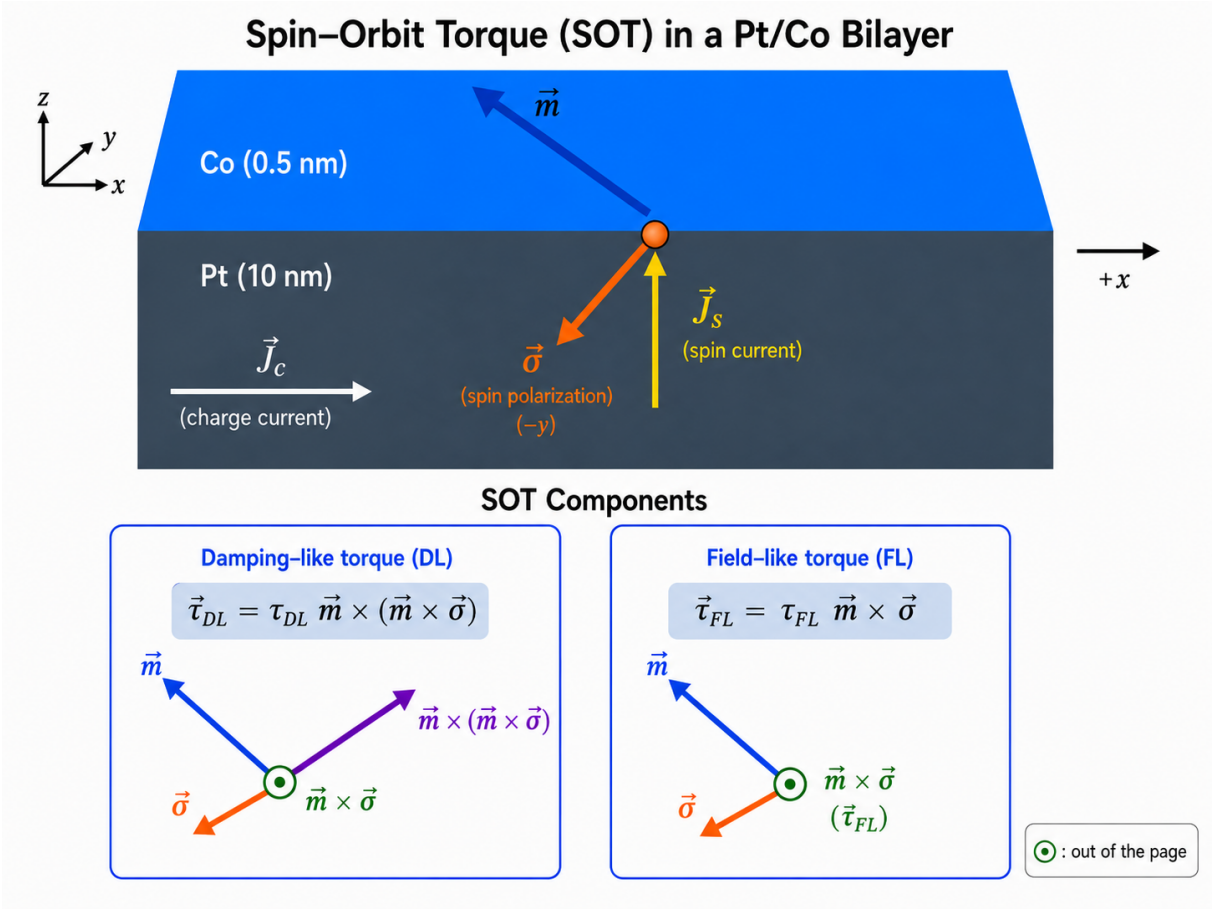


Figure 3: Spin–orbit-torque geometry and vector basis in the Pt(10 nm)/Co(0.5 nm) bilayer. The positive-current reference convention is $\mathbf{J}_c \parallel +x$, spin transport toward Co along $+z$, and $\boldsymbol{\sigma} \parallel -y$. The damping-like arrow shows the basis vector $\mathbf{m}\times(\mathbf{m}\times\boldsymbol{\sigma})$, rather than the implemented torque for positive current. For $J > 0$ and $P > 0$, OOMMF applies the opposite direction, $\boldsymbol{\tau}_{\mathrm{DL}} = -|\gamma_H|\beta\epsilon\, \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma})$. The field-like panel is schematic; the separate field-like SOT term is omitted in the simulations by setting $\epsilon' = 0$.

2.5 Skyrmion Hall Effect and Magnus-Like Force

A current-driven skyrmion does not generally move exactly parallel to the applied driving direction. Its nonzero topological charge gives rise to a gyrotropic response that produces

a velocity component transverse to the driving force. This phenomenon is called the skyrmion Hall effect [2].

Within the rigid-skyrmion approximation, the collective motion may be described by a generalized Thiele equation,

$$\mathbf{G} \times \mathbf{v} + \alpha \boldsymbol{\mathcal{D}} \cdot \mathbf{v} - \mathbf{F}_{\text{drive}} - \mathbf{F}_{\text{conf}} = 0, \tag{18}$$

where $\mathbf{G}$ is the gyrovector, $\mathbf{v}$ is the skyrmion velocity, $\boldsymbol{\mathcal{D}}$ is the dissipative tensor, $\mathbf{F}_{\text{drive}}$ is the current-induced driving force, and $\mathbf{F}_{\text{conf}}$ represents boundary and confinement forces.

The gyrovector is proportional to the topological charge,

$$\mathbf{G} \propto Q\hat{\mathbf{z}}. \tag{19}$$

The term

$$\mathbf{F}_{\text{Magnus}} = \mathbf{G} \times \mathbf{v} \tag{20}$$

is commonly described as the gyrotropic or Magnus-like contribution. Because it is perpendicular to the velocity, it produces transverse deflection.

The skyrmion Hall angle can be defined as

$$\tan \theta_{\text{SkH}} = \frac{v_y}{v_x}, \tag{21}$$

where v_x and v_y are the longitudinal and transverse velocity components.

The direction of the transverse motion depends on the sign of the topological charge, the skyrmion polarity, the current direction, and whether conventional-current or electron-flow conventions are used.

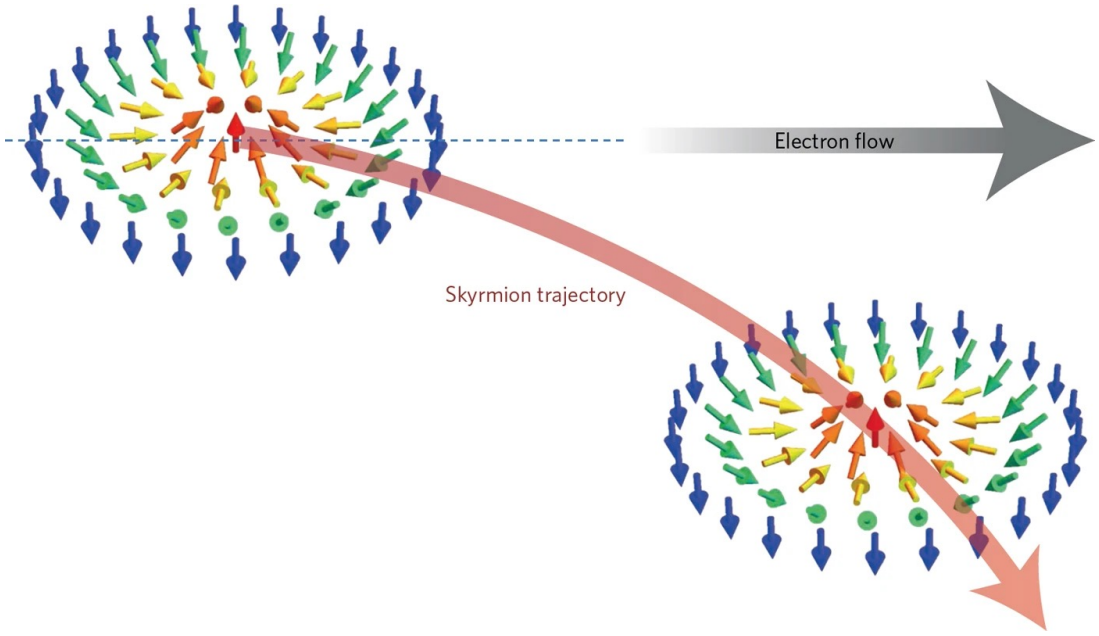


Figure 4: Schematic illustration of the skyrmion Hall effect. The trajectory of the skyrmion is deflected from the longitudinal electron-flow direction because of the transverse gyrotropic, or Magnus-like, contribution. Reproduced from Chen [3].

In a finite-width device, the transverse motion may drive the skyrmion toward one of the boundaries. The boundary interaction then modifies its trajectory and may deform the spin texture. If the confinement is insufficient, the skyrmion can reach the edge and be annihilated.

The observed trajectory is therefore determined by competition among the spin–orbit-torque driving force, Gilbert damping, the gyrotropic deflection, and boundary-induced confinement.

2.6 Geometrical Confinement and Skyrmion Deformation

The dumbbell-shaped structure consists of wider regions connected by a narrow constriction. This geometry produces a spatially nonuniform current-density distribution and changes the strength of the boundary confinement.

For a simple conductor carrying a fixed total current, the average current density may be estimated as

$$J \approx \frac{I}{wt}, \tag{22}$$

where I is the total current, w is the local device width, and t is the conducting-layer thickness. This expression provides only a qualitative approximation for the present nonuniform structure; the actual current-density distribution is obtained from COMSOL.

As the skyrmion approaches the constriction, the local current density generally increases. Since the strength of the damping-like spin–orbit torque is related to current

density, the driving force becomes larger. The skyrmion may therefore reach a higher longitudinal velocity within the narrow region [1].

At the same time, the distance between the upper and lower device boundaries decreases. The skyrmion consequently experiences stronger geometrical confinement and must adjust its internal magnetization configuration.

In the simulations considered in this work, this adjustment appears as a reduction in the effective skyrmion radius. This behavior can be interpreted as confinement-induced compression or deformation caused by the restricted transverse space and the modified micromagnetic-energy balance.

However, radius reduction in a constriction is not a universal rule for all skyrmion systems. The response depends on exchange stiffness, DMI, anisotropy, demagnetizing energy, current density, and the detailed boundary potential. The skyrmion radius is therefore a dynamical state variable rather than a constant material property.

The constriction consequently generates two important effects:

1. the increased current density strengthens the current-induced torque and can increase the longitudinal skyrmion velocity;

2. stronger boundary confinement deforms the skyrmion and, in the present simulations, reduces its effective radius.

These nonlinear, position-dependent responses are useful for physical reservoir computing because different current-pulse sequences can produce distinguishable final skyrmion states.

2.7 Skyrmion Position and Radius as Reservoir States

The skyrmion device maps a sequence of positive and negative current pulses into a final magnetic state. Because the velocity, transverse deflection, radius, and deformation depend on the instantaneous position and preceding pulses, this input-to-state mapping is nonlinear.

The horizontal coordinate x mainly reflects longitudinal displacement under the current-induced driving force. The vertical coordinate y contains information about the skyrmion Hall effect and the interaction with the upper and lower boundaries. The effective radius r captures compression and deformation arising from the spatially varying magnetic energy and geometrical confinement.

These quantities are correlated but describe different aspects of the same magnetic response. They may therefore contain complementary information about the applied four-bit pulse sequence.

The previous skyrmion-reservoir model used the final horizontal skyrmion position as the principal reservoir state [1]. In the present study, the final vertical position and effective radius are additionally extracted and evaluated.

The combined effects of exchange, anisotropy, demagnetizing energy, DMI, Zeeman interaction, the Oersted field, spin–orbit torque, Skyrmion Hall deflection, and geometrical confinement determine the final values of x, y, and r. These features form the physical reservoir-state representation used in the subsequent MNIST classification procedure.

3 Simulation Methodology

This section describes the numerical workflow used to model the skyrmion-based physical reservoir. The device geometry and magnetic material parameters were adopted from the Pt/Co skyrmion reservoir model proposed by Jiang et al. [1]. The spatial distributions of the electric current density and the current-induced magnetic field were first calculated using COMSOL Multiphysics. These quantities were subsequently incorporated into OOMMF for the micromagnetic simulation of current-driven skyrmion dynamics. The final skyrmion states were then extracted and used as physical reservoir features for the classification task.

3.1 Device Geometry and Material Parameters

The physical system considered in this work consists of a Pt/Co bilayer with a laterally confined dumbbell geometry. The general device concept, layer thicknesses, lateral dimensions, and baseline micromagnetic parameters were adopted from the magnetic-skyrmion memristor model proposed by Jiang *et al.* [1]. The electromagnetic properties required for the COMSOL calculation were taken from the COMSOL Multiphysics Material Library [4], while the subsequent micromagnetic simulations were implemented using the Object Oriented MicroMagnetic Framework (OOMMF) [12].

The device consists of a heavy-metal Pt layer underneath an ultrathin ferromagnetic Co layer. The Pt layer provides the current-carrying heavy-metal region associated with spin–orbit-torque generation, whereas the Co layer hosts the magnetic skyrmion. The same lateral dumbbell footprint was used in both the COMSOL electromagnetic model and the OOMMF micromagnetic model.

The coordinate system was defined such that the x direction corresponds to the longitudinal direction of the device and the nominal charge-current direction, the y direction lies transverse to the current within the film plane, and the z direction is normal to the Pt/Co interface.

The overall lateral dimensions followed the device proposed by Jiang *et al.* [1]. The total device length was

$$L_x = 200 \text{ nm}, \tag{23}$$

and the maximum width at the two ends was

$$W_{\text{end}} = 60 \text{ nm}. \tag{24}$$

The minimum width at the central constriction was

$$W_{\text{constriction}} = 30 \text{ nm}. \tag{25}$$

In the present implementation, this dumbbell geometry was represented explicitly using the analytical lateral boundary

$$y(x) = \pm \left[22.5 - 7.5 \cos\left(\frac{\pi x}{100}\right)\right] \text{ nm}, \qquad -100 \leq x \leq 100 \text{ nm}. \tag{26}$$

This analytical representation produces the required 60 nm width at the two ends and the 30 nm minimum width at the center of the device.

The metallic stack consisted of a

$$t_{\text{Pt}} = 10 \text{ nm} \tag{27}$$

Pt layer and a

$$t_{\text{Co}} = 0.5 \text{ nm} \tag{28}$$

Co layer, consistent with the Pt(10 nm)/Co(0.5 nm) structure used in the reference skyrmion-reservoir model [1].

In the three-dimensional COMSOL geometry, the Pt layer occupied

$$0 \leq z \leq 10 \text{ nm}, \tag{29}$$

while the Co layer occupied

$$10 \leq z \leq 10.5 \text{ nm}. \tag{30}$$

Only the magnetic Co layer was represented explicitly in the OOMMF micromagnetic calculation, since the magnetization dynamics and the skyrmion texture reside in the ferromagnetic layer. The effect of the Pt layer entered the OOMMF model through the current-induced spin–orbit torque and the Oersted field calculated from the Pt/Co electromagnetic model.

For the COMSOL electromagnetic calculation, the electrical conductivities were taken from the COMSOL Multiphysics Material Library [4]. The conductivity of Pt was

$$\sigma_{\mathrm{Pt}} = 9.52 \times 10^{6}\ \mathrm{S/m}, \tag{31}$$

while the conductivity of Co was

$$\sigma_{\mathrm{Co}} = 1.60 \times 10^{6}\ \mathrm{S/m}. \tag{32}$$

The surrounding air domain was treated as electrically nonconducting.

The baseline values of M_s, A, D, and α are listed in Table 1 and follow the Pt/Co reservoir model of Jiang *et al.* [1].

Jiang *et al.* employed a uniform perpendicular magnetic anisotropy of $K_u = 0.7\ \mathrm{MJ/m^3}$ in their original model [1]. In the present work, however, the anisotropy profile was modified during model development in order to improve skyrmion confinement.

The interior of the magnetic device was assigned

$$K_{u,\mathrm{center}} = 1.56 \times 10^{6}\ \mathrm{J/m^3}, \tag{33}$$

whereas a 4 nm-wide region adjacent to the device boundaries was assigned the higher anisotropy

$$K_{u,\mathrm{edge}} = 3.0 \times 10^{6}\ \mathrm{J/m^3}. \tag{34}$$

These two anisotropy values are therefore modifications introduced in the present work rather than parameters taken directly from the reference model. The enhanced edge anisotropy was introduced to increase confinement against transverse skyrmion motion and reduce edge annihilation associated with the Skyrmion Hall effect. Its development and implementation are discussed in detail in Sec. 3.3.4.

The principal geometrical, electrical, and micromagnetic parameters used in the final model are summarized in Table 1.

Table 1: Principal geometrical, electrical, and micromagnetic parameters used in the Pt/Co dumbbell model. Parameters adopted from the reference skyrmion-reservoir model are based on Jiang *et al.* [1]; electrical conductivities were taken from the COMSOL Material Library [4].

Parameter	**Symbol**	**Value**	**Source**
Device length	L_x	200 nm	Jiang *et al.*
End width	W_{end}	60 nm	Jiang *et al.*
Constriction width	$W_{\text{constriction}}$	30 nm	Jiang *et al.*
Pt thickness	t_{Pt}	10 nm	Jiang *et al.*
Co thickness	t_{Co}	0.5 nm	Jiang *et al.*
Pt conductivity	σ_{Pt}	9.52×10^6 S/m	COMSOL
Co conductivity	σ_{Co}	1.60×10^6 S/m	COMSOL
Saturation magnetization	M_s	1.4×10^6 A/m	Jiang *et al.*
Exchange stiffness	A	15×10^{-12} J/m	Jiang *et al.*
DMI constant	D	2.9×10^{-3} J/m^2	Jiang *et al.*
Gilbert damping	α	0.3	Jiang *et al.*
Central anisotropy	$K_{u,\text{center}}$	1.56×10^6 J/m^3	Present work
Edge anisotropy	$K_{u,\text{edge}}$	3.0×10^6 J/m^3	Present work
Edge-region width	w_{edge}	4 nm	Present work

3.2 COMSOL Simulation

COMSOL Multiphysics was used to calculate the spatial distribution of the electric current density and the magnetic field generated by the applied current in the Pt/Co device. The COMSOL model was constructed explicitly in three dimensions so that the Pt and Co layers, as well as the surrounding air region, could be represented separately.

3.2.1 Model Initialization and Physics Selection

A new three-dimensional model was first created using the COMSOL *Model Wizard*. Two physics interfaces from the AC/DC Module were added to the model: *Electric Currents (ec)* and *Magnetic Fields (mf)*. The Electric Currents interface was selected to calculate the electric-potential and current-density distributions, whereas the Magnetic Fields interface was included to calculate the magnetic field generated by the applied electric current.

The selected physics interfaces are shown in Fig. 5.

A *Stationary* study was then selected. The purpose of the COMSOL model was to determine the steady-state spatial current-density distribution and the corresponding current-induced magnetic field for a specified applied current. The time-dependent positive and negative current pulses were introduced later in the OOMMF micromagnetic simulations.

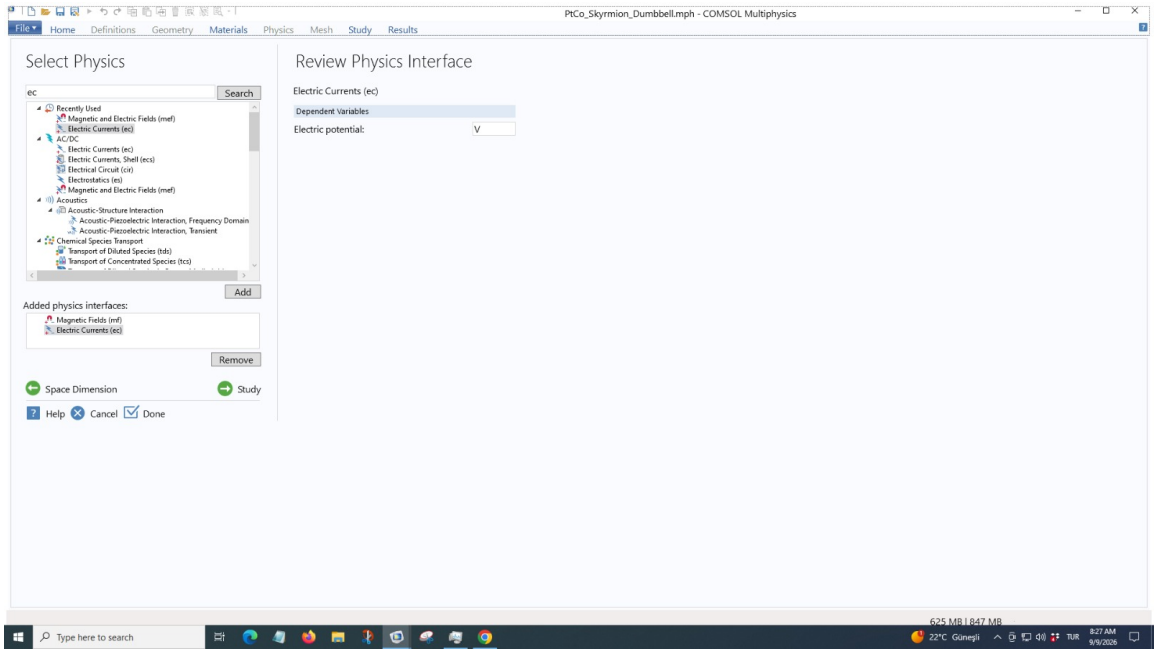


Figure 5: Selection of the Electric Currents (`ec`) and Magnetic Fields (`mf`) physics interfaces in the COMSOL Model Wizard.

3.2.2 Construction of the Device Geometry

The Pt/Co device geometry was constructed directly in COMSOL using two two-dimensional work planes followed by extrusion in the z direction. The lateral shape of both layers was defined as a dumbbell geometry with a total length of 200 nm, a width of 60 nm at the two ends, and a minimum width of 30 nm at the central constriction.

Construction of the Pt footprint. A work plane was first created in the xy plane for the Pt layer. Inside this work plane, the upper and lower boundaries of the dumbbell were constructed using two *Parametric Curve* features.

For the first parametric curve, the curve parameter was defined as s, with the range

$$-100 \leq s \leq 100, \tag{35}$$

where all geometric dimensions were specified in nanometers. The coordinate expressions entered in COMSOL were

$$x_w = s, \tag{36}$$

and

$$y_w = 22.5 - 7.5 \cos\left(\frac{\pi s}{100}\right). \tag{37}$$

This curve defines the upper boundary of the dumbbell structure. The COMSOL

settings used for this curve are shown in Fig. 6.

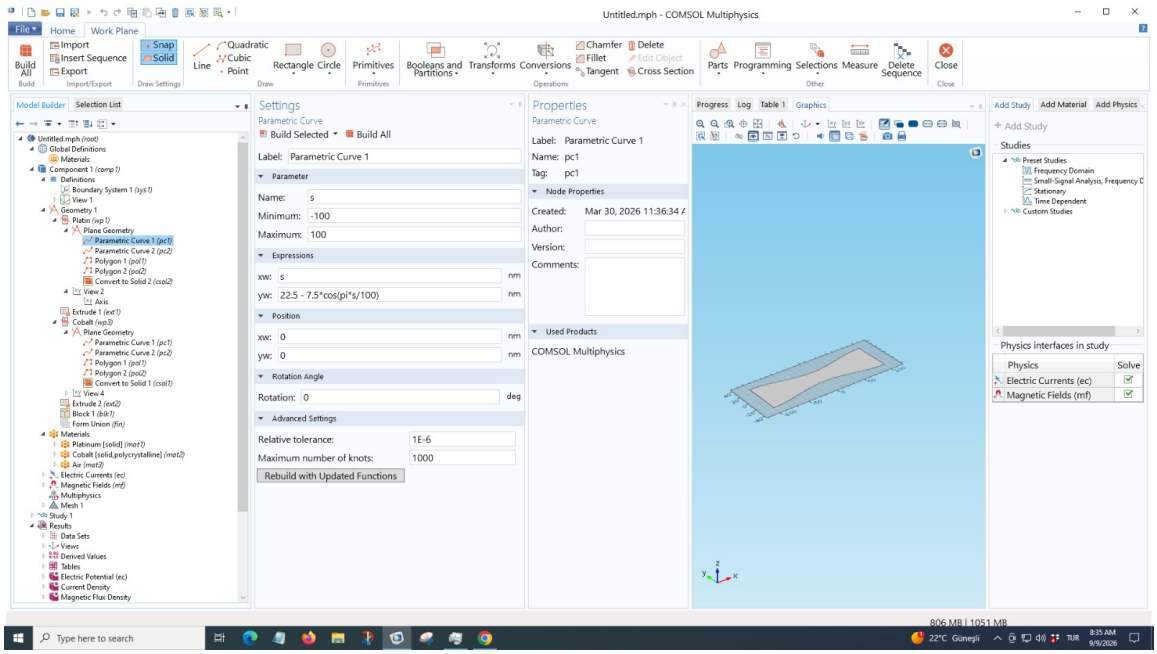


Figure 6: COMSOL settings used for the first parametric curve defining the upper boundary of the dumbbell geometry. The parameter s ranges from -100 to 100 nm, with $x_w = s$ and $y_w = 22.5 - 7.5\cos(\pi s/100)$.

A second *Parametric Curve* feature was then added using the same parameter range,

$$-100 \leq s \leq 100, \tag{38}$$

but with the coordinate expressions

$$x_w = s, \tag{39}$$

and

$$y_w = -22.5 + 7.5\cos\left(\frac{\pi s}{100}\right). \tag{40}$$

The second curve therefore defines the lower boundary of the device, as shown in Fig. 7.

Together, the two curves can be written as

$$y(x) = \pm\left[22.5 - 7.5\cos\left(\frac{\pi x}{100}\right)\right] \text{ nm}, \qquad -100 \leq x \leq 100 \text{ nm}. \tag{41}$$

At the center of the structure, $x = 0$, the upper and lower boundaries are located at $y = \pm 15$ nm, giving a constriction width of

$$W_{\text{center}} = 30 \text{ nm}. \tag{42}$$

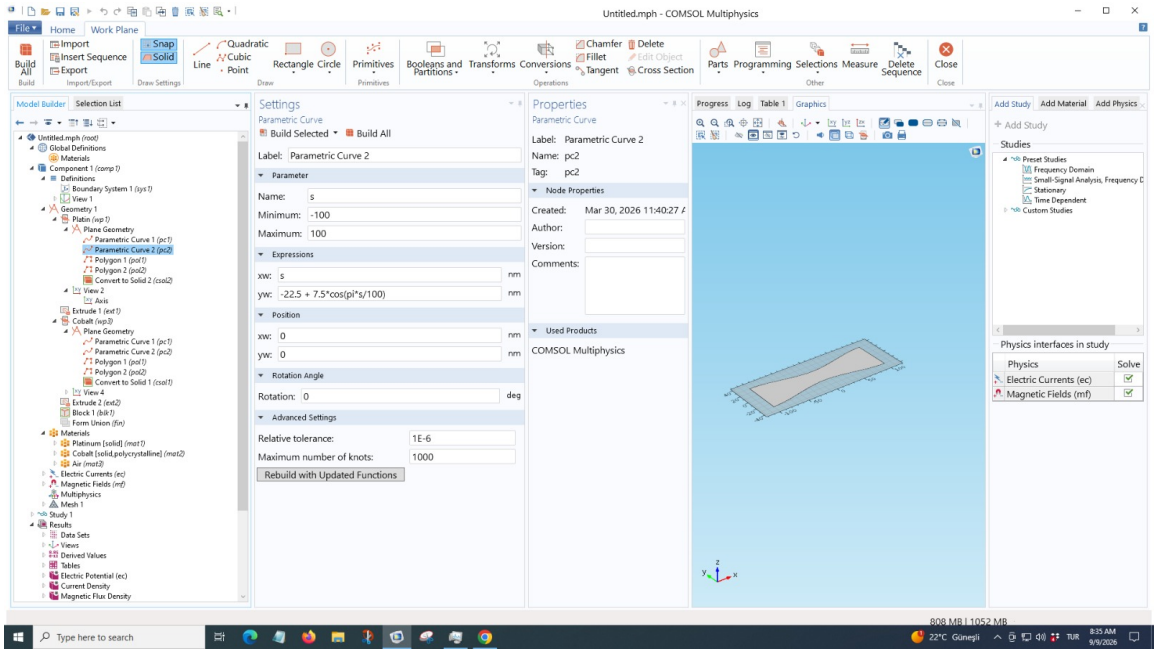


Figure 7: COMSOL settings used for the second parametric curve defining the lower boundary of the dumbbell geometry.

At the two ends, $x = \pm 100$ nm, the boundaries become $y = \pm 30$ nm, giving an end width of

$$W_{\text{end}} = 60 \text{ nm}. \tag{43}$$

The two parametric curves define the upper and lower boundaries but do not form a closed two-dimensional domain by themselves. Two *Polygon* features were therefore added with the object type set to *Open curve*.

The first polygon was used to close the left end of the dumbbell. Its two vertices were entered as

$$(-100, 30) \text{ nm} \quad \text{and} \quad (-100, -30) \text{ nm}. \tag{44}$$

The second polygon was used to close the right end, using

$$(100, 30) \text{ nm} \quad \text{and} \quad (100, -30) \text{ nm}. \tag{45}$$

The polygon settings used to close the left and right ends are shown in Figs. 8 and 9, respectively.

After defining the two parametric curves and the two straight end segments, a *Convert to Solid* operation was added. The four objects `pc1`, `pc2`, `pol1`, and `pol2` were selected as the input objects. This operation converted the closed contour into a two-dimensional solid domain corresponding to the lateral footprint of the Pt layer.

The resulting two-dimensional Pt domain was then converted into a three-dimensional

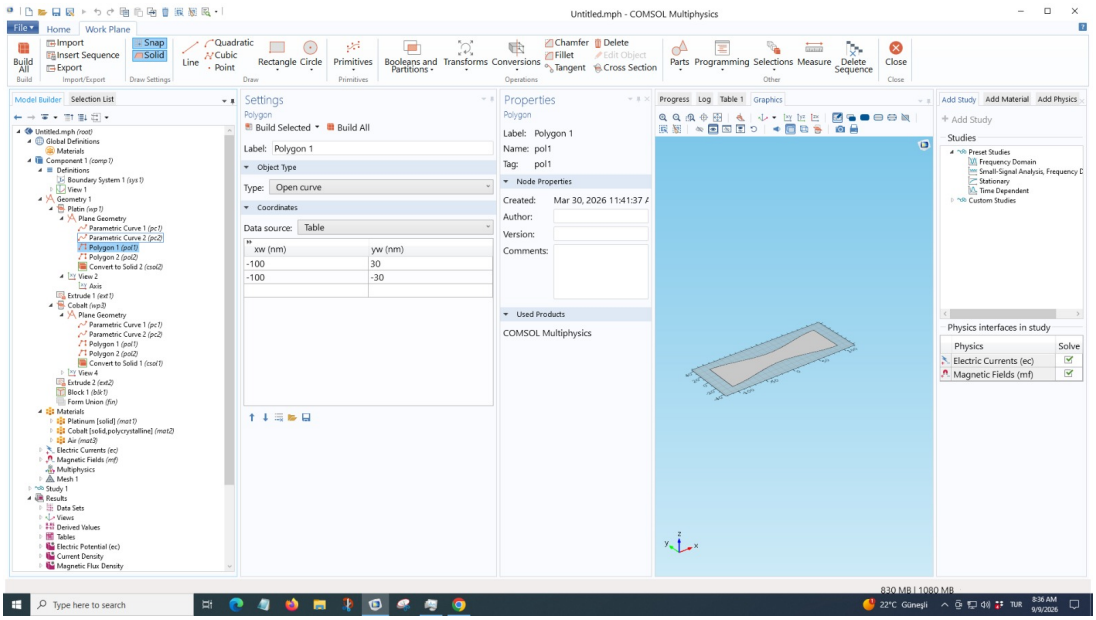


Figure 8: Definition of the open polygon segment used to close the left end of the dumbbell geometry.

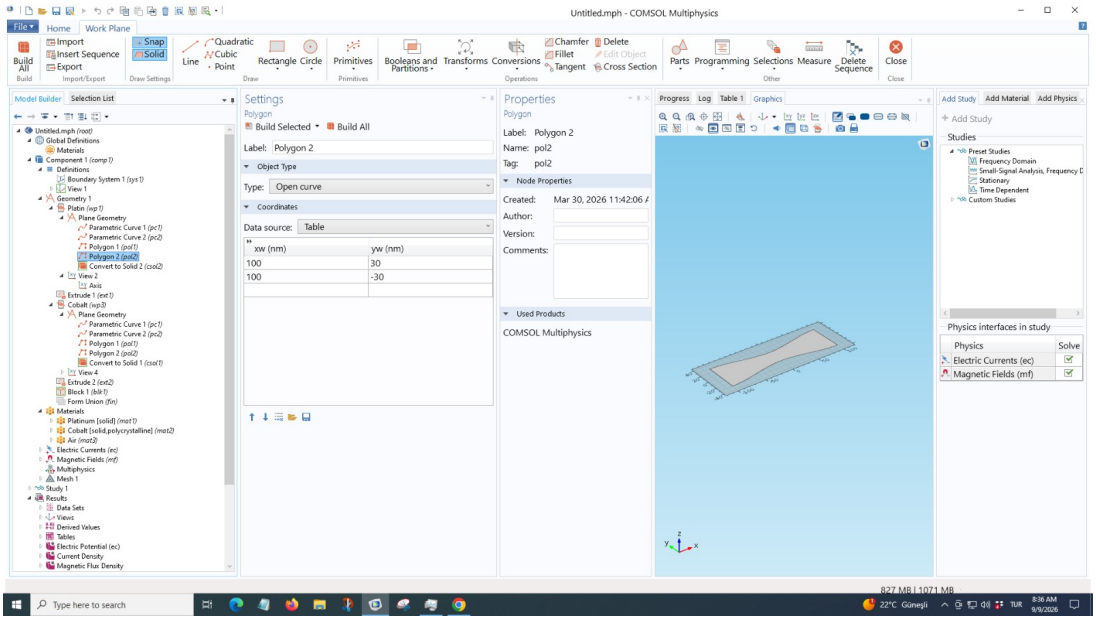


Figure 9: Definition of the open polygon segment used to close the right end of the dumbbell geometry.

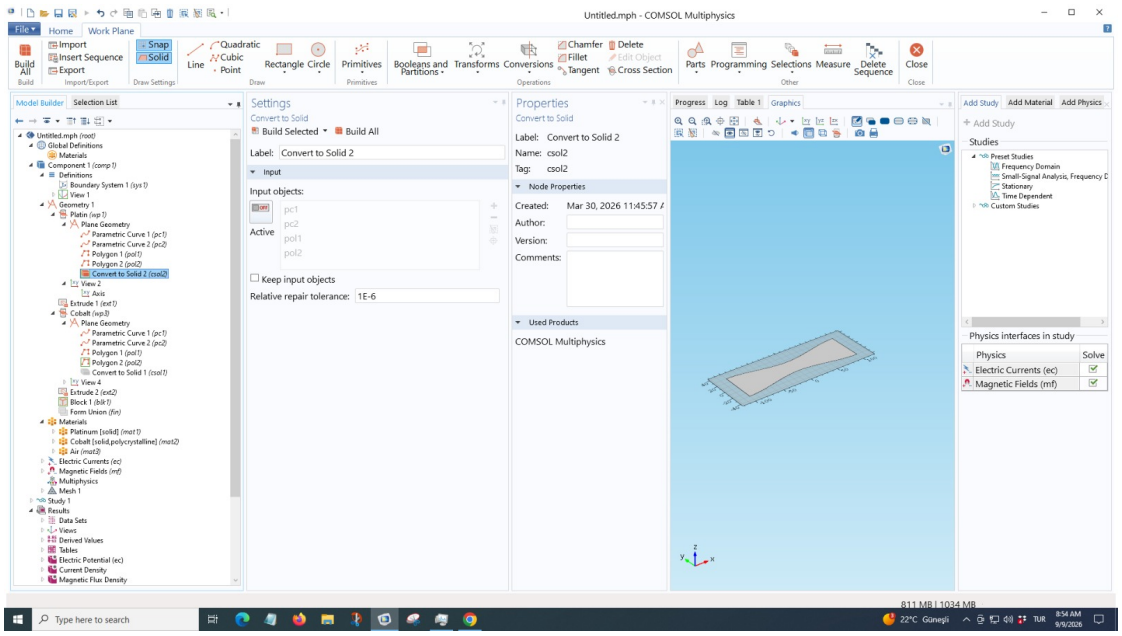


Figure 10: Conversion of the two parametric curves and the two polygon segments into a closed two-dimensional dumbbell domain.

layer using an *Extrude* operation. The Pt work plane was selected as the extrusion source and the extrusion distance was set to

$$t_{\mathrm{Pt}} = 10 \text{ nm}. \tag{46}$$

The Pt layer therefore extended from

$$z = 0 \tag{47}$$

to

$$z = 10 \text{ nm}. \tag{48}$$

The corresponding COMSOL extrusion settings are shown in Fig. 11.

Construction of the Co layer. A second work plane was created for the Co layer. The work plane was placed parallel to the xy plane at

$$z = 10 \text{ nm}, \tag{49}$$

which corresponds to the top surface of the Pt layer. This ensured that the Co layer was placed directly on top of Pt.

The same two parametric curves and the same two end-closing polygon segments were reproduced in this work plane so that the Co layer had the same lateral dumbbell footprint as the Pt layer.

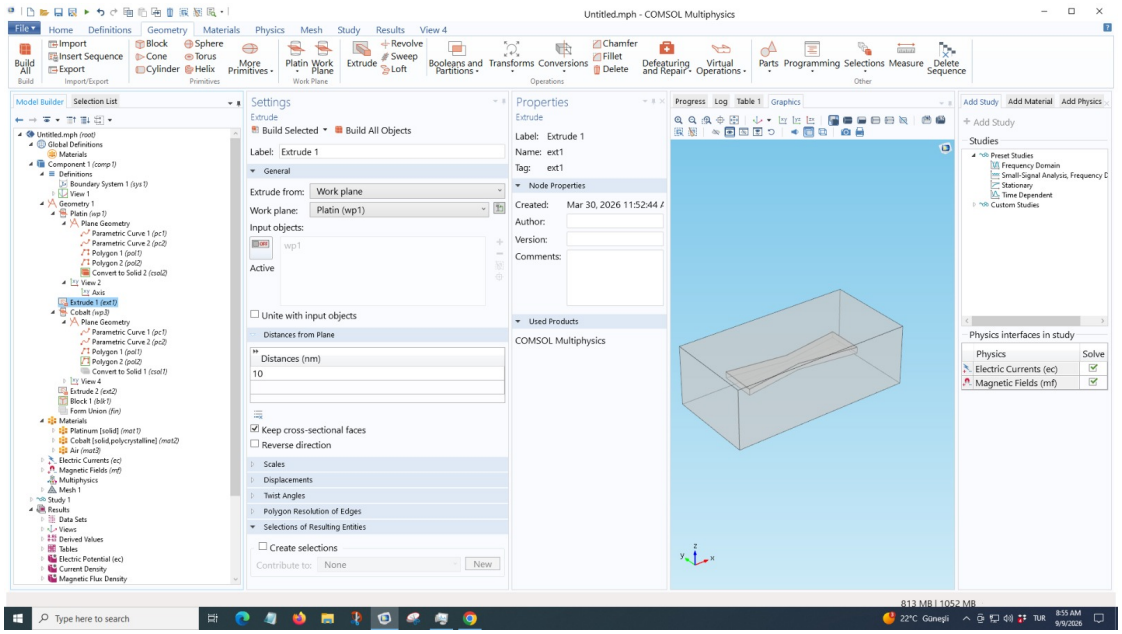


Figure 11: Extrusion of the two-dimensional Pt dumbbell footprint by 10 nm to generate the three-dimensional Pt layer.

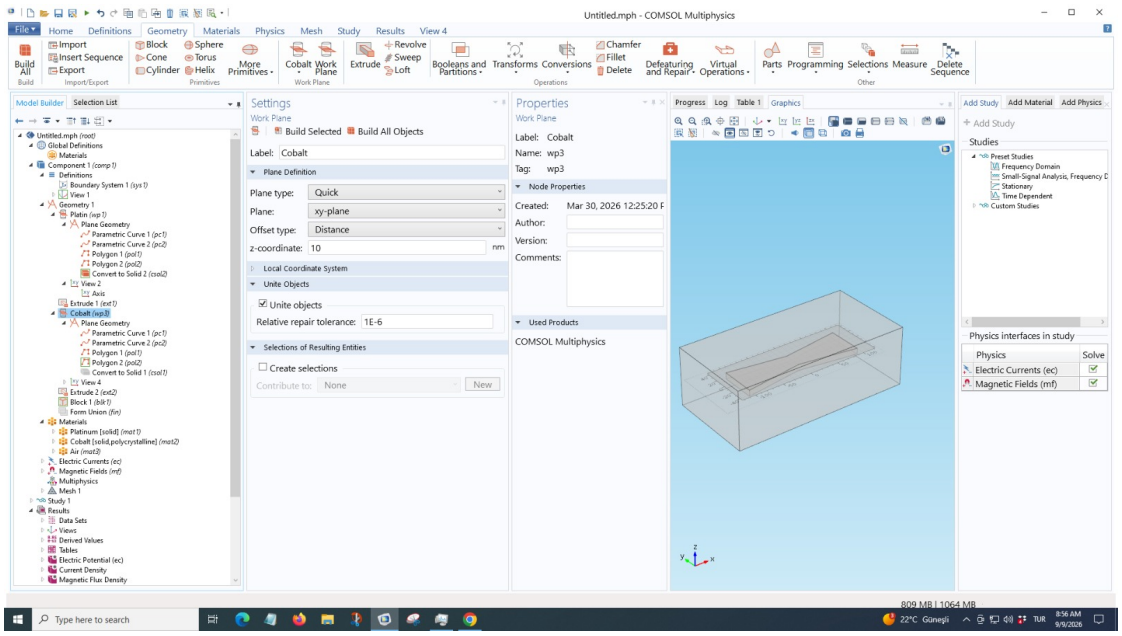


Figure 12: Definition of the Co work plane in the xy plane at $z = 10$ nm, corresponding to the upper surface of the Pt layer.

After converting the closed Co contour into a two-dimensional solid domain, a second *Extrude* operation was applied. The extrusion distance was set to

$$t_{\mathrm{Co}} = 0.5 \text{ nm}. \tag{50}$$

The Co layer therefore extended from

$$z = 10 \text{ nm} \tag{51}$$

to

$$z = 10.5 \text{ nm}. \tag{52}$$

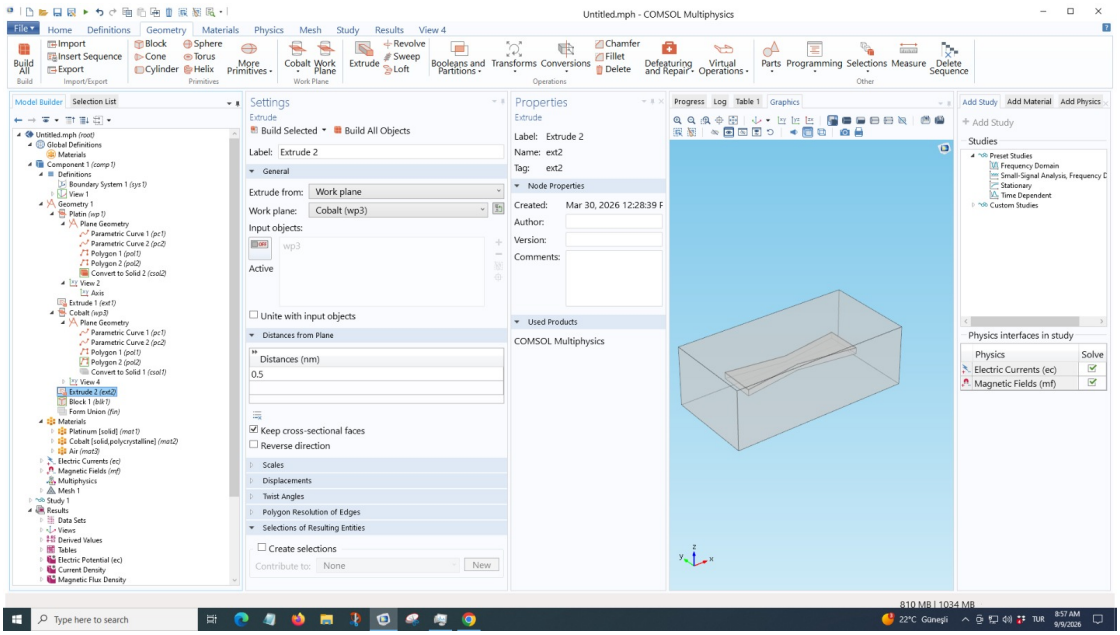


Figure 13: Extrusion of the Co dumbbell footprint by 0.5 nm to generate the three-dimensional Co layer on top of Pt.

The resulting metallic stack therefore consisted of

$$\mathrm{Pt}(10 \text{ nm})/\mathrm{Co}(0.5 \text{ nm}). \tag{53}$$

Construction of the surrounding air domain. A rectangular *Block* primitive was added around the Pt/Co structure to represent the surrounding air region. This domain was included primarily for the Magnetic Fields (`mf`) calculation. The magnetic field generated by the electric current is not confined to the conducting Pt and Co layers, but extends into the surrounding space. Therefore, an additional air domain was required so that the current-induced magnetic-field distribution could be calculated continuously outside the metallic structure instead of being artificially terminated at the Pt/Co boundaries.

The dimensions of the air block were set to

$$300 \times 150 \times 100 \text{ nm}^3. \tag{54}$$

Using the corner of the block as the reference position, its coordinates were set to

$$(x, y, z) = (-150, -75, -50) \text{ nm}. \tag{55}$$

The resulting air region therefore extended over

$$-150 \leq x \leq 150 \text{ nm}, \tag{56}$$

$$-75 \leq y \leq 75 \text{ nm}, \tag{57}$$

and

$$-50 \leq z \leq 50 \text{ nm}. \tag{58}$$

These dimensions completely enclosed the Pt/Co device and provided a surrounding computational region in which the current-induced magnetic field and its associated field lines could extend through the space around the device.

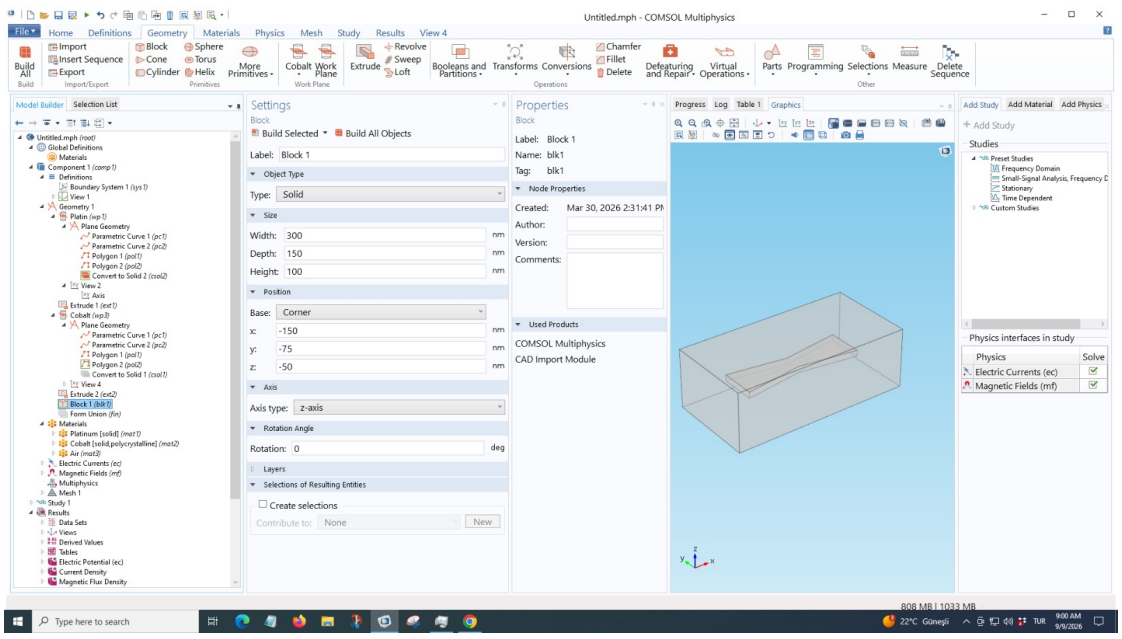


Figure 14: Definition of the surrounding air block used for the magnetic field calculation. The block dimensions were $300 \times 150 \times 100$ nm^3, with the corner positioned at $(-150, -75, -50)$ nm.

Finally, the geometry sequence was completed using the *Form Union* operation with a relative repair tolerance of 10^{-6}. This operation generated the final computational

geometry while retaining the internal boundaries required to distinguish the Pt, Co, and air domains during the subsequent material assignment and physics setup.

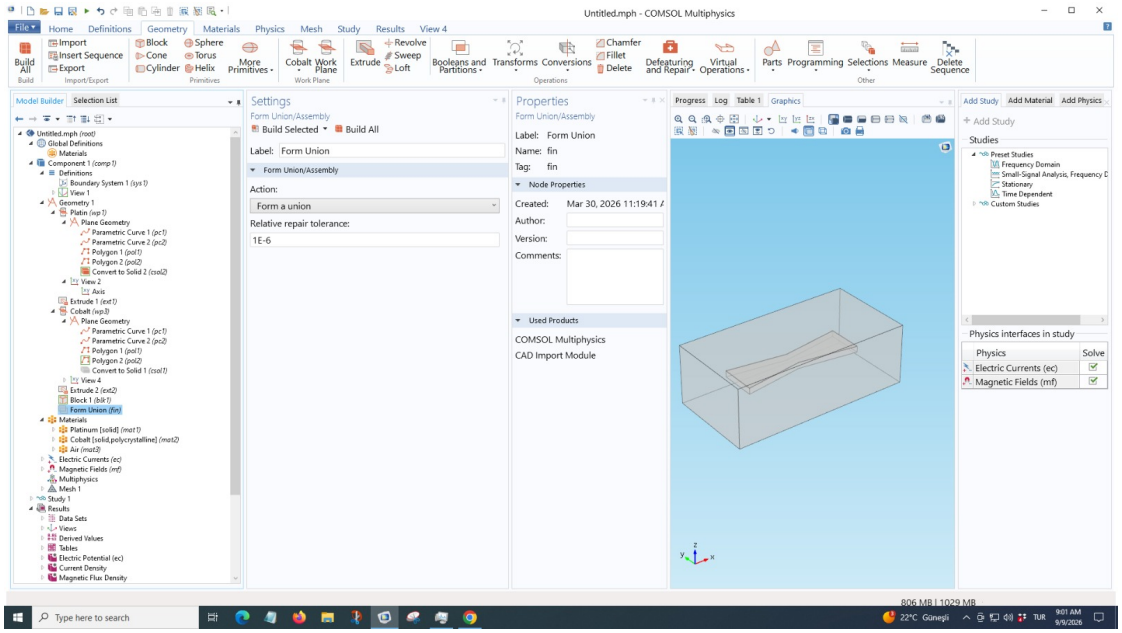


Figure 15: Finalization of the Pt/Co/air computational geometry using the Form Union operation.

3.2.3 Material Assignment

After completing the geometry, material properties were assigned separately to the Pt, Co, and air domains. The material definitions used for the electromagnetic calculation were taken from the COMSOL Multiphysics Material Library [4].

The Pt layer was assigned to Domain 2 using the *Platinum [solid]* material from the COMSOL Material Library. The electrical conductivity used in the model was

$$\sigma_{\mathrm{Pt}} = 9.52 \times 10^{6}\ \mathrm{S/m}. \tag{59}$$

The relative permittivity and relative permeability were taken as

$$\varepsilon_r = 1, \qquad \mu_r = 1. \tag{60}$$

The corresponding Pt material assignment is shown in Fig. 16.

The Co layer was assigned to Domain 3 using the *Cobalt [solid, polycrystalline]* material from the COMSOL Material Library. The electrical conductivity used for Co was

$$\sigma_{\mathrm{Co}} = 1.60 \times 10^{6}\ \mathrm{S/m}, \tag{61}$$

with

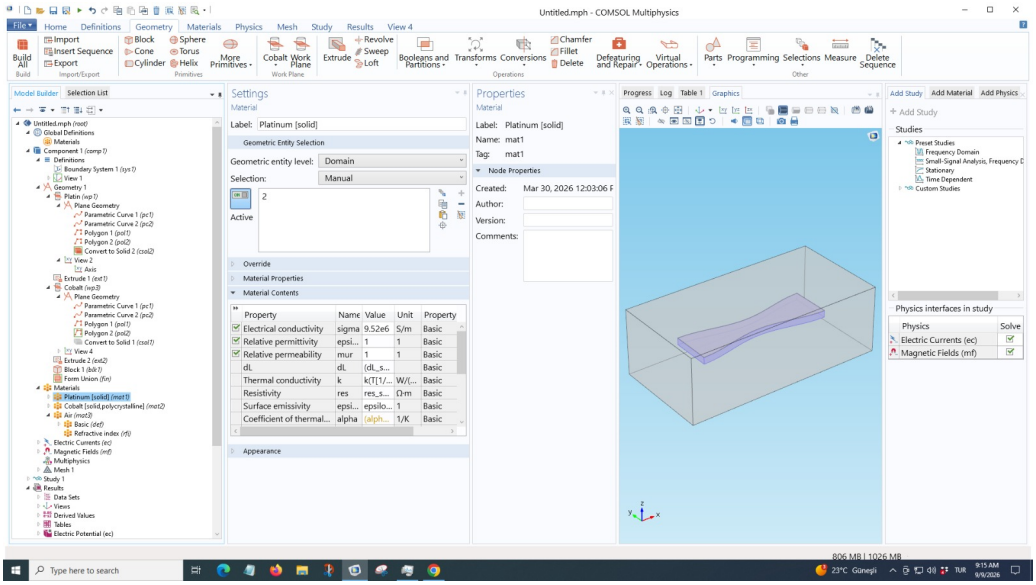


Figure 16: Assignment of the *Platinum [solid]* material to Domain 2. The electrical conductivity used in the model was 9.52×10^6 S/m, taken from the COMSOL Material Library.

$$\varepsilon_r = 1, \qquad \mu_r = 1. \tag{62}$$

The corresponding Co material assignment is shown in Fig. 17.

The surrounding air region was assigned to Domain 1. For the electromagnetic calculation, air was treated as nonconducting, with

$$\sigma_{\text{air}} = 0 \text{ S/m}, \tag{63}$$

and with relative permittivity and relative permeability both equal to unity,

$$\varepsilon_r = 1, \qquad \mu_r = 1. \tag{64}$$

The corresponding air-domain assignment is shown in Fig. 18.

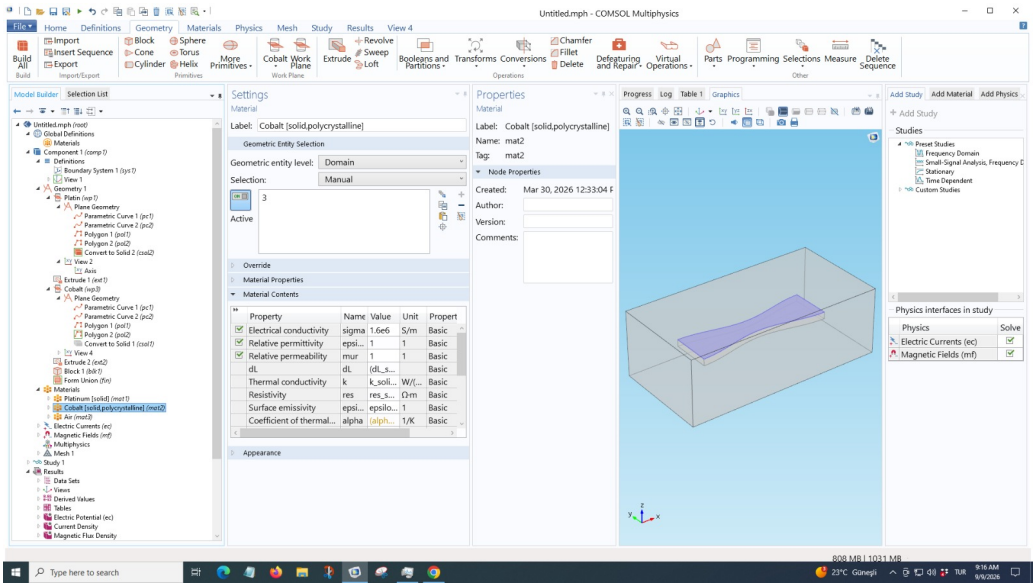


Figure 17: Assignment of the *Cobalt [solid, polycrystalline]* material to Domain 3. The electrical conductivity used in the model was 1.60×10^6 S/m, taken from the COMSOL Material Library.

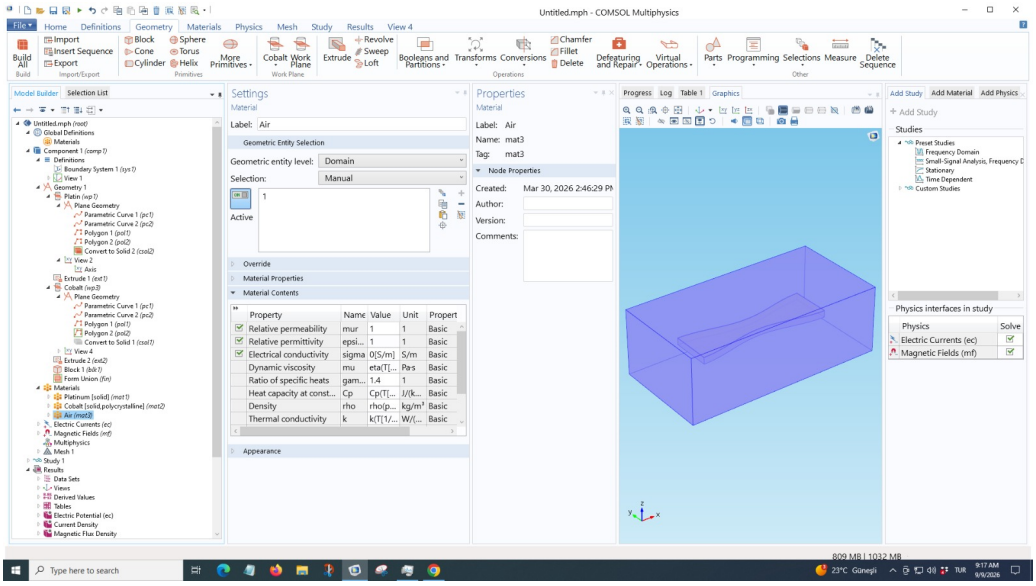


Figure 18: Assignment of the air material to Domain 1. The surrounding region was treated as nonconducting, with $\sigma_{\text{air}} = 0$ S/m.

3.2.4 Electric Currents Physics Setup

After assigning the material properties, the *Electric Currents (ec)* physics interface was configured to calculate the electric-potential and current-density distributions in the metallic Pt/Co structure.

The Electric Currents interface was applied only to Domains 2 and 3, corresponding to the Pt and Co layers, respectively. The surrounding air domain was therefore excluded from the electric-current calculation. Within the conducting domains, COMSOL solves the stationary current-conservation equation together with Ohm's law,

$$\nabla \cdot \mathbf{J} = 0, \tag{65}$$

$$\mathbf{J} = \sigma \mathbf{E}, \tag{66}$$

and

$$\mathbf{E} = -\nabla V, \tag{67}$$

where $\mathbf{J}$ is the electric current density, σ is the electrical conductivity, $\mathbf{E}$ is the electric field, and V is the electric potential.

A *Terminal* boundary condition was then added to one end of the Pt/Co structure. Boundaries 15 and 16 were selected for this condition, corresponding to the end faces of the Pt and Co layers. The terminal type was set to *Current*, and the applied current was specified as

$$I_0 = 145 \times 10^{-6}\ \text{A} = 145\ \mu\text{A}. \tag{68}$$

At the opposite end of the device, a *Ground* boundary condition was applied to Boundaries 6 and 9. The electric potential on these boundaries was therefore fixed to

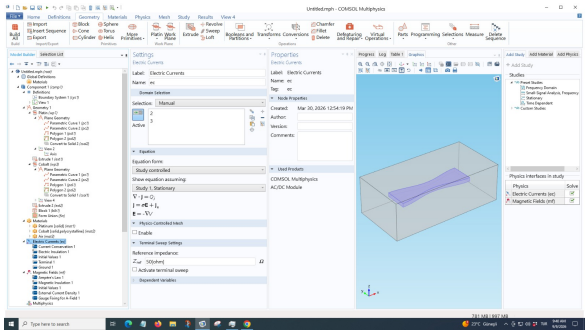
$$V = 0. \tag{69}$$

The remaining applicable external boundaries were assigned the default *Electric Insulation* condition,

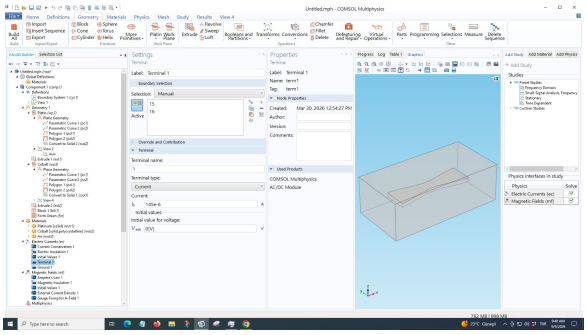
$$\mathbf{n} \cdot \mathbf{J} = 0, \tag{70}$$

where $\mathbf{n}$ is the outward surface-normal vector. This condition prevents electric current from flowing through the side surfaces of the conducting structure. The terminal boundaries are automatically overridden by the Terminal condition and are therefore excluded from the insulation condition.

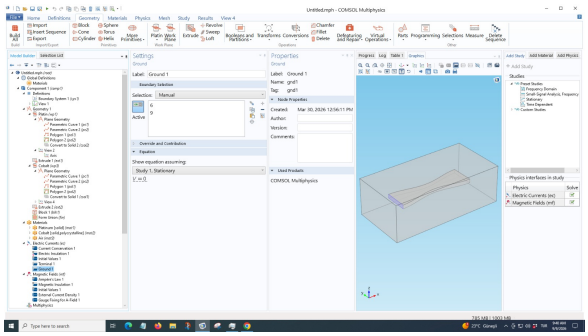
The complete Electric Currents configuration is summarized in Fig. 19.



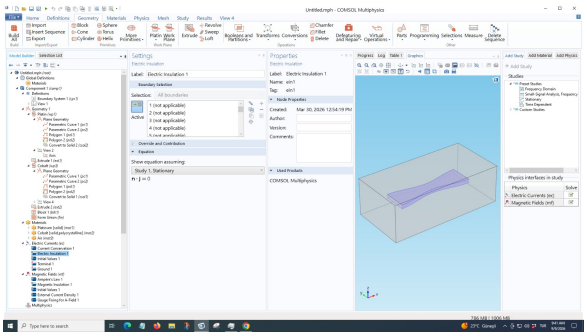
(a) Electric Currents physics applied to Domains 2 and 3, corresponding to the Pt and Co layers.



(b) Current terminal applied to Boundaries 15 and 16 with $I_0 = 145\ \mu\text{A}$.



(c) Ground condition applied to Boundaries 6 and 9, imposing $V = 0$.



(d) Electric insulation condition applied to the remaining applicable boundaries, imposing $\mathbf{n} \cdot \mathbf{J} = 0$.

Figure 19: Configuration of the Electric Currents (`ec`) physics in COMSOL: (a) conducting-domain selection, (b) applied current terminal, (c) ground boundary condition, and (d) electric insulation condition.

3.2.5 Magnetic Fields Interface

After defining the geometry and assigning the material properties, the *Magnetic Fields* (`mf`) interface was added in order to calculate the magnetic field distribution generated by the electric current flowing through the Pt/Co structure. The current density obtained from the *Electric Currents* (`ec`) interface was used as the source term in the magnetic-field calculation. Figure 20 summarizes the main settings used in the `mf` interface. The `Magnetic Fields` node was applied to all domains, including the surrounding air region, the platinum layer, and the cobalt layer. Solving the magnetic field in the surrounding air domain is important because the magnetic field is not confined to the conducting layers; it extends into the surrounding space. Therefore, the air domain was included so that the field lines around the device could be captured properly.

Within the `mf` interface, the `Ampère's Law` feature was also applied to all domains. This formulation solves for the magnetic vector potential and the resulting magnetic flux density according to

$$\nabla \times \mathbf{H} = \mathbf{J}_e, \qquad \mathbf{B} = \nabla \times \mathbf{A}, \tag{71}$$

where $\mathbf{J}_e$ is the externally supplied current density. In the present model, this current density was not entered manually; instead, it was imported directly from the electric-current solution through the `External Current Density` feature.

All outer boundaries in the magnetic-field model were assigned the `Magnetic Insulation` condition. This imposes

$$\mathbf{n} \times \mathbf{A} = 0, \tag{72}$$

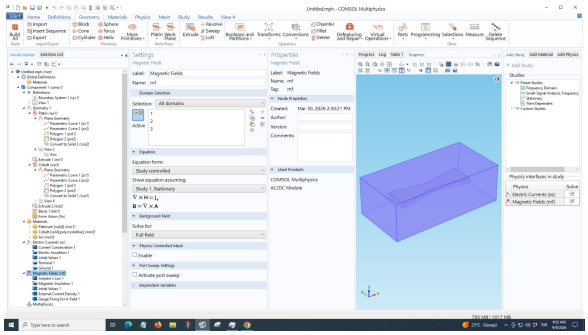
which is the default boundary condition in the `mf` interface and provides a closed magnetic boundary for the finite computational domain.

The `External Current Density` feature was applied to the platinum and cobalt domains, and its components were defined as

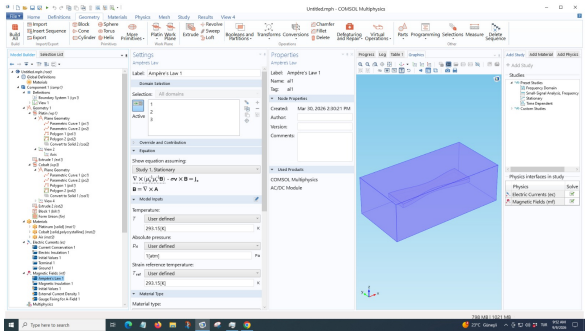
$$J_{e,x} = \texttt{ec.Jx}, \qquad J_{e,y} = \texttt{ec.Jy}, \qquad J_{e,z} = \texttt{ec.Jz}. \tag{73}$$

This step is essential because it couples the `ec` and `mf` physics interfaces: the current distribution first computed electrically is then used to generate the magnetic field responsible for the Oersted contribution.

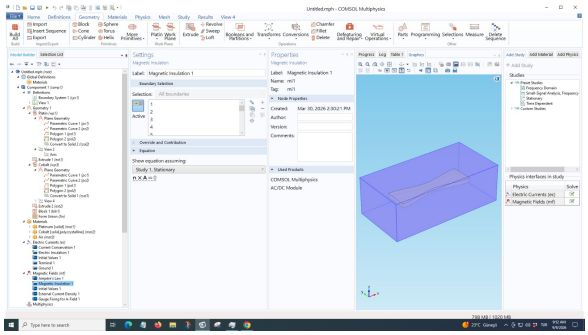
Finally, the `Gauge Fixing for A-Field` feature was enabled for all domains. This setting was important for obtaining a stable solution, since the magnetic vector potential $\mathbf{A}$ is not unique unless a gauge condition is imposed. In practice, including gauge fixing prevented solver difficulties and allowed the stationary magnetic-field problem to converge reliably.



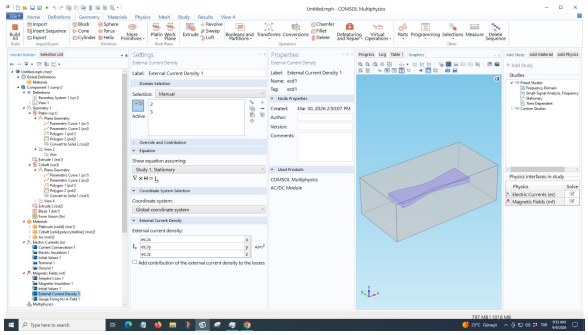
(a) Magnetic Fields domain selection.



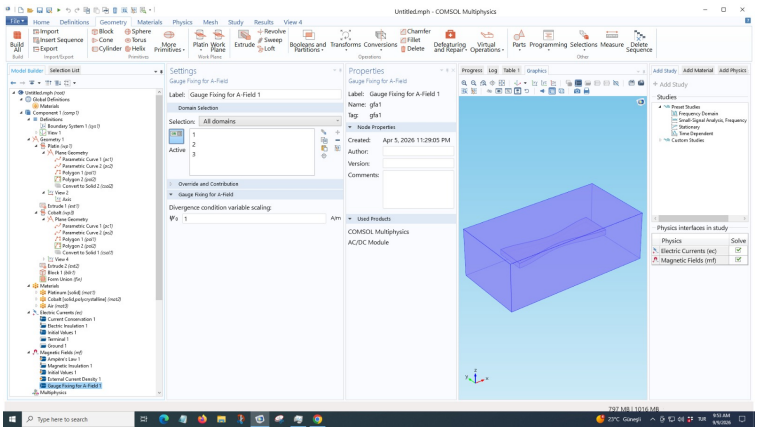
(b) Ampère's Law settings.



(c) Magnetic insulation boundary condition.



(d) External current density coupling from the `ec` interface.



(e) Gauge fixing for the magnetic vector potential.

Figure 20: Configuration of the *Magnetic Fields* (`mf`) interface in COMSOL Multiphysics. The magnetic field problem was solved in all domains, using the current density obtained from the *Electric Currents* (`ec`) interface as the source term.

3.2.6 Mesh Generation

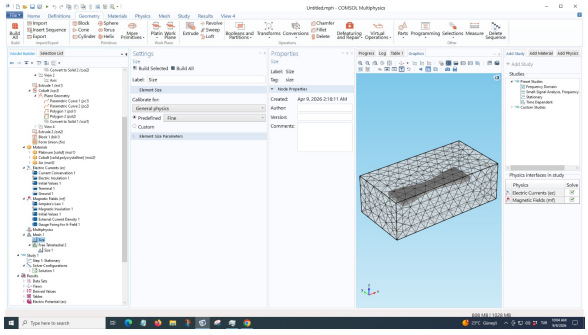
After completing the physics setup, the computational domain was discretized using a user-controlled mesh. A user-controlled mesh was preferred so that different mesh resolutions could be assigned to the metallic Pt/Co device and to the surrounding air region.

The default element size was first set to the predefined *Fine* setting, as shown in Fig. 21(a). This mesh size was retained for the Pt and Co domains, where a relatively higher spatial resolution was required to resolve the variation of the electric current density and the corresponding current-induced magnetic field. This is particularly important in the dumbbell structure because the device width varies along the x direction and the current distribution changes significantly around the central constriction.

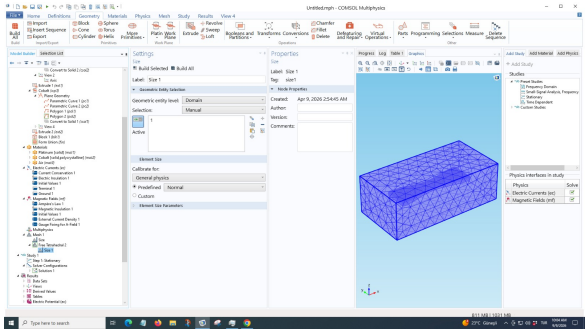
A local mesh-size override was then introduced for Domain 1, corresponding to the surrounding air region. For this domain, the predefined mesh size was changed to *Normal*, as shown in Fig. 21(b). Consequently, the Pt and Co domains retained the finer discretization, whereas a relatively coarser mesh was used in the air region.

This difference in mesh resolution was introduced to balance numerical accuracy and computational cost. Reducing the characteristic element size increases the number of mesh elements and degrees of freedom, which in turn increases the required memory and computational time. Therefore, the finer mesh was retained in the Pt/Co region, where the quantities of interest vary more strongly, while a coarser mesh was considered sufficient in the surrounding air domain, whose primary purpose was to allow the magnetic field to extend outside the metallic device.

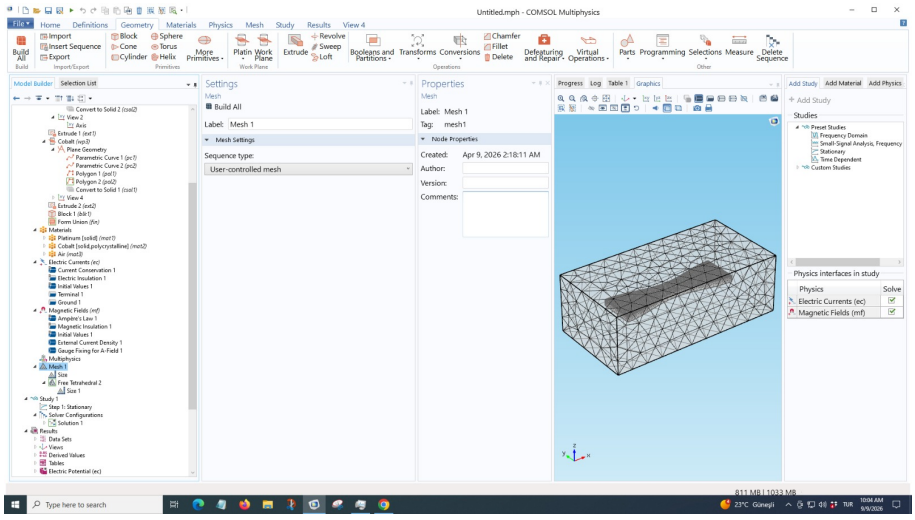
The three-dimensional domains were discretized using the *Free Tetrahedral* mesh feature. After defining the global and local element-size settings, the complete mesh was generated using the *Build All* command. The resulting tetrahedral mesh is shown in Fig. 21(c).



(a) Global mesh-size setting defined as *Fine*.



(b) Local mesh-size override for the air domain (Domain 1), defined as *Normal*.



(c) Final tetrahedral mesh generated after selecting *Build All*.

Figure 21: Mesh configuration used in the COMSOL model: (a) the global predefined *Fine* mesh retained for the Pt/Co device, (b) the local *Normal* mesh-size override applied to the surrounding air domain, and (c) the final tetrahedral mesh after building the complete mesh sequence.

3.2.7 Stationary Study and Solver Configuration

After the mesh was generated, the coupled electric and magnetic field problem was solved using a stationary study. Under *Study 1*, a *Stationary* study step was used, with both the *Electric Currents* (`ec`) and *Magnetic Fields* (`mf`) physics interfaces included in the solution. The previously generated *Mesh 1* was used for the calculation.

A stationary formulation was appropriate because the objective of the COMSOL calculation was to determine the steady-state spatial distributions of the current density and the corresponding current-induced magnetic field for a fixed applied terminal current. The time dependence of the binary current pulses was introduced later in the OOMMF simulations.

The stationary study configuration is shown in Fig. 22(a).

The stationary solver was configured with a relative tolerance of

$$\mathrm{tol}_{\mathrm{rel}} = 10^{-3}. \tag{74}$$

This tolerance controls the convergence criterion used by the stationary solver. The corresponding solver settings are shown in Fig. 22(b).

Rather than solving all dependent variables simultaneously in a single fully coupled system, a *Segregated* solution strategy was used. The coupled problem was divided into separate groups of dependent variables and solved sequentially. The termination technique was set to *Tolerance*, with a maximum of

$$N_{\max} = 100 \tag{75}$$

segregated iterations. The tolerance factor and residual factor were set to 1 and 1000, respectively, while no additional stabilization or acceleration method was applied.

The segregated-solver settings are shown in Fig. 22(c).

The first segregated step solved the electrical variables associated with the *Electric Currents* interface, including the electric potential and the terminal voltage. Once the electric-current distribution had been updated, the second segregated step solved the variables associated with the *Magnetic Fields* interface, including the magnetic vector potential and the additional variable associated with the gauge condition.

The two groups can therefore be represented schematically as

$$\text{Segregated Step 1:} \qquad V,\ V_{\mathrm{terminal}}, \tag{76}$$

followed by

$$\text{Segregated Step 2:} \qquad \mathbf{A},\ \psi, \tag{77}$$

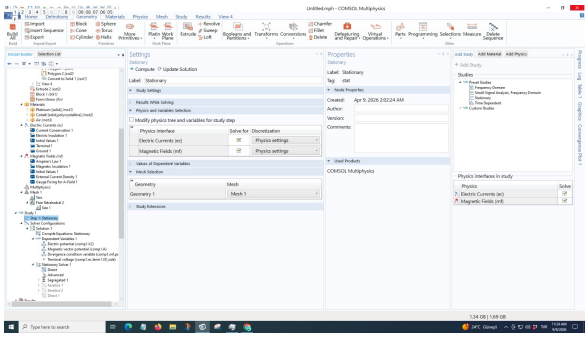
where V is the electric potential, $\mathbf{A}$ is the magnetic vector potential, and ψ denotes

the additional scalar variable introduced by the magnetic-vector-potential formulation and gauge condition.

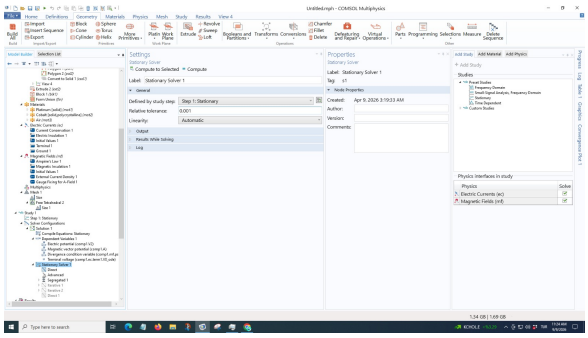
Because the magnetic-field calculation uses the current density obtained from the `ec` solution through `ec.Jx`, `ec.Jy`, and `ec.Jz`, the segregated solver evaluates the electrical variables first and the magnetic variables subsequently within the stationary solution sequence. The electromagnetic coupling used here is therefore one-way from the Electric Currents interface to the Magnetic Fields interface. The corresponding dependent-variable grouping is shown in Fig. 22(d).

For the direct solution of the sparse linear systems generated during the stationary calculation, the *PARDISO* direct solver was selected. The preordering algorithm was set to the multithreaded nested-dissection option available in COMSOL. The corresponding linear-solver configuration is shown in Fig. 22(e).

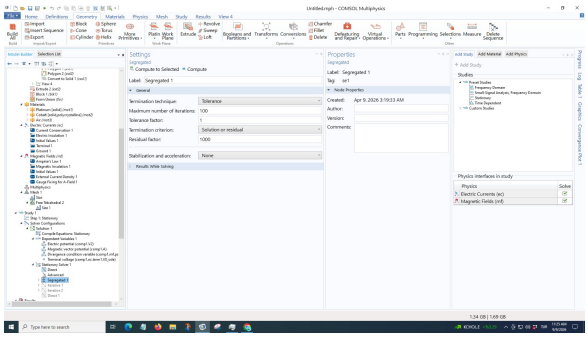
After the study and solver settings were defined, the stationary problem was computed to obtain the converged electric-current and magnetic-field solutions.



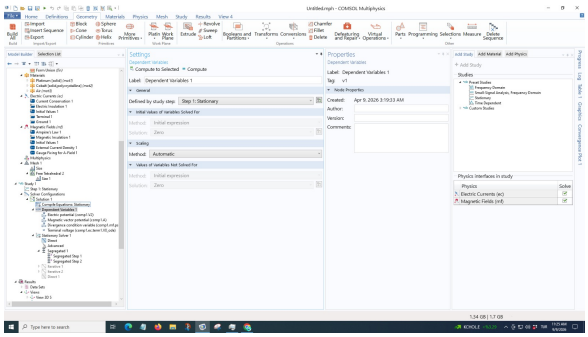
(a) Stationary study including the Electric Currents and Magnetic Fields interfaces.



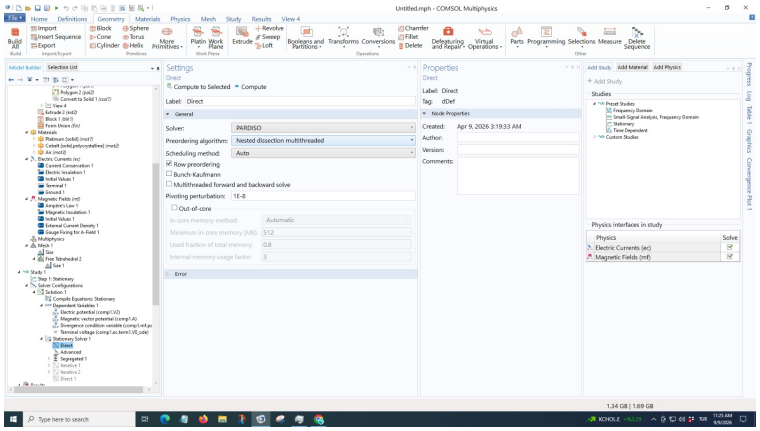
(b) Stationary solver configuration with a relative tolerance of 10^{-3}.



(c) Segregated-solver settings, including the tolerance-based termination criterion and maximum number of iterations.



(d) Dependent-variable grouping used for the segregated electric and magnetic solution steps.



(e) PARDISO direct linear-solver configuration.

Figure 22: Stationary study and solver configuration used in COMSOL: (a) stationary study definition, (b) stationary-solver tolerance, (c) segregated-solver settings, (d) dependent-variable grouping, and (e) PARDISO direct linear solver.

3.2.8 Post-Processing and Data Export

After the stationary solution converged, the COMSOL results were post-processed to visualize the spatial distributions of the electric current density and the current-induced magnetic field. The same solution was then sampled on two-dimensional cut planes and exported on a regular Cartesian grid for subsequent use in the OOMMF micromagnetic simulations.

Visualization of the current-density distribution. The spatial distribution of the electric current density was visualized using the current-density magnitude

$$|\mathbf{J}| = \sqrt{J_x^2 + J_y^2 + J_z^2}, \tag{78}$$

which corresponds to the COMSOL expression

$$\texttt{ec.normJ}. \tag{79}$$

The calculated current-density distribution is shown in Fig. 23. The current density is spatially nonuniform because the lateral width of the dumbbell structure changes along the transport direction. In particular, the current density increases toward the central constriction, where the available conducting cross section becomes smaller.

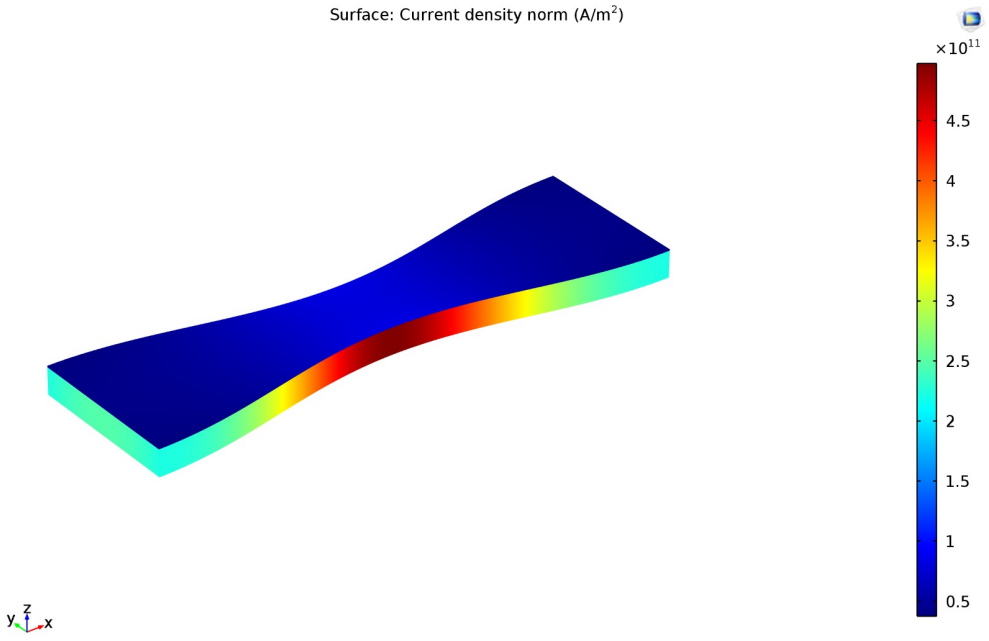


Figure 23: Spatial distribution of the current-density magnitude $|\mathbf{J}|$ in the Pt/Co dumbbell structure obtained from the stationary COMSOL solution at $I = 145\,\mu$A. The current density increases toward the central constriction because of the reduced conducting cross section.

Visualization of the current-induced magnetic field. The magnetic field generated by the applied electric current was obtained from the *Magnetic Fields* (`mf`) solution. The three field components were

$$\mathbf{H} = (H_x, H_y, H_z), \tag{80}$$

corresponding to the COMSOL variables

$$\texttt{mf.Hx}, \qquad \texttt{mf.Hy}, \qquad \texttt{mf.Hz}. \tag{81}$$

For visualization, the field magnitude

$$|\mathbf{H}| = \sqrt{H_x^2 + H_y^2 + H_z^2} \tag{82}$$

was displayed using the COMSOL expression

$$\texttt{mf.normH}. \tag{83}$$

The direction of the magnetic field was simultaneously represented using three-dimensional vector arrows based on `mf.Hx`, `mf.Hy`, and `mf.Hz`. The resulting current-induced magnetic-field distribution is shown in Fig. 24. This field represents the Oersted-field contribution generated by the electric current flowing through the metallic structure.

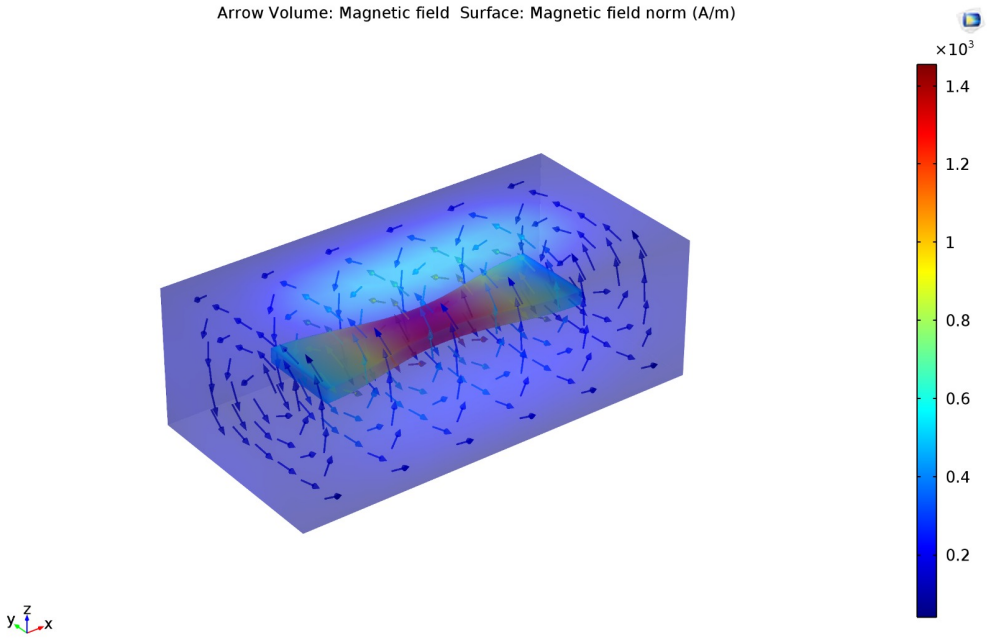


Figure 24: Current-induced magnetic-field distribution obtained from COMSOL at $I = 145\,\mu$A. The surface color represents the field magnitude $|\mathbf{H}|$, while the arrows indicate the local direction of (H_x, H_y, H_z).

Definition of the cut planes. The COMSOL finite-element solution is defined on an irregular tetrahedral mesh, whereas the OOMMF micromagnetic model uses a regular Cartesian mesh. Therefore, the COMSOL fields were not exported directly at the original finite-element nodes. Instead, two cut-plane data sets were introduced so that the required quantities could be sampled at physically relevant positions and subsequently evaluated on a regular grid.

For the electric current density, the cut plane was positioned at

$$z_J = 5 \text{ nm}. \tag{84}$$

The Pt layer occupied the region

$$0 \leq z \leq 10 \text{ nm}, \tag{85}$$

and therefore $z_J = 5$ nm corresponds to the mid-plane of the Pt layer. This plane was selected because the charge-current distribution in Pt is the quantity used to represent the current responsible for the spin–orbit-torque drive.

For the current-induced magnetic field, a second cut plane was positioned at

$$z_H = 10.25 \text{ nm}. \tag{86}$$

The Co layer extended from

$$10 \leq z \leq 10.5 \text{ nm}, \tag{87}$$

so that $z_H = 10.25$ nm corresponds to the mid-plane of the 0.5 nm Co layer. The Oersted field was sampled at this position because the skyrmion magnetization resides in the Co layer.

The two cut-plane definitions are shown in Figs. 25a and 25b, respectively.

Matching the COMSOL sampling grid to the OOMMF mesh. The lateral OOMMF discretization used in the micromagnetic model was

$$\Delta x = \Delta y = 2 \text{ nm}. \tag{88}$$

To minimize additional interpolation during the COMSOL-to-OOMMF data transfer, the export grid was therefore chosen to coincide with the centers of the OOMMF cells.

Along the x direction, the sampling coordinates were defined in COMSOL as

$$x = \texttt{range(-99[nm],2[nm],99[nm])}, \tag{89}$$

which gives

$$x = -99, -97, \ldots, 97, 99 \text{ nm}. \tag{90}$$

This corresponds to 100 sampling positions.

Similarly, along the y direction,

$$y = \texttt{range(-29[nm],2[nm],29[nm])}, \tag{91}$$

giving

$$y = -29, -27, \ldots, 27, 29 \text{ nm}, \tag{92}$$

corresponding to 30 sampling positions.

The complete export grid therefore contained

$$100 \times 30 = 3000 \tag{93}$$

sampling points.

The half-cell offset relative to the physical boundaries of the 200 nm $\times$ 60 nm lateral geometry was selected so that the COMSOL sampling coordinates coincided with the centers of the 2 nm $\times$ 2 nm OOMMF cells. Thus, the mapping was constructed according to

$$\text{COMSOL sampling point} \longleftrightarrow \text{OOMMF cell center}. \tag{94}$$

Export of the current-density and magnetic-field data. For the current-density data set, the Pt mid-plane cut plane at $z = 5$ nm was selected and the expression

$$\texttt{ec.normJ} \tag{95}$$

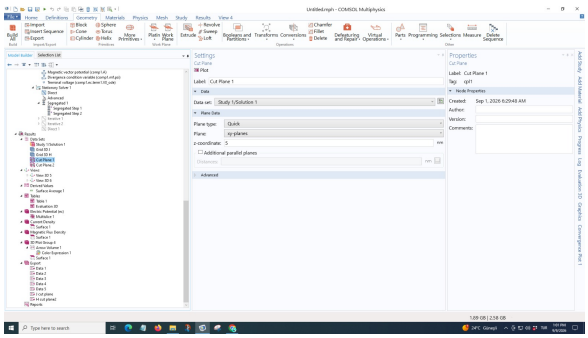
was evaluated on the 2 nm Cartesian grid described above.

For the magnetic-field data set, the Co mid-plane cut plane at $z = 10.25$ nm was selected. The three field components

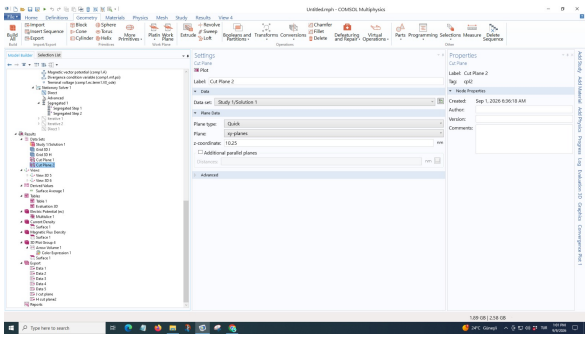
$$\texttt{mf.Hx}, \qquad \texttt{mf.Hy}, \qquad \texttt{mf.Hz} \tag{96}$$

were exported at the same lateral grid positions.

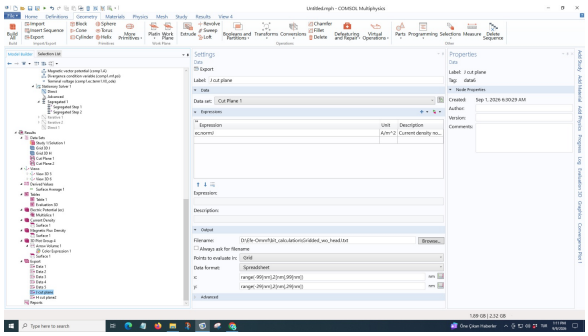
The corresponding cut-plane and export settings are summarized in Fig. 25. Using the same lateral sampling coordinates for both data sets ensured a consistent spatial correspondence between the current-density and magnetic-field maps before their transfer to OOMMF.



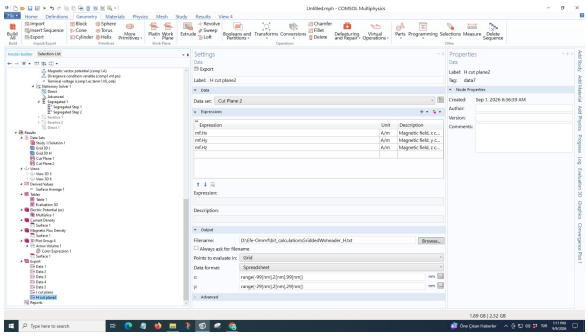
(a) Cut plane at $z = 5$ nm used to sample the current density at the mid-plane of the Pt layer.



(b) Cut plane at $z = 10.25$ nm used to sample the magnetic field at the mid-plane of the Co layer.



(c) Grid-based export of `ec.normJ` using a 2 nm lateral sampling interval.



(d) Grid-based export of `mf.Hx`, `mf.Hy`, and `mf.Hz` using the same 2 nm lateral sampling grid.

Figure 25: Extraction of the COMSOL data used in the OOMMF simulations: (a) Pt mid-plane cut plane for the current-density distribution, (b) Co mid-plane cut plane for the current-induced magnetic field, (c) regular-grid export of the current-density magnitude, and (d) regular-grid export of the three magnetic-field components.

3.3 OOMMF Micromagnetic Simulation

3.3.1 Overview of the OOMMF Simulation Workflow

Following the electromagnetic calculations performed in COMSOL, the time-dependent magnetization dynamics of the Co layer were simulated using the Object Oriented Micro-Magnetic Framework (OOMMF). Whereas COMSOL was used to determine the spatial distributions of the electric current density and the corresponding current-induced magnetic field, OOMMF was used to model the response of the magnetic skyrmion to these quantities.

The OOMMF simulation procedure consisted of a sequence of preparation, relaxation, and driven-dynamics steps. The overall workflow can be summarized as follows:

$$\text{geometry} \rightarrow \text{material parameters} \rightarrow \text{initial skyrmion} \rightarrow \text{relaxation}$$
$$\Downarrow$$
$$\text{COMSOL-derived } J(x,y),\ \mathbf{H}_{\mathrm{Oe}}(x,y) \rightarrow \text{SOT pulse sequence} \rightarrow \text{final skyrmion state}$$

First, the magnetic Co layer was represented using the same lateral dumbbell geometry employed in the COMSOL model, but discretized on a regular Cartesian micromagnetic mesh. The magnetic material parameters and the micromagnetic energy terms governing the skyrmion configuration were then defined.

An initial Néel-type skyrmion texture was subsequently introduced into the wider section of the device. Because this analytically constructed magnetization texture was not assumed to represent the equilibrium state of the complete micromagnetic system, it was first relaxed by energy minimization. The resulting relaxed skyrmion was then used as a common initial state for the current-driven simulations.

The spatially dependent current-density magnitude obtained from COMSOL,

$$J(x,y), \tag{97}$$

and the current-induced Oersted-field distribution,

$$\mathbf{H}_{\mathrm{Oe}}(x,y) = \left[H_x(x,y), H_y(x,y), H_z(x,y)\right], \tag{98}$$

were subsequently mapped onto the corresponding OOMMF cells. As described in Sec. 3.2.8, the COMSOL data were exported on a 2 nm $\times$ 2 nm lateral grid selected to match the centers of the OOMMF computational cells. This allowed the electromagnetic fields calculated in COMSOL to be incorporated directly into the micromagnetic model without requiring an additional lateral interpolation step.

The imported current-density distribution was used to generate the spin–orbit-torque

drive, while the imported magnetic-field distribution was included as the corresponding current-induced Oersted field. The binary input was encoded using bipolar current pulses,

$$1 \rightarrow +J(x, y), \qquad 0 \rightarrow -J(x, y), \tag{99}$$

with the sign of the Oersted field reversed consistently when the current direction was reversed.

Each four-bit input sequence therefore generated a time-dependent skyrmion trajectory determined by the combined effects of spin–orbit torque, the Oersted field, the micromagnetic energy landscape, the Skyrmion Hall effect, and geometrical confinement.

Although the complete transient magnetization dynamics were calculated throughout each four-pulse sequence, the reservoir-computing readout used in this work required only the state of the skyrmion after completion of the full input sequence. The final skyrmion position and size were therefore extracted and subsequently used as the physical reservoir-state variables.

The following subsections describe the OOMMF implementation in detail, including the computational geometry and mesh, magnetic material parameters, edge-confinement strategy, preparation and relaxation of the initial skyrmion, implementation of spin–orbit torque, incorporation of the COMSOL data, optimization of the driving conditions, and definition of the final-state readout.

3.3.2 Micromagnetic Geometry and Mesh

The magnetic Co layer was represented in OOMMF using a finite-difference micromagnetic model based on a regular Cartesian mesh [12]. Unlike the three-dimensional COMSOL model, which explicitly contained the Pt layer, the Co layer, and the surrounding air region, the OOMMF model contained only the ferromagnetic Co layer in which the magnetization dynamics and the skyrmion texture evolve. The influence of the Pt layer was incorporated separately through the spin–orbit-torque drive and the current-induced Oersted field.

Representation of the dumbbell geometry. The lateral magnetic geometry was defined using an `Oxs_ScriptAtlas`. The same analytical dumbbell boundary used in the COMSOL model was reproduced in OOMMF so that the electromagnetic and micromagnetic calculations described the same lateral device shape. The upper and lower boundaries were defined as

$$y_{\mathrm{b}}(x) = \pm \left[22.5 - 7.5 \cos\left(\frac{\pi x}{100}\right)\right] \text{ nm}, \qquad -100 \leq x \leq 100 \text{ nm}. \tag{100}$$

A point was therefore assigned to the magnetic region when

$$|y| \leq 22.5 - 7.5 \cos\left(\frac{\pi x}{100}\right). \tag{101}$$

This produces a 200 nm-long dumbbell with a width of 60 nm at the two ends and a minimum width of 30 nm at the central constriction, consistent with the skyrmion-reservoir geometry of Jiang *et al.* [1].

The corresponding OOMMF atlas was defined over the rectangular bounding region

$$-100 \leq x \leq 100 \text{ nm}, \tag{102}$$

$$-40 \leq y \leq 40 \text{ nm}, \tag{103}$$

and

$$0 \leq z \leq 0.5 \text{ nm}. \tag{104}$$

The y extent of the atlas is deliberately larger than the maximum physical half-width of the magnetic device. It defines only the rectangular computational bounding box; the actual magnetic region is determined by the analytical dumbbell function. Points outside the dumbbell boundary are therefore excluded from the magnetic region.

The relevant MIF definition can be written schematically as

```
proc dumbbell_shape { x y z } {
    set x_nm [expr {$x*1e9}]
    set y_nm [expr {$y*1e9}]
    set pi 3.141592653589793

    set y_bound \
        [expr {22.5 - 7.5*cos($x_nm*$pi/100.0)}]

    if {abs($y_nm) <= $y_bound} {
        return 1
    }
    return 0
}

Specify Oxs_ScriptAtlas:atlas {
    xrange { -100e-9 100e-9 }
    yrange { -40e-9 40e-9 }
    zrange { 0 0.5e-9 }
    regions { magnet }
```

```
    script dumbbell_shape
    script_args { rawpt }
}
```

Using a scripted atlas rather than approximating the constriction with a small number of rectangles allowed the continuously varying lateral boundary to be retained in the micromagnetic model.

Finite-difference discretization. The magnetic region was discretized using an `Oxs_RectangularMe` with cell dimensions

$$\Delta x = 2 \text{ nm}, \qquad \Delta y = 2 \text{ nm}, \qquad \Delta z = 0.5 \text{ nm}. \tag{105}$$

The corresponding OOMMF definition was

```
set dx 2e-9
set dy 2e-9
set dz 0.5e-9

Specify Oxs_RectangularMesh:mesh {
    cellsize { 2e-9 2e-9 0.5e-9 }
    atlas :atlas
}
```

The resulting discretization therefore used

$$\boxed{2 \times 2 \times 0.5 \text{ nm}^3} \tag{106}$$

micromagnetic cells. This is the same cell size employed in the original Pt/Co skyrmion-reservoir simulations of Jiang *et al.* [1].

In a finite-difference micromagnetic model, the magnetization is represented by a single magnetization vector within each computational cell. The spatial variation of the magnetization texture is therefore resolved through differences between neighboring cells. The cell dimensions must consequently be sufficiently small compared with the shortest characteristic magnetic length scales of the system. If the mesh is too coarse, the rapid magnetization rotation forming the skyrmion wall cannot be represented accurately, which can lead to errors in the exchange energy, skyrmion radius, edge interaction, and current-driven trajectory [9].

A useful characteristic scale for evaluating the discretization is the magnetostatic exchange length,

$$\ell_{\mathrm{ex}} = \sqrt{\frac{2A}{\mu_0 M_s^2}}, \tag{107}$$

which measures the length scale over which exchange interaction competes with magnetostatic energy [9]. Using the parameters employed in the present model,

$$A = 15 \times 10^{-12}\ \mathrm{J/m}, \tag{108}$$

and

$$M_s = 1.4 \times 10^{6}\ \mathrm{A/m}, \tag{109}$$

gives

$$\ell_{\mathrm{ex}} = \sqrt{\frac{2(15 \times 10^{-12})}{\mu_0 (1.4 \times 10^{6})^2}} \approx 3.49\ \mathrm{nm}. \tag{110}$$

The lateral cell dimensions therefore satisfy

$$\Delta x = \Delta y = 2\ \mathrm{nm} < \ell_{\mathrm{ex}} \approx 3.49\ \mathrm{nm}. \tag{111}$$

Thus, the lateral discretization is smaller than the exchange length and is sufficiently fine to resolve magnetization variations on the exchange-dominated length scale. This consideration provides a physical justification for the selected discretization in addition to its consistency with the mesh used in the reference skyrmion-reservoir model [1].

The exchange-length criterion should not be interpreted as an absolute proof of numerical convergence. Micromagnetic accuracy also depends on the DMI strength, anisotropy, skyrmion-wall width, local deformation near the constriction, and the interaction of the skyrmion with the device boundaries. Nevertheless, choosing a lateral cell size below the exchange length provides an important baseline requirement for resolving the relevant magnetic texture.

Discretization through the Co thickness. The magnetic Co layer has a physical thickness of only

$$t_{\mathrm{Co}} = 0.5\ \mathrm{nm}. \tag{112}$$

The mesh thickness was therefore chosen as

$$\Delta z = 0.5\ \mathrm{nm}, \tag{113}$$

so that the complete Co layer was represented by a single computational cell through the thickness.

This approximation is appropriate for the present ultrathin-film model, where the primary spatial variation of the skyrmion magnetization occurs in the xy plane and the magnetic texture is treated as effectively uniform through the 0.5 nm Co thickness. It also avoids introducing artificial subdivisions along z that would not represent additional physically resolved structure while substantially increasing the number of degrees of freedom.

Because the OOMMF coordinate system was defined locally for the magnetic Co layer, its z coordinate extended from 0 to 0.5 nm. The center of the single magnetic cell was therefore located at

$$z_{\mathrm{OOMMF}} = 0.25 \text{ nm}. \tag{114}$$

In the full Pt/Co geometry used in COMSOL, the Co layer occupied $10 \leq z \leq 10.5$ nm. The corresponding physical mid-plane was therefore

$$z_{\mathrm{COMSOL}} = 10.25 \text{ nm}. \tag{115}$$

Thus, the local OOMMF cell center and the COMSOL magnetic-field sampling plane represent the same physical position within the Co layer, with the difference arising only from the choice of coordinate origin.

Correspondence with the COMSOL export grid. The 2 nm lateral OOMMF discretization was also chosen to match the regular Cartesian export grid defined in Sec. 3.2.8. The COMSOL sampling coordinates were positioned at the centers of the corresponding OOMMF cells, providing a direct one-to-one spatial correspondence between the electromagnetic data and the micromagnetic mesh.

Consequently, the COMSOL-derived current-density and Oersted-field values could be assigned directly to the corresponding OOMMF cells without an additional lateral interpolation step. This correspondence is especially useful near the central constriction, where both the current density and the current-induced magnetic field vary appreciably over short lateral distances.

Numerical accuracy and computational cost. The mesh resolution represents a compromise between spatial accuracy and computational cost. Decreasing the cell dimensions improves the representation of the skyrmion wall, deformation, boundary interactions, and spatially varying effective fields. However, it also increases the number of magnetization degrees of freedom and consequently increases the computational cost of evaluating the exchange, anisotropy, DMI, demagnetizing, and current-driven terms at every integration step.

For the present geometry, the rectangular atlas covers 200 nm $\times$ 80 nm $\times$ 0.5 nm and

is discretized into a $100 \times 40 \times 1$ Cartesian bounding mesh. Only cells whose centers satisfy the dumbbell boundary condition belong to the magnetic region. The regular finite-difference structure therefore provides sufficient lateral resolution of the skyrmion texture while remaining computationally efficient for the repeated simulations required for all 16 four-bit input sequences.

The selected $2 \times 2 \times 0.5$ nm^3 mesh therefore provided a consistent numerical discretization for the present model. It reproduced the cell size used in the reference Pt/Co reservoir simulation, remained smaller than the characteristic exchange length of the present parameter set, and was directly compatible with the regular COMSOL export grid described in Sec. 3.2.8.

3.3.3 Magnetic Material Parameters and Energy Terms

The energy terms introduced in Sec. 2 were implemented in OOMMF using the parameters in Table 1.

The total magnetic energy considered in the model can be written as

$$E_{\mathrm{tot}} = E_{\mathrm{ex}} + E_{\mathrm{ani}} + E_{\mathrm{demag}} + E_{\mathrm{DMI}} + E_{\mathrm{Z}}, \tag{116}$$

where the terms denote the exchange, perpendicular-anisotropy, magnetostatic, interfacial Dzyaloshinskii–Moriya, and Zeeman contributions, respectively. These energy contributions collectively determine the effective magnetic field defined in Eqs. (11) and (12), which enters the LLG dynamics of Eq. (13). The values of the saturation magnetization, exchange stiffness, DMI constant, and Gilbert damping coefficient were selected consistently with the Pt/Co skyrmion-reservoir model of Jiang *et al.* [1]. The perpendicular anisotropy was subsequently modified in the present work to improve lateral skyrmion confinement, as discussed separately in Sec. 3.3.4.

Saturation magnetization. The value of M_s in Table 1 was assigned only to the magnetic dumbbell region:

```
Specify Oxs_AtlasScalarField:Ms_field {
    atlas :atlas
    multiplier 1
    default_value 0
    values {
        magnet 1.4e6
    }
}
```

The `magnet` region therefore carries the finite saturation magnetization, whereas the default value outside the magnetic atlas is zero. This ensures that the rectangular compu-

tational bounding box does not introduce artificial magnetic material outside the actual dumbbell geometry.

The magnitude of M_s is particularly important because it sets the characteristic strength of the magnetostatic interaction. For the present parameter set, the thin-film demagnetizing-energy scale is approximately

$$K_d = \frac{\mu_0 M_s^2}{2}, \tag{117}$$

which gives

$$K_d \approx 1.23 \times 10^6 \text{ J/m}^3. \tag{118}$$

This value is comparable to the perpendicular anisotropy used in the central region and therefore confirms that the perpendicular state is determined by a nontrivial balance between anisotropy and magnetostatic energy rather than by the anisotropy constant alone.

Exchange interaction. The exchange stiffness in Table 1 was implemented uniformly over the magnetic region:

```
Specify Oxs_UniformExchange:exch {
    A 15e-12
}
```

This value is identical to that used in the reference Pt/Co reservoir model [1]. In the present simulation, the exchange term provides the energetic penalty associated with spatial variation of the magnetization and is particularly relevant when the skyrmion is compressed or distorted while approaching the central constriction.

The numerical adequacy of the 2 nm lateral discretization relative to the corresponding exchange length was established in Sec. 3.3.2.

Perpendicular magnetic anisotropy. The easy axis was chosen along the film normal,

$$\mathbf{e}_u = \hat{\mathbf{z}}, \tag{119}$$

and the anisotropy was implemented as a spatially dependent scalar field rather than as a single uniform constant. The corresponding OOMMF definitions were

```
Specify Oxs_ScriptScalarField:K1_field {
    atlas atlas
    script edge_K1
```

```
    script_args { rawpt }
}

Specify Oxs_UniaxialAnisotropy:ani {
    K1 :K1_field
    axis { 0 0 1 }
}
```

The command

```
axis { 0 0 1 }
```

defines the perpendicular easy axis, while `K1_field` supplies the local anisotropy value at each micromagnetic cell.

The central part of the dumbbell was assigned

$$K_{u,\mathrm{center}} = 1.56 \times 10^6 \ \mathrm{J/m^3}, \tag{120}$$

whereas the outer edge strip was assigned

$$K_{u,\mathrm{edge}} = 3.0 \times 10^6 \ \mathrm{J/m^3}. \tag{121}$$

The complete definition of the `edge_K1` function, including the 4 nm boundary width, is presented in Sec. 3.3.4 because this modification forms part of the skyrmion-confinement strategy rather than the baseline material model.

A useful estimate of the effective perpendicular anisotropy in the central region is

$$K_{\mathrm{eff}} \approx K_{u,\mathrm{center}} - \frac{\mu_0 M_s^2}{2}. \tag{122}$$

Using the values above gives

$$K_{\mathrm{eff}} \approx 1.56 \times 10^6 - 1.23 \times 10^6 = 3.28 \times 10^5 \ \mathrm{J/m^3}. \tag{123}$$

Thus,

$$K_{\mathrm{eff}} > 0, \tag{124}$$

indicating that the central region retains an effective preference for out-of-plane magnetization under the usual thin-film approximation.

The same parameter set gives the characteristic domain-wall length

$$\Delta = \sqrt{\frac{A}{K_{\mathrm{eff}}}} \approx 6.8 \ \mathrm{nm}. \tag{125}$$

Since the lateral computational cell size is only 2 nm, several cells span this characteristic rotation length, providing additional support for the spatial resolution selected in Sec. 3.3.2. The quantities K_{eff} and Δ are used here only as characteristic thin-film estimates; the actual finite dumbbell geometry is solved using the full micromagnetic energy terms rather than these approximations.

Magnetostatic interaction. The demagnetizing interaction was included explicitly through

```
Specify Oxs_Demag:demag { }
```

Although this MIF definition contains no explicit numerical coefficient, `Oxs_Demag` evaluates the magnetostatic field generated by the complete discretized magnetization distribution [12].

This distinction is important for the present geometry. The dumbbell width varies continuously along the longitudinal direction and decreases to 30 nm at the central constriction. Consequently, the demagnetizing environment experienced by the skyrmion changes as a function of position. The magnetostatic contribution is therefore calculated from the actual finite geometry instead of being replaced by the uniform thin-film approximation used above only to estimate K_{eff}.

Interfacial Dzyaloshinskii–Moriya interaction. The DMI constant in Table 1 was implemented using the six-neighbor DMI exchange extension:

```
Specify Oxs_DMExchange6Ngbr:dmi {
    default_D 2.9e-3
    atlas :atlas
    D {
        magnet magnet 2.9e-3
    }
}
```

The block assigns the same DMI coefficient between neighboring cells belonging to the magnetic region. The physical role of interfacial DMI and its connection with the Néel-type skyrmion chirality were discussed in Sec. 2; here, its main significance is that the value is incorporated directly into the discrete micromagnetic energy evaluation.

A useful scale for comparison is the approximate critical interfacial DMI [10],

$$D_c = \frac{4}{\pi}\sqrt{AK_{\text{eff}}}, \tag{126}$$

which gives, using Eq. (123),

$$D_c \approx 2.83 \text{ mJ/m}^2. \tag{127}$$

The adopted value,

$$D = 2.9 \text{ mJ/m}^2, \tag{128}$$

is therefore close to this characteristic scale. This comparison shows that DMI is sufficiently strong to compete significantly with the exchange and effective-anisotropy contributions. However, D_c is an analytical thin-film estimate and should not be interpreted as an exact stability threshold for the finite, spatially nonuniform dumbbell structure.

Zeeman contribution from the Oersted field. The final energy contribution in Eq. (116) is the Zeeman interaction associated with the current-induced magnetic field. Rather than prescribing an idealized uniform external field, the present model uses the spatially resolved Oersted-field distribution calculated in COMSOL,

$$\mathbf{H}_{\text{Oe}}(x, y) = [H_x(x, y), H_y(x, y), H_z(x, y)]\,. \tag{129}$$

This field was loaded into OOMMF as a spatial vector field. Its detailed file-reading procedure, cell-by-cell mapping, and polarity reversal during the four-bit current sequence are described in Sec. 3.3.7. The Zeeman term is therefore retained here as part of the total driven-system energy, while its numerical implementation is deferred to the subsection devoted specifically to the COMSOL–OOMMF coupling.

During the initial relaxation stage, no current-induced Oersted field was applied. The corresponding energy landscape therefore consisted of

$$E_{\text{relax}} = E_{\text{ex}} + E_{\text{ani}} + E_{\text{demag}} + E_{\text{DMI}}. \tag{130}$$

During the driven simulations, the imported Oersted field adds the Zeeman contribution,

$$E_{\text{driven}} = E_{\text{relax}} + E_{\text{Z}}. \tag{131}$$

Combined parameter regime. The selected parameter set places the system in a regime in which no single interaction determines the magnetic configuration independently. For the final parameter set, the characteristic scales were $K_d \approx 1.23$ MJ/m^3, $K_{\text{eff}} \approx 0.328$ MJ/m^3, $\ell_{\text{ex}} \approx 3.49$ nm, $\Delta \approx 6.8$ nm, and $D_c \approx 2.83$ mJ/m^2.

These values show quantitatively that exchange, perpendicular anisotropy, magnetostatic energy, and DMI operate on comparable magnetic scales in the present ultrathin Pt/Co system. The stable skyrmion used as the reservoir state therefore emerges from

the complete numerical minimization of the combined micromagnetic energy rather than from a prescribed skyrmion size or shape.

Gilbert damping. The Gilbert damping coefficient was set to

$$\alpha = 0.3, \tag{132}$$

consistent with the reference model [1]. This parameter is intentionally separated from the energy terms above because it does not contribute to the static energy functional. Instead, it enters the dynamical evolution equation and controls the dissipation of magnetic energy during time evolution.

The corresponding parameter appears later in the `Oxs_SpinXferEvolve` block. Since that block also contains the spin–orbit-torque parameters, its complete implementation is discussed in Sec. 3.3.6 rather than repeated here.

3.3.4 Edge Anisotropy and Skyrmion Confinement

During the development of the micromagnetic model, the lateral confinement of the current-driven skyrmion was found to be an important numerical and physical consideration. As discussed in Sec. 2.5, the Skyrmion Hall effect produces a transverse component of motion in addition to the longitudinal current-driven displacement. In a finite-width structure, this transverse motion can bring the skyrmion into close proximity with the device boundary, where sufficiently strong interaction with the edge may lead to severe deformation or annihilation [2, 10].

In early simulations, the magnetic device did not contain the enhanced edge-anisotropy strip used in the final model. Under current-driven motion, the transverse displacement of the skyrmion was then insufficiently confined in some trajectories, particularly as the skyrmion approached regions where the lateral device width decreased. To increase the energy barrier against boundary contact, a region of enhanced perpendicular magnetic anisotropy was therefore introduced adjacent to the upper and lower lateral boundaries of the dumbbell.

The final anisotropy profile consisted of a central value

$$K_{u,\mathrm{center}} = 1.56 \times 10^6 \ \mathrm{J/m^3}, \tag{133}$$

and an enhanced edge value

$$K_{u,\mathrm{edge}} = 3.0 \times 10^6 \ \mathrm{J/m^3}. \tag{134}$$

The enhanced region extended approximately

$$w_{\text{edge}} = 4 \text{ nm} \tag{135}$$

in the transverse direction from the upper and lower lateral boundaries. Since the lateral OOMMF cell size was 2 nm, this corresponds approximately to two computational cells near each lateral edge.

For the analytical dumbbell boundary

$$y_b(x) = 22.5 - 7.5 \cos\left(\frac{\pi x}{100}\right) \quad \text{nm}, \tag{136}$$

the spatially dependent anisotropy profile may be written as

$$K_u(x, y) = \begin{cases} K_{u,\text{edge}}, & y_b(x) - w_{\text{edge}} \leq |y| \leq y_b(x), \\ K_{u,\text{center}}, & |y| < y_b(x) - w_{\text{edge}}, \\ 0, & |y| > y_b(x). \end{cases} \tag{137}$$

The corresponding implementation in the OOMMF MIF file was

```
set edge_thickness_nm 4.0
set K_center 1.56e6
set K_edge   3.0e6

proc edge_K1 { x y z } {

    global edge_thickness_nm K_center K_edge

    set x_nm [expr {$x * 1e9}]
    set y_nm [expr {$y * 1e9}]
    set ay_nm [expr {abs($y_nm)}]

    set pi 3.141592653589793

    set y_bound \
        [expr {22.5 - 7.5*cos($x_nm*$pi/100.0)}]

    if {$ay_nm > $y_bound} {
        return 0.0
    }

    if {$ay_nm >= ($y_bound - $edge_thickness_nm)} {
        return $K_edge
```

```
    }

    return $K_center
}
```

This function was used by the `Oxs_ScriptScalarField:K1_field` definition introduced in Sec. 3.3.3. For every magnetic cell, OOMMF first evaluated the local half-width $y_b(x)$ of the dumbbell and then compared the absolute transverse coordinate $|y|$ with the edge-region threshold. Cells lying within the outer 4 nm of the upper or lower lateral boundary were therefore assigned $K_{u,\mathrm{edge}}$, while the remaining magnetic cells were assigned $K_{u,\mathrm{center}}$.

It should be noted that the 4 nm distance in this implementation is defined in the local transverse y direction rather than as an exact normal distance from the continuously curved boundary. In addition, no separate enhanced-anisotropy strip was introduced along the vertical end faces at $x = \pm 100$ nm; the modification was applied specifically to the upper and lower lateral boundaries relevant to transverse skyrmion motion.

The resulting anisotropy distribution on the 2×2 nm^2 lateral OOMMF grid is shown in Fig. 26.

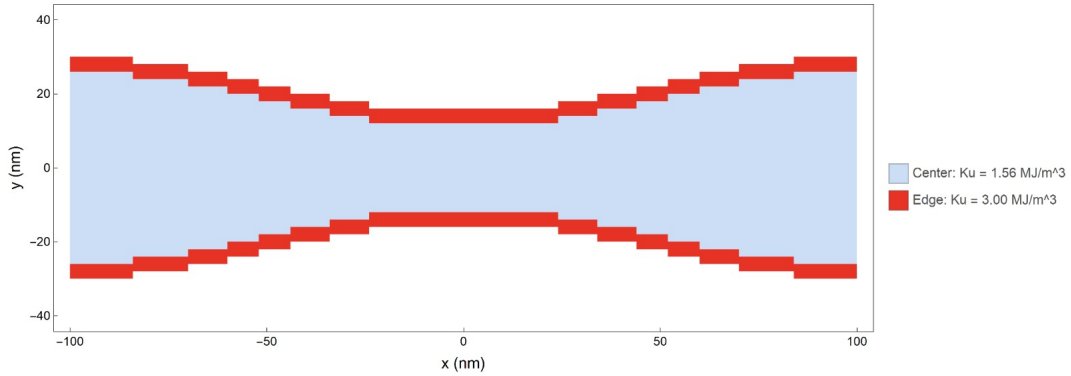


Figure 26: Spatial distribution of the perpendicular anisotropy used in the OOMMF model. The central magnetic region was assigned $K_{u,\mathrm{center}} = 1.56 \times 10^6$ J/m^3, whereas an approximately 4 nm-wide strip adjacent to the upper and lower lateral boundaries was assigned $K_{u,\mathrm{edge}} = 3.0 \times 10^6$ J/m^3 to enhance skyrmion confinement. The step-like appearance of the boundary reflects the 2 nm finite-difference discretization used in OOMMF.

The physical purpose of the enhanced edge anisotropy is to increase the energy cost of bringing the nonuniform skyrmion texture into the boundary region. The nearly uniform out-of-plane background magnetization remains approximately aligned with the perpendicular easy axis, whereas the skyrmion wall contains substantial in-plane magnetization components. Consequently, penetration of the skyrmion wall into a region with larger K_u increases the anisotropy-energy cost and produces an effective repulsive contribution to the lateral confinement potential.

This additional confinement does not remove the Skyrmion Hall effect or eliminate the intrinsic transverse dynamics of the skyrmion. Instead, it modifies the boundary-energy landscape so that the current-driven skyrmion can undergo transverse displacement while remaining less likely to contact and annihilate at the device edge. The enhanced anisotropy therefore acts together with the geometrical confinement of the dumbbell rather than replacing it.

The values $K_{u,\mathrm{center}} = 1.56 \times 10^6$ J/m^3, $K_{u,\mathrm{edge}} = 3.0 \times 10^6$ J/m^3, and $w_{\mathrm{edge}} = 4$ nm were modifications introduced in the present work during model development. They should therefore be regarded as parameters of the final confinement strategy rather than as values adopted directly from the original reservoir model of Jiang et al. The subsequent current-amplitude and pulse-duration optimization was performed using this final anisotropy profile.

3.3.5 Initial Skyrmion State and Energy Relaxation

Before applying the current-pulse sequences, a common initial skyrmion state was prepared and relaxed in OOMMF. The purpose of this procedure was to generate a reproducible low-energy magnetic configuration that could be used as the same starting point for all subsequent four-bit input simulations.

Construction of the initial skyrmion seed. The initial skyrmion was placed in the wider left section of the dumbbell structure, with its center defined as

$$(x_0, y_0) = (-70, 0) \text{ nm}. \tag{138}$$

Two characteristic radii were used to construct the initial magnetization texture,

$$r_{\mathrm{in}} = 8 \text{ nm}, \tag{139}$$

and

$$r_{\mathrm{out}} = 16 \text{ nm}. \tag{140}$$

For each micromagnetic cell, the squared radial distance from the prescribed skyrmion center was calculated as

$$\rho^2 = (x - x_0)^2 + (y - y_0)^2. \tag{141}$$

The magnetization was then initialized in three regions. Inside the inner radius, the skyrmion core was oriented along the negative film-normal direction,

$$\mathbf{m} = (0, 0, -1), \qquad \rho < r_{\mathrm{in}}. \tag{142}$$

Outside the outer radius, the background magnetization was initialized along the positive film-normal direction,

$$\mathbf{m} = (0, 0, +1), \qquad \rho > r_{\text{out}}. \tag{143}$$

Within the annular region $r_{\text{in}} \leq \rho \leq r_{\text{out}}$, the initial magnetization was defined to lie entirely in the film plane and to point radially outward from the skyrmion center. The corresponding MIF implementation was

```
set skx -70e-9
set sky 0.0
set r_inner 8e-9
set r_outer 16e-9

set rsq_inner [expr {$r_inner*$r_inner}]
set rsq_outer [expr {$r_outer*$r_outer}]

proc SkyrmionSeed { x y z } {
    global skx sky rsq_inner rsq_outer

    set xoff [expr {$skx-$x}]
    set yoff [expr {$sky-$y}]
    set rsq [expr {$xoff*$xoff + $yoff*$yoff}]

    if {$rsq < $rsq_inner} {
        return {0.0 0.0 -1.0}
    }

    if {$rsq > $rsq_outer} {
        return {0.0 0.0 1.0}
    }

    return [list [expr {-$xoff}] [expr {-$yoff}] 0.0]
}
```

The initial vector field was normalized cell by cell using

```
Specify Oxs_ScriptVectorField:m0 {
    atlas :atlas
    script SkyrmionSeed
    script_args { rawpt }
```

```
    norm 1
}
```

This construction provides a simple Néel-type seed with a negative out-of-plane core, a positive out-of-plane background, and a radially oriented in-plane transition region. The prescribed values r_{in} and r_{out} define only the initial seed and do not constrain the final relaxed skyrmion radius or wall structure.

Energy minimization. The analytically constructed seed is not an equilibrium configuration of the complete micromagnetic model. It was therefore relaxed using the same device geometry, $2\times2\times0.5\ \mathrm{nm}^3$ mesh, material parameters, DMI interaction, and spatially varying anisotropy profile described in the preceding subsections.

No current-driven torque or current-induced Oersted field was included during this stage. The relaxed configuration was therefore determined by

$$E_{\mathrm{relax}} = E_{\mathrm{ex}} + E_{\mathrm{ani}} + E_{\mathrm{demag}} + E_{\mathrm{DMI}}. \tag{144}$$

Energy minimization was performed using the OOMMF conjugate-gradient minimization evolver `Oxs_CGEvolve` together with an `Oxs_MinDriver` [12]. The implementation used in the relaxation MIF file was

```
Specify Oxs_CGEvolve:evolve {}

Specify Oxs_MinDriver:relax {
    evolver :evolve
    stopping_mxHxm 1e-5
    mesh :mesh
    Ms :Ms_field
    m0 :m0
}
```

The stopping criterion was therefore set to

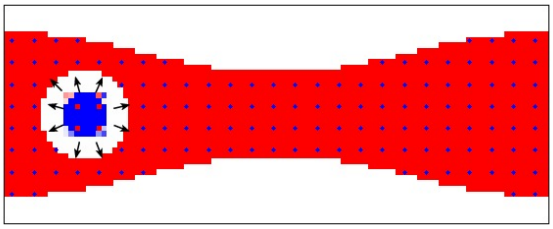
$$\mathrm{stopping_mxHxm} = 1\times10^{-5}, \tag{145}$$

which controls termination of the minimization according to the residual magnetic torque measure used by the minimization driver. Minimization evolvers in OOMMF are designed to move the magnetization toward a local minimum of the micromagnetic energy rather than to reproduce a physical time trajectory. [12]

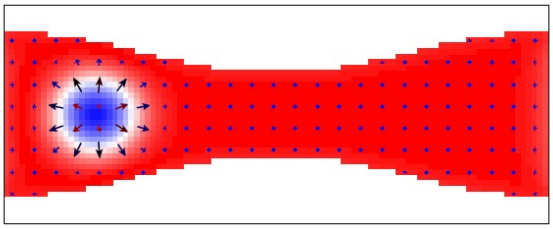
During relaxation, the initially prescribed skyrmion texture was allowed to adjust freely under the exchange, anisotropy, demagnetizing, and DMI energy contributions. As

a result, the relaxed state was no longer fixed by the prescribed seed geometry, but instead represented a lower-energy configuration of the complete micromagnetic model.

The initial and relaxed configurations are compared in Fig. 27.



(a) Analytically initialized Néel-type skyrmion before energy minimization.



(b) Skyrmion configuration after energy minimization in OOMMF.

Figure 27: Initial and relaxed magnetization configurations used in the preparation of the common reservoir state. The color map represents the out-of-plane magnetization component, while the arrows indicate the local in-plane magnetization direction. In the analytical seed, the core is initialized along $-\hat{z}$, the surrounding background along $+\hat{z}$, and the intermediate annular region with a radial in-plane magnetization. After energy minimization, the sharp prescribed seed develops into a smoother micromagnetic skyrmion profile determined by the exchange, anisotropy, demagnetizing, and DMI energy contributions.

The relaxed magnetization state was archived at the end of the minimization using

```
Destination archive mmArchive

Schedule DataTable archive Done 1
Schedule Oxs_MinDriver:relax:Magnetization archive Done 1
```

and the resulting relaxed configuration was subsequently retained as

```
ComsolData_Relaxed.omf
```

for use as the common initial magnetization state in the current-driven simulations. The corresponding pre-relaxation configuration was retained separately as

```
ComsolData_Relaxed_before_run.omf
```

for comparison in Fig. 27.

Using a single relaxed state as the starting configuration for all 16 four-bit input sequences ensures that differences among the final reservoir states arise from the applied pulse sequences rather than from variations in the initial magnetic configuration.

3.3.6 Spin–Orbit-Torque Implementation in OOMMF

Current-driven dynamics were calculated using `Oxs_SpinXferEvolve`, which integrates the Gilbert equation with spin-transfer source terms [12]. The corresponding Gilbert equation is

$$\frac{d\mathbf{m}}{dt} = -|\gamma_H|\,\mathbf{m}\times\mathbf{H}_{\text{eff}} + \alpha\left(\mathbf{m}\times\frac{d\mathbf{m}}{dt}\right) + |\gamma_H|\beta\epsilon\left[(\mathbf{m}\times\mathbf{m}_p)\times\mathbf{m}\right] - |\gamma_H|\beta\epsilon'\left(\mathbf{m}\times\mathbf{m}_p\right), \tag{146}$$

where $\mathbf{m}$ is the reduced magnetization and $\mathbf{m}_p$ is the unit reference polarization. Here γ_H is the Gilbert coefficient in $\mathrm{m\,A^{-1}\,s^{-1}}$, so that $|\gamma_H| = \mu_0\gamma_T$ with γ_T as defined in Eq. (13); these symbols use the same Gilbert convention. The signed current-dependent field coefficient is [12]

$$\beta = \left|\frac{\hbar}{\mu_0 e}\right|\frac{J}{t_{\text{Co}}M_s}. \tag{147}$$

The angular dependence of the primary spin-transfer term is described by

$$\epsilon = \frac{P\Lambda^2}{(\Lambda^2+1)+(\Lambda^2-1)(\mathbf{m}\cdot\mathbf{m}_p)}, \tag{148}$$

where P controls the strength of the spin-transfer contribution and Λ controls its angular asymmetry [12].

Effective representation of the spin–orbit torque. The `Oxs_SpinXferEvolve` module is formulated in terms of a Slonczewski-type spin-transfer torque and is not, by itself, a dedicated spin-Hall-effect solver. In the present Pt/Co model, it was used as an effective representation of the damping-like spin–orbit torque generated by the spin Hall effect in the Pt layer.

For the coordinate convention adopted in this work, a positive charge current is directed along $+\hat{x}$. The spin current generated in Pt is taken to flow toward the Co layer along $+\hat{z}$, with the corresponding spin polarization along $-\hat{y}$. The polarization direction in OOMMF was therefore specified as

$$\mathbf{m}_p = (0,-1,0), \tag{149}$$

through the MIF entry

```
mp { 0 -1 0 }
```

Thus, $\mathbf{m}_p$ in the OOMMF implementation represents the effective spin-polarization direction associated with the spin Hall current injected from Pt into Co; no separate fixed ferromagnetic polarizer layer was explicitly included in the simulated geometry.

The asymmetry parameter was selected as

$$\Lambda = 1. \tag{150}$$

With this choice, the OOMMF angular factor simplifies to

$$\epsilon = \frac{P}{2}, \tag{151}$$

independent of $\mathbf{m}\cdot\mathbf{m}_p$; the additional Slonczewski angular asymmetry is removed [12].

The torque-efficiency parameter was set to

$$P = 0.1. \tag{152}$$

For $\Lambda = 1$, the coefficient of the primary current-induced torque therefore becomes proportional to

$$\beta\epsilon = \left|\frac{\hbar}{\mu_0 e}\right| \frac{J}{t_{\mathrm{Co}} M_s} \frac{P}{2}. \tag{153}$$

For $\Lambda = 1$, the product is

$$\beta\epsilon = \frac{\hbar P J}{2\mu_0 |e| M_s t_{\mathrm{Co}}}.$$

This is the standard damping-like SOT field scale when P is used as an effective torque efficiency; there is no additional factor of one half. For $P = \theta_{\mathrm{H}} = 0.1$, the ratio of the two coefficient magnitudes, expressed in the same units, is 1.000000. The choice matches the numerical value used by Jiang et al. [1]; identifying this effective efficiency with a bulk spin Hall angle assumes the corresponding interfacial spin transmission and does not constitute an experimental calibration [11].

The torque basis must also be matched explicitly:

$$(\mathbf{m} \times \mathbf{m}_p) \times \mathbf{m} = -\mathbf{m} \times (\mathbf{m} \times \mathbf{m}_p).$$

For $\mathbf{m}_p = \boldsymbol{\sigma}$ and $\epsilon' = 0$, the OOMMF source term is therefore

$$\boldsymbol{\tau}_{\mathrm{DL}} = |\gamma_H|\beta\epsilon(\mathbf{m} \times \boldsymbol{\sigma}) \times \mathbf{m} = -|\gamma_H|\beta\epsilon\mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}).$$

Consequently, the signed scalar in Eq. (17) is $\tau_{\mathrm{DL}}(J) = -|\gamma_H|\beta(J)\epsilon$. For positive J and P, this source term tends to align $\mathbf{m}$ with $\mathbf{m}_p$, as specified by OOMMF [12]. The vector identity changes the scalar sign when the basis is changed; it does not determine the physical polarization direction, which is defined separately by the spin Hall and coordinate convention. The fixed $\mathbf{m}_p$ and the signed J used in the simulations are unchanged.

The secondary spin-transfer term was neglected by setting

$$\epsilon' = 0. \tag{154}$$

In OOMMF this corresponds to

```
eps_prime 0
```

so that the last term in the `Oxs_SpinXferEvolve` equation vanishes. In the effective SOT interpretation used here, this means that only the damping-like current-induced torque was retained, while a separate field-like torque contribution was not included.

Gilbert damping and spatial current-density field. The Gilbert damping coefficient in the driven simulations was

$$\alpha = 0.3, \tag{155}$$

consistent with the Pt/Co parameter set used by Jiang et al. [1]. As discussed in Sec. 3.3.3, this coefficient does not form part of the static micromagnetic energy functional but controls dissipative magnetization dynamics during time evolution.

Rather than assigning a single uniform value of J, the `Oxs_SpinXferEvolve` block was supplied with the spatial current-density field

```
J J_field
```

where `J_field` contains the position-dependent current-density magnitude obtained from the COMSOL calculation. The construction of this field from the exported COMSOL data is described separately in Sec. 3.3.7.

The resulting spin-transfer evolver used in the final model was

```
Specify Oxs_SpinXferEvolve:evolve [subst {
    alpha 0.3
    J J_field
    mp { 0 -1 0 }
    P 0.1
    Lambda 1
    eps_prime 0
    J_profile FourBitPulse
    J_profile_args { stage stage_time total_time }
}]
```

Time-dependent current-pulse sign. The spatial field `J_field` defines the magnitude and spatial distribution of the current density, whereas the temporal sign of the applied current was controlled through the `J_profile` option. OOMMF allows the specified current-density field to be multiplied by a time-dependent scalar function, with `stage`, `stage_time`, and `total_time` passed to the corresponding Tcl procedure [12].

In the present implementation,

```
J_profile FourBitPulse
J_profile_args { stage stage_time total_time }
```

allowed each stage of the four-bit sequence to reverse the sign of the same spatial current-density distribution. The effective current density can therefore be written schematically as

$$J_{\mathrm{eff}}(x, y, t) = f_{\mathrm{bit}}(t) J_{\mathrm{COMSOL}}(x, y), \tag{156}$$

where

$$f_{\mathrm{bit}} = \begin{cases} +1, & \text{for bit 1,} \\ -1, & \text{for bit 0.} \end{cases} \tag{157}$$

Reversing the pulse sign reverses the spin-transfer contribution while retaining the spatial current-density magnitude profile and the fixed reference polarization [12].

The pulse duration and total sequence length are discussed in Sec. 3.3.8, while the final-state archiving procedure is described in Sec. 3.3.9.

Treatment of the Oersted field. The current-generated Oersted field is not calculated internally by `Oxs_SpinXferEvolve`. The OOMMF documentation explicitly states that such a field must be introduced separately as a Zeeman energy contribution [12]. Therefore, the COMSOL-derived Oersted field was incorporated independently from the spin-transfer evolver and was assigned the same stage-dependent sign as the applied current pulse.

The spatial mapping and stage-dependent implementation of both $J_{\mathrm{COMSOL}}(x, y)$ and $\mathbf{H}_{\mathrm{Oe}}(x, y)$ are described in Sec. 3.3.7.

3.3.7 Integration of COMSOL Current-Density and Oersted-Field Data

The COMSOL fields from Sec. 3.2.8 were imported at the OOMMF cell centers defined in Sec. 3.3.2 using the following lookup procedures.

COMSOL data files. The current-density and magnetic-field data exported from COMSOL were processed into the two text files used by the final OOMMF simulations:

```
Gridded_J_2nm_clean.txt
GriddedWoheader_H_2nm.txt
```

They were referenced in the MIF file through

```
set data_dir "D:/Efe-Ommf/SOT_P=0.1"

set jfile "$data_dir/Gridded_J_2nm_clean.txt"
set hfile "$data_dir/GriddedWoheader_H_2nm.txt"
```

The current-density file contained entries of the form

$$(x_{\mathrm{nm}}, y_{\mathrm{nm}}, J), \tag{158}$$

whereas the magnetic-field file contained

$$(x_{\mathrm{nm}}, y_{\mathrm{nm}}, H_x, H_y, H_z). \tag{159}$$

The sampling planes, exported quantities, and coordinate ranges used to generate these files are given in Sec. 3.2.8.

Conversion of spatial coordinates to grid indices. Using the COMSOL sampling coordinates defined in Sec. 3.2.8, each coordinate pair was converted into the integer array indices required by the OOMMF lookup procedures:

$$i_x = \mathrm{round}\left(\frac{x_{\mathrm{nm}} + 99}{2}\right), \tag{160}$$

and

$$i_y = \mathrm{round}\left(\frac{y_{\mathrm{nm}} + 29}{2}\right). \tag{161}$$

The corresponding MIF implementation was

```
set ix [expr {int(round(($xnm + 99.0)/2.0))}]
set iy [expr {int(round(($ynm + 29.0)/2.0))}]
```

These expressions convert the known cell-center coordinates into array indices. The use of `round` accommodates small floating-point differences in the numerical representation of the coordinates; it does not introduce interpolation between neighboring cells.

The current-density values were first loaded into a Tcl array:

```
array set Jmap {}
```

```
set jraw [ReadFile $jfile]
foreach line [split $jraw "\n"] {
    set line [string trim $line]
    if {$line eq ""} { continue }

    if {[scan $line "%lf %lf %lf" xnm ynm Jval] != 3} {
        continue
    }

    set ix [expr {int(round(($xnm + 99.0)/2.0))}]
    set iy [expr {int(round(($ynm + 29.0)/2.0))}]

    if {$ix < 0 || $ix > 99 || $iy < 0 || $iy > 29} {
        continue
    }

    set Jmap($ix,$iy) $Jval
}
unset jraw
```

Similarly, the three components of the COMSOL-derived Oersted field were stored in separate Tcl arrays:

```
array set Hxmap {}
array set Hymap {}
array set Hzmap {}

set hraw [ReadFile $hfile]
foreach line [split $hraw "\n"] {
    set line [string trim $line]
    if {$line eq ""} { continue }

    if {[scan $line "%lf %lf %lf %lf %lf" \
        xnm ynm Hx Hy Hz] != 5} {
        continue
    }

    set ix [expr {int(round(($xnm + 99.0)/2.0))}]
    set iy [expr {int(round(($ynm + 29.0)/2.0))}]
```

```
    if {$ix < 0 || $ix > 99 || $iy < 0 || $iy > 29} {
        continue
    }

    set Hxmap($ix,$iy) $Hx
    set Hymap($ix,$iy) $Hy
    set Hzmap($ix,$iy) $Hz
}
unset hraw
```

Cell-by-cell current-density lookup. During the micromagnetic calculation, the COMSOL current density corresponding to a particular OOMMF cell was retrieved through the `J_from_comsol` procedure:

```
proc J_from_comsol { x y z } {
    global Jmap

    set xnm [expr {$x*1e9}]
    set ynm [expr {$y*1e9}]

    set ix [expr {int(round(($xnm + 99.0)/2.0))}]
    set iy [expr {int(round(($ynm + 29.0)/2.0))}]

    if {$ix < 0 || $ix > 99 || $iy < 0 || $iy > 29} {
        return 0.0
    }

    set key "$ix,$iy"

    if {[info exists Jmap($key)]} {
        return $Jmap($key)
    }

    return 0.0
}
```

A zero value was returned when the requested point lay outside the imported data range or when no corresponding entry was available. The spatially dependent current-density field was then defined as

```
Specify Oxs_ScriptScalarField:J_field {
    atlas :atlas
    script J_from_comsol
    script_args { rawpt }
}
```

The resulting `J_field` was supplied to `Oxs_SpinXferEvolve` as described in Sec. 3.3.6. Consequently, the local current-induced torque amplitude was determined by the COMSOL-derived current density associated with each magnetic cell rather than by a spatially uniform value.

Cell-by-cell Oersted-field lookup. The COMSOL-derived Oersted field was transferred in an analogous manner, except that all three vector components were retained. The lookup procedure used in the final MIF file was

```
proc H_from_comsol { x y z } {
    global Hxmap Hymap Hzmap

    set xnm [expr {$x*1e9}]
    set ynm [expr {$y*1e9}]

    set ix [expr {int(round(($xnm + 99.0)/2.0))}]
    set iy [expr {int(round(($ynm + 29.0)/2.0))}]

    if {$ix < 0 || $ix > 99 || $iy < 0 || $iy > 29} {
        return [list 0.0 0.0 0.0]
    }

    set key "$ix,$iy"

    if {[info exists Hxmap($key)] && \
        [info exists Hymap($key)] && \
        [info exists Hzmap($key)]} {

        return [list \
            $Hxmap($key) \
            $Hymap($key) \
            $Hzmap($key)]
    }
```

```
    return [list 0.0 0.0 0.0]
}
```

The resulting vector field was defined as

```
Specify Oxs_ScriptVectorField:H_comsol {
    atlas :atlas
    script H_from_comsol
    script_args { rawpt }
}
```

This field represents the spatial Oersted-field distribution associated with the reference positive-current polarity imported from COMSOL.

Stage-dependent reversal of the Oersted field. As discussed in Sec. 3.3.6, the current polarity changes according to the binary input sequence. Because the Oersted field must reverse together with the current that generates it, the imported magnetic-field map was assigned the same stage-dependent sign.

This was implemented using `Oxs_TransformZeeman`, which allows a time-dependent transformation to be applied to a supplied vector field [12]. The transformation function used in the final MIF file was

```
proc HsignProfile { stage stage_time total_time } {
    global bit_time Hscale bit_string

    if {$stage < 0 || \
        $stage >= [string length $bit_string]} {

        return [list 0 0 0  0 0 0]
    }

    if {$stage_time < 0 || $stage_time >= $bit_time} {
        return [list 0 0 0  0 0 0]
    }

    set bit [string index $bit_string $stage]

    if {$bit eq "1"} {
        set s $Hscale
    } else {
        set s [expr {-$Hscale}]
```

```
    }

    return [list $s $s $s  0 0 0]
}
```

and was applied through

```
Specify Oxs_TransformZeeman:comsol_field {
    field :H_comsol
    type diagonal
    script HsignProfile
    script_args { stage stage_time total_time }
}
```

The diagonal transformation multiplies each imported field component by the same stage-dependent factor, so that the complete Oersted-field vector changes polarity without altering its spatial distribution.

For the final simulations,

$$J_{\text{scale}} = 1, \qquad H_{\text{scale}} = 1. \tag{162}$$

Therefore, no additional amplitude rescaling was applied to the imported COMSOL distributions. The current-pulse polarity was defined through the four-bit profile described in Sec. 3.3.6, while the Oersted field was assigned the same temporal sign. The applied magnetic field can therefore be written as

$$\mathbf{H}_{\text{Oe}}(x, y, t) = f_{\text{bit}}(t)\mathbf{H}_{\text{COMSOL}}(x, y), \tag{163}$$

where $f_{\text{bit}}(t)$ is the bipolar pulse factor defined in Sec. 3.3.6. Thus, reversal of the applied current is accompanied by the corresponding reversal of the current-induced magnetic field.

Final COMSOL–OOMMF coupling. The direct cell-center correspondence between the COMSOL export grid and the OOMMF mesh was established in Sec. 3.3.2. The implementation described above provides the numerical mechanism by which the exported electromagnetic data are retrieved for each magnetic cell during the micromagnetic calculation.

The two imported quantities enter the magnetization dynamics through separate mechanisms. The current-density map determines the local current-induced torque through `Oxs_SpinXferEvolve`, whereas the Oersted-field map contributes independently through `Oxs_TransformZeeman`. The selected current amplitude and pulse duration used for the four-bit simulations are discussed in the following subsection.

3.3.8 Current-Amplitude and Pulse-Duration Selection

The current amplitude and pulse duration were selected empirically to obtain stable skyrmion transport through the dumbbell structure while preserving sufficient dependence of the final state on the complete four-bit input sequence.

At relatively low applied currents, the skyrmion could move within the wider left section of the device but was unable to reliably pass through the central constriction. The constriction therefore imposed a practical lower limit on the current amplitude required for useful reservoir operation.

Increasing the driving current improved transport through the constriction. For the final COMSOL model, a terminal current of

$$I = 145\ \mu\text{A} \tag{164}$$

provided sufficient driving strength for the skyrmion to traverse a substantial fraction of the device without requiring an excessively long pulse sequence.

The temporal scale of the input was chosen together with the current amplitude. If the individual pulses were made too long, the skyrmion underwent a large displacement during each bit. In this regime, the final magnetic state became increasingly dominated by the most recently applied pulse, reducing the number of distinguishable final reservoir states and therefore weakening the memory of the preceding bits.

Conversely, very short pulses produced smaller state changes and did not provide sufficient displacement through the constricted geometry. A bit duration of

$$t_{\text{bit}} = 2\ \text{ns} \tag{165}$$

was therefore selected for the final simulations. For a four-bit input, the total driven interval was consequently

$$t_{\text{sequence}} = 4t_{\text{bit}} = 8\ \text{ns}. \tag{166}$$

At the selected current amplitude, this timescale corresponded approximately to the time required for the skyrmion to travel across the device length under sustained driving. The final operating point was therefore chosen not to maximize the skyrmion velocity, but to balance reliable passage through the constriction with the generation of distinguishable final states for different four-bit input sequences.

The final parameters used for the reservoir simulations were thus

$$I = 145\ \mu\text{A}, \qquad t_{\text{bit}} = 2\ \text{ns}, \qquad t_{\text{sequence}} = 8\ \text{ns}. \tag{167}$$

3.3.9 Final-State Output

Each four-bit input was evolved over four consecutive 2 ns stages and sampled at $t = 8$ ns. No finite current-free relaxation interval was included after the driven sequence. The pulse functions return zero at the stage endpoint, but this instantaneous convention does not provide a relaxed state. The extracted features represent a transient endpoint of the driven response, rather than a demonstrated nonvolatile memory state.

Only the magnetization state at the end of the four-bit sequence was archived for subsequent reservoir-state analysis. The corresponding output was generated using

```
Schedule DataTable archive Done 1
Schedule Oxs_TimeDriver:init:Magnetization archive Done 1
```

Thus, each of the 16 possible four-bit input patterns produced one final micromagnetic state. These final OMF files were subsequently processed to extract the skyrmion position and size used as reservoir features, as described in Sec. 4.

3.4 Model limitations and physical realizability

The simulations are deterministic: no stochastic thermal field is included, corresponding to the $T = 0$ K formulation of the evolver [12]. Thermal fluctuations and readout noise may reduce the separability of nearby states, but their quantitative effects have not been evaluated. A separate field-like SOT is omitted by setting $\epsilon' = 0$, while the Oersted field is retained. The effect of a nonzero field-like term on deformation and the extracted features remains untested.

The reported features are sampled at the end of the 8 ns driven sequence without an additional current-free relaxation interval. Their persistence under delayed readout has not been established. The damping constant $\alpha = 0.3$ follows the reference model [1], but has not been calibrated for a specific experimental sample or tested in a sensitivity analysis. Its effects on relaxation times and classification performance cannot be quantified from the present data.

The enhanced-anisotropy regions are prescribed numerical material profiles, not fabricated structures demonstrated in this work. They extend 4 nm in the transverse direction from the upper and lower boundaries; no separate high-anisotropy strip is assigned to the end faces at $x = \pm 100$ nm. Their fabrication tolerance and the effects of a smooth or disordered profile have not been tested. The $2 \times 2 \times 0.5$ nm^3 mesh is consistent with the characteristic length-scale estimates, but no mesh-refinement study has established convergence of the extracted states. The nominal edge strip spans only about two lateral cells, and the small final y differences require particular caution.

The COMSOL data are sampled at OOMMF cell centers, avoiding an additional interpolation step during import; finite-element evaluation and finite-difference discretization

errors remain. Using the local current-density magnitude with fixed $\mathbf{m}_p = (0, -1, 0)$ represents spatial variations of torque amplitude while approximating its orientation by the longitudinal reference direction. Sensitivity to a polarization following the local current direction has not been quantified. Thermal and parameter sensitivity, mesh refinement, delayed current-free readout, and trajectory comparisons are future work.

4 Reservoir State Extraction and MNIST Classification

The micromagnetic simulations described in Sec. 3 produced one final magnetization state for each of the 16 possible four-bit input sequences. These final states were converted into numerical reservoir features and subsequently used for handwritten-digit classification.

The OMF states were processed through the following classification pipeline:

final OMF state $\rightarrow (x, y, r) \rightarrow$ four-bit lookup table $\rightarrow$ MNIST feature vector $\rightarrow$ trained readout.

The physical reservoir remained fixed throughout the classification procedure. Only the software readout was trained.

4.1 Extraction of the Final Skyrmion State

The final magnetization states generated using OOMMF [12] were stored in OMF format and post-processed using Wolfram Mathematica [13]. For each four-bit input sequence, the magnetization vector at every magnetic mesh cell was read from the corresponding final OMF file.

The out-of-plane magnetization component was normalized as

$$m_{z,i} = \frac{M_{z,i}}{\sqrt{M_{x,i}^2 + M_{y,i}^2 + M_{z,i}^2}}, \tag{168}$$

where i denotes an individual mesh cell. The reversed skyrmion core was identified as the set of cells satisfying

$$\mathcal{C} = \{ i \mid m_{z,i} < 0 \}. \tag{169}$$

If N_c denotes the number of cells in this region, the final skyrmion position was calculated from the centroid of the selected cells,

$$x_\mathrm{f} = \frac{1}{N_\mathrm{c}} \sum_{i \in \mathcal{C}} x_i, \qquad y_\mathrm{f} = \frac{1}{N_\mathrm{c}} \sum_{i \in \mathcal{C}} y_i. \tag{170}$$

The skyrmion size was represented by an area-equivalent core radius. The core area was calculated as

$$A_{\text{core}} = N_{\text{c}} \Delta x \Delta y, \tag{171}$$

where Δx and Δy are the lateral OOMMF cell dimensions. These values were read directly from the OMF header. For the final simulations,

$$\Delta x = \Delta y = 2 \text{ nm}.$$

The corresponding radius was then calculated from

$$r_{\text{f}} = \sqrt{\frac{A_{\text{core}}}{\pi}}. \tag{172}$$

Each final micromagnetic configuration was reduced to the reservoir state

$$\mathbf{s} = (x_{\text{f}}, y_{\text{f}}, r_{\text{f}}) \,. \tag{173}$$

Here, r_{f} represents the radius of a circle having the same area as the reversed skyrmion core; it is not obtained from a direct circular fit to the magnetization boundary. The reported coordinates and radii are displayed to two decimal places in nanometers. Rounding is applied only to their presentation; the numerical values used to construct `STATE_TABLE` in Appendix A.3 are retained. Subcell centroid values can result from averaging cell-center coordinates, but the number of displayed digits does not establish physical accuracy or numerical convergence.

Experimental readout. In this study, the position is calculated as the centroid of the $m_z < 0$ cells in the OMF data using Eqs. (169) and (170). The radius is the area-equivalent radius in Eq. (172), rather than a circle fitted to the core boundary. Experimental imaging methods include magnetic force microscopy [14], magneto-optical Kerr microscopy [15], scanning transmission X-ray microscopy [16], and Lorentz transmission electron microscopy [17]. Their suitability depends on resolution, magnetic contrast, and sample geometry; Kerr microscopy has been used to image larger skyrmion bubbles [15], but conventional diffraction-limited optical imaging would not directly resolve the nanometer-scale core considered here. Electrical detection of individual skyrmions has also been demonstrated through the anomalous Hall effect [14] and magnetic tunnel junctions [18]. These measurements provide an electrical response to the magnetic texture, with a size-dependent anomalous Hall contribution linked to the spatially averaged out-of-plane magnetization [16]; a single scalar signal cannot, in general, uniquely determine x, y, and r without additional constraints, calibration, or spatially distinct measurements. No experimental readout is demonstrated here; these methods are possible routes for a

future device.

4.2 Four-Bit Reservoir-State Lookup

A four-bit input has

$$2^4 = 16 \tag{174}$$

possible binary sequences, ranging from 0000 to 1111. Each sequence was simulated from the same relaxed initial skyrmion state using the final pulse configuration described in Sec. 3.

After post-processing, every four-bit input was associated with one final physical reservoir state,

$$b_1 b_2 b_3 b_4 \longrightarrow (x_{\mathrm{f}}, y_{\mathrm{f}}, r_{\mathrm{f}})\,. \tag{175}$$

The resulting 16 states formed a fixed lookup table,

$$\mathcal{L}(c) = (x_c, y_c, r_c)\,, \qquad c \in \{0, 1, \ldots, 15\}, \tag{176}$$

where c is the decimal representation of the corresponding four-bit word.

This approach avoids rerunning a micromagnetic simulation whenever the same four-bit sequence appears in an MNIST image. Instead, the physically simulated response associated with each of the 16 possible inputs is reused during construction of the image-level reservoir representation. The extracted state values are presented in Sec. 5.

4.3 MNIST Preprocessing and Four-Bit Encoding

The classification task was based on the MNIST handwritten-digit dataset [19]. MNIST contains grayscale images of handwritten digits from 0 to 9, with each image consisting of

$$28 \times 28 = 784 \tag{177}$$

pixels. Representative examples are shown in Fig. 28.

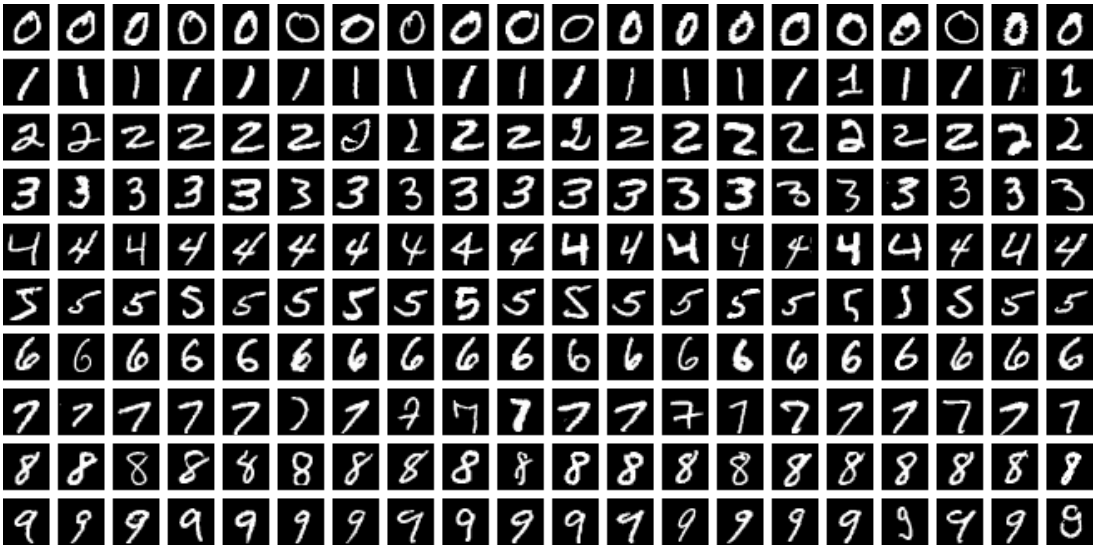


Figure 28: Representative handwritten-digit samples from the MNIST dataset used for classification. Each original image contains 28×28 grayscale pixels.

The grayscale pixel values range from 0 to 255. Before four-bit encoding, the images were binarized using the threshold

$$b = \begin{cases} 1, & p > 127, \\ 0, & p \leq 127, \end{cases} \tag{178}$$

where p is the original grayscale pixel value.

Each binarized image was flattened in row-major order into a 784-element binary vector. Consecutive pixels were then divided into groups of four, producing

$$\frac{784}{4} = 196 \tag{179}$$

four-bit words per image.

For the kth group, the corresponding integer code was defined as

$$c_k = 8b_{4k-3} + 4b_{4k-2} + 2b_{4k-1} + b_{4k}, \qquad k = 1, \ldots, 196. \tag{180}$$

Thus,

$$c_k \in \{0, 1, \ldots, 15\}.$$

Each code was subsequently replaced by the corresponding simulated reservoir state,

$$c_k \longrightarrow \mathcal{L}(c_k) = (x_k, y_k, r_k). \tag{181}$$

Consequently, each MNIST image was represented by a sequence of 196 physical skyrmion states.

Table 2: Input representations used for MNIST classification.

Representation	Feature dimension
Raw binary pixels	784
Four-bit integer codes	196
Skyrmion x	196
Skyrmion y	196
Skyrmion r	196
Skyrmion $x + y$	392
Skyrmion $x + r$	392
Skyrmion $y + r$	392
Skyrmion $x + y + r$	588

4.4 Reservoir Feature Construction

Different combinations of the extracted physical variables were used to evaluate the information carried by the final skyrmion state.

Using only the longitudinal coordinate gives

$$\mathbf{F}_x = (x_1, x_2, \ldots, x_{196}), \tag{182}$$

which contains 196 features.

For two physical observables, the values were stored consecutively for each four-bit word. For example,

$$\mathbf{F}_{xy} = (x_1, y_1, x_2, y_2, \ldots, x_{196}, y_{196}), \tag{183}$$

which gives 392 features.

The complete three-variable representation was

$$\mathbf{F}_{xyr} = (x_1, y_1, r_1, x_2, y_2, r_2, \ldots, x_{196}, y_{196}, r_{196}), \tag{184}$$

resulting in 588 features per image.

The complete set of evaluated input representations is summarized in Table 2. In addition to the physical reservoir features, two digital reference representations were included: the original 784 binarized pixels and the 196 four-bit integer codes.

4.5 Logistic-Regression Readout and Evaluation

The standard MNIST dataset split was used, consisting of 60,000 training images and 10,000 test images. The same training and test samples were used for every feature representation so that their classification performances could be compared directly.

Before classification, each feature dimension was standardized using `StandardScaler`. The scaling parameters were determined from the training data and the same transfor-

mation was then applied to the test data.

Classification was performed using a logistic-regression readout implemented with `scikit-learn` [20]. The `lbfgs` solver was used with a maximum of 2000 iterations. A separate classifier was trained for each representation listed in Table 2.

The skyrmion reservoir itself was not optimized or retrained during this stage. The 16-state physical lookup remained fixed, and training was restricted entirely to the software readout.

Classification performance was quantified by the test-set accuracy,

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{test}}}, \tag{185}$$

where N_{correct} is the number of correctly classified test images and

$$N_{\text{test}} = 10\,000. \tag{186}$$

The same evaluation procedure was applied to the raw-binary baseline, the four-bit-code baseline, and all single and combined physical reservoir representations. The resulting classification performances are presented in Sec. 5.

5 Results

The 16 four-bit inputs produce the states and classification results below.

5.1 Final Skyrmion States for the Four-Bit Inputs

Figure 29 shows the final magnetization configurations obtained after applying each of the 16 possible four-bit pulse sequences. The panels are arranged in ascending binary order from 0000 to 1111.

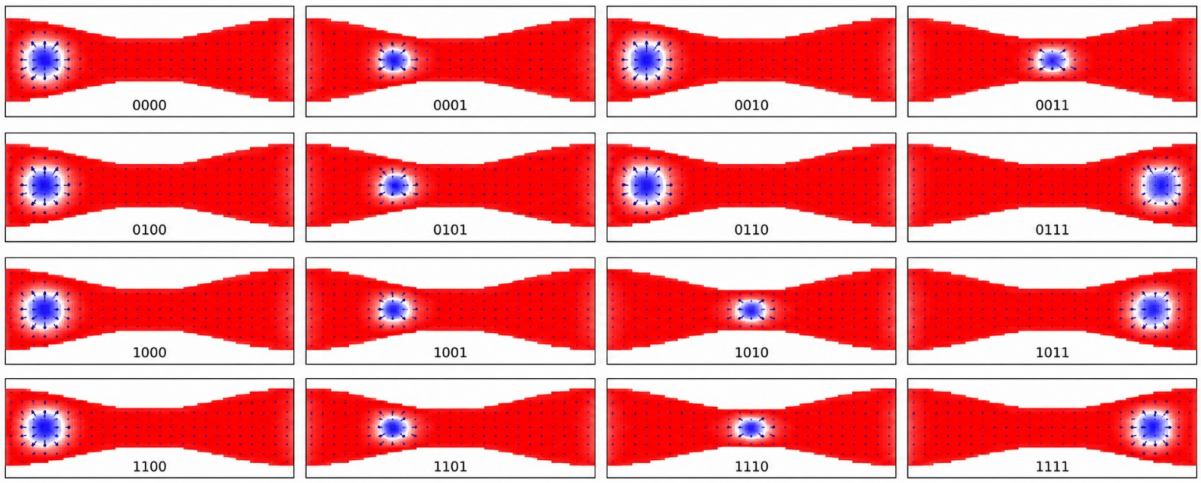


Figure 29: Final OOMMF magnetization states for the 16 four-bit input patterns. Each run starts from the same relaxed skyrmion state and applies four consecutive bipolar pulses with amplitude $|I| = 145\,\mu$A and $t_{\mathrm{bit}} = 2\,$ns ($4 \times 2\,$ns in total). Bits 1 and 0 correspond to positive and negative current, respectively. The states are sampled at $t = 8\,$ns without an additional current-free relaxation interval.

The final configurations demonstrate that the applied pulse history changes both the position and the size of the skyrmion. Several sequences leave the skyrmion in the wider left part of the device, whereas other sequences drive it toward the central constriction or into the wider right-hand region. The strongest longitudinal displacement is observed for sequences such as 0111, 1110, and 1111, for which the skyrmion crosses the central constriction.

The four-bit input mapping is not one-to-one in the extracted (x, y, r) representation. At the reported numerical precision, 0000, 0100, 1000, and 1100 share the same feature vector; 0001 and 1001 form a second identical pair, and 0010 and 1010 form a third. The 16 inputs therefore give 11 distinct reported feature vectors. Equality of these three observables does not establish equality of the full magnetization fields. Without trajectory or full-field comparisons, the repeated endpoints cannot be assigned specifically to left-boundary pinning, edge collisions, or another mechanism. The repeated vectors limit the distinguishability of individual input words in this representation, while the remaining states can still carry information useful for classification.

5.2 Extracted Reservoir States

The final horizontal position x, vertical position y, and area-equivalent radius r were extracted from each magnetization configuration using the procedure described in Sec. 4.1. The resulting reservoir states are listed in Table 3.

Table 3: Extracted final skyrmion states for the 16 four-bit input patterns. Coordinates and area-equivalent radii are displayed in nanometers to two decimal places; this formatting does not imply an uncertainty of 0.01 nm. The numerical values used for classification are retained in Appendix A.3.

Input	x (nm)	y (nm)	r (nm)
0000	-73.10	0.00	12.67
0001	-38.59	0.41	9.71
0010	-73.00	-0.17	12.67
0011	-21.43	0.83	7.65
0100	-73.10	0.00	12.67
0101	-38.87	0.39	9.84
0110	-72.15	-0.29	12.57
0111	78.27	-0.48	11.94
1000	-73.10	0.00	12.67
1001	-38.59	0.41	9.71
1010	-73.00	-0.17	12.67
1011	-21.74	0.78	7.65
1100	-73.10	0.00	12.67
1101	-39.00	0.36	9.77
1110	37.78	-0.42	9.57
1111	73.10	0.00	12.67

The horizontal coordinate exhibits the largest variation among the three extracted variables, ranging from approximately -73 nm in the left part of the device to approximately $+78$ nm in the right part. The vertical displacement is considerably smaller, remaining below approximately 1 nm in magnitude for the final states. In contrast, the radius changes from approximately 7.65 nm to 12.67 nm, indicating substantial deformation as the skyrmion moves through different regions of the dumbbell geometry.

In particular, the smallest radii are obtained for states located close to the central constriction, such as 0011 and 1011. This is consistent with the increased geometrical confinement in the narrow region discussed in Sec. 2.6. Using the original numerical values in Appendix A.3 for all 16 input patterns in Table 3 with equal weights, including repeated states, the Pearson coefficient is $\rho(x, r) = -0.225$. For the absolute longitudinal position, $\rho(|x|, r) = 0.984$. The local width is $W(x) = 2y_b(x) = 45 - 15\cos(\pi x/100)$ nm, with x expressed in nm, and gives $\rho(W(x), r) = 0.982$. The radius therefore has a weak linear association with signed x but a strong association with $|x|$ and the local width over these sampled states. The smallest radii occur for 0011 and 1011 near the constriction, at $x \approx -21$ nm. This descriptive association is consistent with confinement, but it does not establish causality or prove that radius is determined uniquely by position under other driving histories. The 16 inputs contain only 11 distinct reported feature vectors and are not independent experimental trials, so no significance test or p-value is assigned.

Table 4: MNIST test-set classification accuracy obtained using the digital baseline representations and the different physical skyrmion feature combinations.

Representation	Test accuracy (%)
Raw binary pixels	91.29
Four-bit integer codes	89.39
Skyrmion x	87.48
Skyrmion y	50.85
Skyrmion r	75.08
Skyrmion $x + y$	88.36
Skyrmion $x + r$	89.35
Skyrmion $y + r$	81.70
Skyrmion $x + y + r$	**90.27**

5.3 MNIST Classification Performance

The extracted reservoir states were subsequently used to construct the MNIST feature representations described in Sec. 4. The classification accuracies obtained with the logistic-regression readout are summarized in Table 4.

Among the individual physical variables, the horizontal skyrmion position provided the highest classification accuracy, reaching 87.48%. The radius alone achieved 75.08%, whereas the vertical coordinate alone produced a substantially lower accuracy of 50.85%.

Combining multiple physical observables generally improved the classification performance. The $x + r$ representation reached 89.35%, while the complete $x + y + r$ representation achieved the highest accuracy among all physical reservoir representations,

$$\boxed{\text{Accuracy}_{x+y+r} = 90.27\%}.$$

To determine whether this improvement could be explained simply by the increase in feature-vector dimension, an additional feature-duplication control was performed. The longitudinal coordinate x was repeated two and three times while the same preprocessing and logistic-regression procedure were retained. The resulting accuracies were

$$\text{Accuracy}_{x} = 87.48\%, \qquad \text{Accuracy}_{x+x} = 87.46\%, \qquad \text{Accuracy}_{x+x+x} = 87.42\%.$$

Repeating x did not improve accuracy under the reported training procedure, despite increasing the feature-vector dimension from 196 to 392 and 588. This supports the usefulness of the distinct observables, but does not establish statistically independent physical information or a unique mechanism for the improvement. As shown in Sec. 5.2, r is strongly associated with $|x|$ and the local width, so it may provide a nonlinear representation of position to the linear classifier. Repeated-x controls alone do not exclude this explanation. Feature duplication can also change the effective L2 penalty at fixed

regularization strength; a matched-regularization comparison remains future work.

The complete $x + y + r$ representation exceeded the four-bit integer-code baseline of 89.39% by 0.88 percentage points. Its performance was also only 1.02 percentage points below the 91.29% accuracy obtained directly from the 784 binarized MNIST pixels.

Overall, the results indicate that using multiple components of the final skyrmion state provides a more informative representation for the readout than using the longitudinal coordinate alone. Although the vertical coordinate gives weak classification performance when used independently, the highest accuracy among the evaluated physical representations was obtained when the longitudinal position, transverse position, and area-equivalent radius were used together.

A Simulation and Analysis Codes

A.1 Representative OOMMF Input File

The following MIF file shows the final OOMMF configuration for the four-bit input pattern 1100. The remaining input patterns used the same model parameters and differed only in the four-bit pulse sequence and the corresponding output basename.

Listing 1: Representative OOMMF MIF file for the 1100 input pattern.

```
# MIF 2.2

SetOptions {
    basename skyrmion_1100_Ms1400_Kuc156_Kue300_edge4nm_P0p1_JH1_bit2ns
}

set bit_string "1100"
set nbits [string length $bit_string]
set bit_time 2e-9

set Jscale 1.0
set Hscale 1.0

set data_dir "D:/Efe-Ommf/SOT_P=0.1"
set jfile "$data_dir/Gridded_J_2nm_clean.txt"
set hfile "$data_dir/GriddedWoheader_H_2nm.txt"
set mOfile "$data_dir/ComsolData_Relaxed.omf"


# Load COMSOL current-density map
array set Jmap {}

set jraw [ReadFile $jfile]

foreach line [split $jraw "\n"] {
    set line [string trim $line]

    if {$line eq ""} {
        continue
    }
```

```
    if {[scan $line "%lf %lf %lf" xnm ynm Jval] != 3} {
        continue
    }

    set ix [expr {int(round(($xnm + 99.0)/2.0))}]
    set iy [expr {int(round(($ynm + 29.0)/2.0))}]

    if {$ix < 0 || $ix > 99 || $iy < 0 || $iy > 29} {
        continue
    }

    set Jmap($ix,$iy) $Jval
}

unset jraw


# Load COMSOL Oersted-field map
array set Hxmap {}
array set Hymap {}
array set Hzmap {}

set hraw [ReadFile $hfile]

foreach line [split $hraw "\n"] {
    set line [string trim $line]

    if {$line eq ""} {
        continue
    }

    if {[scan $line "%lf %lf %lf %lf %lf" \
         xnm ynm Hx Hy Hz] != 5} {
        continue
    }

    set ix [expr {int(round(($xnm + 99.0)/2.0))}]
    set iy [expr {int(round(($ynm + 29.0)/2.0))}]

    if {$ix < 0 || $ix > 99 || $iy < 0 || $iy > 29} {
        continue
    }

    set Hxmap($ix,$iy) $Hx
    set Hymap($ix,$iy) $Hy
    set Hzmap($ix,$iy) $Hz
}

unset hraw


proc J_from_comsol { x y z } {
    global Jmap

    set xnm [expr {$x*1e9}]
    set ynm [expr {$y*1e9}]

    set ix [expr {int(round(($xnm + 99.0)/2.0))}]
```

```
    set iy [expr {int(round(($ynm + 29.0)/2.0))}]

    if {$ix < 0 || $ix > 99 || $iy < 0 || $iy > 29} {
        return 0.0
    }

    set key "$ix,$iy"

    if {[info exists Jmap($key)]} {
        return $Jmap($key)
    }

    return 0.0
}


proc H_from_comsol { x y z } {
    global Hxmap Hymap Hzmap

    set xnm [expr {$x*1e9}]
    set ynm [expr {$y*1e9}]

    set ix [expr {int(round(($xnm + 99.0)/2.0))}]
    set iy [expr {int(round(($ynm + 29.0)/2.0))}]

    if {$ix < 0 || $ix > 99 || $iy < 0 || $iy > 29} {
        return [list 0.0 0.0 0.0]
    }

    set key "$ix,$iy"

    if {[info exists Hxmap($key)] && \
        [info exists Hymap($key)] && \
        [info exists Hzmap($key)]} {

        return [list \
            $Hxmap($key) \
            $Hymap($key) \
            $Hzmap($key)]
    }

    return [list 0.0 0.0 0.0]
}


proc FourBitPulse { stage stage_time total_time } {
    global bit_time Jscale bit_string

    if {$stage < 0 || \
        $stage >= [string length $bit_string]} {
        return 0.0
    }

    if {$stage_time < 0 || $stage_time >= $bit_time} {
        return 0.0
    }

    set bit [string index $bit_string $stage]
```

```
    if {$bit eq "1"} {
        return $Jscale
    } else {
        return [expr {-$Jscale}]
    }
}


# Geometry and mesh
set dx 2e-9
set dy 2e-9
set dz 0.5e-9

set edge_thickness_nm 4.0
set K_center 1.56e6
set K_edge 3.0e6


proc dumbbell_shape { x y z } {
    set x_nm [expr {$x*1e9}]
    set y_nm [expr {$y*1e9}]
    set pi 3.141592653589793

    set y_bound \
        [expr {22.5 - 7.5*cos($x_nm*$pi/100.0)}]

    if {abs($y_nm) <= $y_bound} {
        return 1
    }

    return 0
}


Specify Oxs_ScriptAtlas:atlas {
    xrange { -100e-9 100e-9 }
    yrange { -40e-9 40e-9 }
    zrange { 0 0.5e-9 }

    regions { magnet }

    script dumbbell_shape
    script_args { rawpt }
}


Specify Oxs_RectangularMesh:mesh [subst {
    cellsize { $dx $dy $dz }
    atlas :atlas
}]


# Material parameters
Specify Oxs_AtlasScalarField:Ms_field {
    atlas :atlas
    multiplier 1
    default_value 0

    values {
```

```
        magnet 1.4e6
    }
}


Specify Oxs_UniformExchange:exch {
    A 15e-12
}


proc edge_K1 { x y z } {
    global edge_thickness_nm K_center K_edge

    set x_nm [expr {$x*1e9}]
    set y_nm [expr {$y*1e9}]
    set ay_nm [expr {abs($y_nm)}]
    set pi 3.141592653589793

    set y_bound \
        [expr {22.5 - 7.5*cos($x_nm*$pi/100.0)}]

    if {$ay_nm > $y_bound} {
        return 0.0
    }

    if {$ay_nm >= ($y_bound - $edge_thickness_nm)} {
        return $K_edge
    }

    return $K_center
}


Specify Oxs_ScriptScalarField:K1_field {
    atlas :atlas
    script edge_K1
    script_args { rawpt }
}


Specify Oxs_UniaxialAnisotropy:ani {
    K1 :K1_field
    axis { 0 0 1 }
}


Specify Oxs_Demag:demag {}


Specify Oxs_DMExchange6Ngbr:dmi {
    default_D 2.9e-3
    atlas :atlas

    D {
        magnet magnet 2.9e-3
    }
}
```

```
# Relaxed initial state
Specify Oxs_FileVectorField:m0 [subst {
    file "$m0file"
    atlas :atlas
}]


# COMSOL current-density field
Specify Oxs_ScriptScalarField:J_field {
    atlas :atlas
    script J_from_comsol
    script_args { rawpt }
}


# COMSOL Oersted field
Specify Oxs_ScriptVectorField:H_comsol {
    atlas :atlas
    script H_from_comsol
    script_args { rawpt }
}


proc HsignProfile { stage stage_time total_time } {
    global bit_time Hscale bit_string

    if {$stage < 0 || \
        $stage >= [string length $bit_string]} {
        return [list 0 0 0 0 0 0]
    }

    if {$stage_time < 0 || $stage_time >= $bit_time} {
        return [list 0 0 0 0 0 0]
    }

    set bit [string index $bit_string $stage]

    if {$bit eq "1"} {
        set s $Hscale
    } else {
        set s [expr {-$Hscale}]
    }

    return [list $s $s $s 0 0 0]
}


Specify Oxs_TransformZeeman:comsol_field {
    field :H_comsol
    type diagonal
    script HsignProfile
    script_args { stage stage_time total_time }
}


# Current-induced torque
Specify Oxs_SpinXferEvolve:evolve [subst {
    alpha 0.3
```

```
    J :J_field

    mp { 0 -1 0 }

    P 0.1
    Lambda 1
    eps_prime 0

    J_profile FourBitPulse
    J_profile_args { stage stage_time total_time }
}]


# Four 2-ns stages
Specify Oxs_TimeDriver:init [subst {
    evolver :evolve
    mesh :mesh
    Ms :Ms_field
    m0 :m0

    stopping_time $bit_time
    stage_count $nbits
}]


Destination archive mmArchive

Schedule DataTable archive Done 1
Schedule Oxs_TimeDriver:init:Magnetization archive Done 1
```

A.2 Skyrmion-State Extraction in Mathematica

The following Mathematica code was used to extract the final skyrmion centroid and area-equivalent radius from the 16 OOMMF magnetization files.

Listing 2: Mathematica code used to extract the final (x, y, r) reservoir states.

```
ClearAll["Global`*"];

dir = "D:/Efe-Ommf/SOT_P=0.1/";

outputCSV =
 FileNameJoin[
  {dir, "skyrmion_states_2ns.csv"}
 ];


parseNumber[s_String] :=
 ToExpression[
  StringReplace[
   StringTrim[s],
   {"e" -> "*^", "E" -> "*^"}
  ]
 ];


headerValue[header_String, key_String] :=
```

```
 Module[{hit},

  hit =
   StringCases[
    header,
    RegularExpression[
      "(?m)^#\\s*" <> key <>
      "\\s*:\\s*([^\\r\\n]+)"
    ] -> "$1"
   ];

  If[
   Length[hit] == 0,
   Missing["NotFound", key],
   parseNumber[First[hit]]
  ]
 ];


ReadOMF[file_String] :=
 Module[
  {
   bytes,
   text,
   headerEndPos,
   header,
   markerPos,
   afterMarker,
   newlinePos,
   dataStartChar,
   stream,
   xnodes,
   ynodes,
   znodes,
   xbase,
   ybase,
   xstep,
   ystep,
   count,
   vals,
   data
  },

  bytes =
   BinaryReadList[
    file,
    "Byte"
   ];

  text =
   FromCharacterCode[
    bytes,
    "ISO8859-1"
   ];

  headerEndPos =
   StringPosition[
    text,
    "# End: Header"
```

```
 ][[1, 1]];

header =
 StringTake[
  text,
  headerEndPos - 1
 ];

markerPos =
 StringPosition[
  text,
  "# Begin: Data Binary 8"
 ];

afterMarker =
 markerPos[[1, 2]] + 1;

newlinePos =
 afterMarker - 1 +
  StringPosition[
    StringDrop[
     text,
     afterMarker - 1
    ],
    "\n"
  ][[1, 1]];

dataStartChar =
 newlinePos + 1;

xnodes =
 Round[
  headerValue[
   header,
   "xnodes"
  ]
 ];

ynodes =
 Round[
  headerValue[
   header,
   "ynodes"
  ]
 ];

znodes =
 Round[
  headerValue[
   header,
   "znodes"
  ]
 ];

xbase =
 headerValue[
  header,
  "xbase"
 ];
```

```
ybase =
 headerValue[
  header,
  "ybase"
 ];

xstep =
 headerValue[
  header,
  "xstepsize"
 ];

ystep =
 headerValue[
  header,
  "ystepsize"
 ];

count =
 xnodes*ynodes*znodes*3;

stream =
 OpenRead[
  file,
  BinaryFormat -> True
 ];

SetStreamPosition[
 stream,
 dataStartChar - 1
];

(* Binary-8 check value *)
BinaryRead[
 stream,
 "Real64",
 ByteOrdering -> -1
];

vals =
 BinaryReadList[
  stream,
  "Real64",
  count,
  ByteOrdering -> -1
 ];

Close[stream];

data =
 ArrayReshape[
  vals,
  {
   znodes,
   ynodes,
   xnodes,
   3
  }
```

```
   ];

  <|
   "Data" -> data,
   "xnodes" -> xnodes,
   "ynodes" -> ynodes,
   "xbase" -> xbase,
   "ybase" -> ybase,
   "xstep" -> xstep,
   "ystep" -> ystep
  |>
 ];


extractPattern[file_String] :=
 Module[{hit},

  hit =
   StringCases[
    FileNameTake[file],
    RegularExpression[
      "skyrmion_([01]{4})"
    ] -> "$1"
   ];

  First[hit]
 ];


files =
 FileNames[
  "skyrmion_*_Ms1400_Kuc156_Kue300_edge4nm_" <>
   "POp1_JH1_bit2ns-Oxs_TimeDriver-init-" <>
   "Magnetization-03-*.omf",
  dir
 ];


If[
 Length[files] != 16,
 Print[
  "Expected 16 final OMF files, found ",
  Length[files]
 ];
 Abort[];
];


files =
 SortBy[
  files,
  FromDigits[
    ToExpression /@ Characters[
     extractPattern[#]
    ],
    2
   ] &
 ];
```

```
SkyrmionState[file_String] :=
 Module[
  {
   omf,
   layer,
   mx,
   my,
   mz,
   norm,
   mzNorm,
   xnm,
   ynm,
   coreMask,
   coords,
   centroid,
   coreCount,
   dxnm,
   dynm,
   radiusNm,
   pattern
  },

  omf =
   ReadOMF[file];

  layer =
   omf["Data"][[1]];

  mx =
   layer[[All, All, 1]];

  my =
   layer[[All, All, 2]];

  mz =
   layer[[All, All, 3]];

  norm =
   Sqrt[
    mx^2 +
    my^2 +
    mz^2
   ];

  mzNorm =
   MapThread[
    If[
      #2 > 0,
      #1/#2,
      Indeterminate
    ] &,
    {mz, norm},
    2
   ];

  xnm =
   10^9*
    (
```

```
   omf["xbase"] +
    Range[
     0,
     omf["xnodes"] - 1
    ]*
    omf["xstep"]
  );

ynm =
 10^9*
  (
   omf["ybase"] +
    Range[
     0,
     omf["ynodes"] - 1
    ]*
    omf["ystep"]
  );

coreMask =
 Map[
  TrueQ[# < 0] &,
  mzNorm,
  {2}
 ];

coords =
 Flatten[
  Table[
   If[
    coreMask[[j, i]],
    {
     xnm[[i]],
     ynm[[j]]
    },
    Nothing
   ],
   {
    j,
    omf["ynodes"]
   },
   {
    i,
    omf["xnodes"]
   }
  ],
  1
 ];

If[
 Length[coords] == 0,
 Return[$Failed]
];

centroid =
 Mean[coords];

coreCount =
 Length[coords];
```

```
  dxnm =
   10^9*
    omf["xstep"];

  dynm =
   10^9*
    omf["ystep"];

  radiusNm =
   Sqrt[
    coreCount*
     dxnm*
     dynm/Pi
   ];

  pattern =
   extractPattern[file];

  <|
   "Pattern" -> pattern,

   "Code" ->
 FromDigits[
  ToExpression /@ Characters[pattern],
  2
 ],

   "x_nm" ->
    centroid[[1]],

   "y_nm" ->
    centroid[[2]],

   "r_nm" ->
    radiusNm
  |>
 ];


results =
 DeleteCases[
  SkyrmionState /@ files,
  $Failed
 ];


If[
 Length[results] != 16,
 Print[
  "Could not extract all 16 states."
 ];
 Abort[];
];


results =
 SortBy[
  results,
```

```
  #["Code"] &
 ];


Dataset[results]


headers = {
 "Pattern",
 "Code",
 "x_nm",
 "y_nm",
 "r_nm"
};


rows =
 Table[
  Lookup[
   result,
   headers
  ],
  {result, results}
 ];


Export[
 outputCSV,
 Prepend[
  rows,
  headers
 ],
 "CSV"
 ];
```

A.3 MNIST Classification Code

The following Python code was used to construct the MNIST reservoir representations and train the logistic-regression readout.

Listing 3: Python code used for MNIST classification with the extracted skyrmion states.

```
import numpy as np
import pandas as pd

from tensorflow.keras.datasets import mnist

from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score


# Final reservoir states: (x, y, r) in nm
state_dict = {
    "0000": (-73.0952, 0.0000, 12.6660),
    "0001": (-38.5946, 0.4054, 9.7067),
```

```
    "0010": (-73.0000, -0.1746, 12.6660),
    "0011": (-21.4348, 0.8261, 7.6530),

    "0100": (-73.0952, 0.0000, 12.6660),
    "0101": (-38.8684, 0.3947, 9.8370),
    "0110": (-72.1452, -0.2903, 12.5651),
    "0111": ( 78.2679, -0.4821, 11.9416),

    "1000": (-73.0952, 0.0000, 12.6660),
    "1001": (-38.5946, 0.4054, 9.7067),
    "1010": (-73.0000, -0.1746, 12.6660),
    "1011": (-21.7391, 0.7826, 7.6530),

    "1100": (-73.0952, 0.0000, 12.6660),
    "1101": (-39.0000, 0.3600, 9.7721),
    "1110": ( 37.7778, -0.4167, 9.5746),
    "1111": ( 73.0952, 0.0000, 12.6660),
}


STATE_TABLE = np.zeros(
    (16, 3),
    dtype=np.float64
)

for bits, values in state_dict.items():
    STATE_TABLE[int(bits, 2)] = values


# Load and binarize MNIST
(X_train, y_train), \
(X_test, y_test) = mnist.load_data()

X_train_bin = (
    X_train > 127
).astype(
    np.uint8
).reshape(
    -1,
    784
)

X_test_bin = (
    X_test > 127
).astype(
    np.uint8
).reshape(
    -1,
    784
)


# Four-bit encoding
bit_weights = np.array(
    [8, 4, 2, 1],
    dtype=np.uint8
)
```

```
def make_4bit_codes(X_binary):
    groups = X_binary.reshape(
        -1,
        196,
        4
    )

    return np.sum(
        groups * bit_weights,
        axis=2
    ).astype(
        np.uint8
    )


train_codes = make_4bit_codes(
    X_train_bin
)

test_codes = make_4bit_codes(
    X_test_bin
)


# Physical reservoir lookup
train_states = STATE_TABLE[
    train_codes
]

test_states = STATE_TABLE[
    test_codes
]


def physical_features(states, columns):
    selected = states[:, :, columns]

    return selected.reshape(
        states.shape[0],
        -1
    )


feature_sets = {

    "Raw Binary (784 px)": (
        X_train_bin.astype(np.float64),
        X_test_bin.astype(np.float64)
    ),

    "4-bit Code (196 vals)": (
        train_codes.astype(np.float64),
        test_codes.astype(np.float64)
    ),

    "Skyrmion x only": (
        physical_features(
            train_states,
            [0]
```

```
        ),
        physical_features(
            test_states,
            [0]
        )
    ),

    "Skyrmion y only": (
        physical_features(
            train_states,
            [1]
        ),
        physical_features(
            test_states,
            [1]
        )
    ),

    "Skyrmion r only": (
        physical_features(
            train_states,
            [2]
        ),
        physical_features(
            test_states,
            [2]
        )
    ),

    "Skyrmion x+y": (
        physical_features(
            train_states,
            [0, 1]
        ),
        physical_features(
            test_states,
            [0, 1]
        )
    ),

    "Skyrmion x+r": (
        physical_features(
            train_states,
            [0, 2]
        ),
        physical_features(
            test_states,
            [0, 2]
        )
    ),

    "Skyrmion y+r": (
        physical_features(
            train_states,
            [1, 2]
        ),
        physical_features(
            test_states,
            [1, 2]
```

```
        )
    ),

    "Skyrmion x+y+r": (
        physical_features(
            train_states,
            [0, 1, 2]
        ),
        physical_features(
            test_states,
            [0, 1, 2]
        )
    ),
}


def run_logistic_regression(Xtr, Xte):

    model = make_pipeline(
        StandardScaler(),

        LogisticRegression(
            max_iter=2000,
            solver="lbfgs",
            random_state=42
        )
    )

    model.fit(
        Xtr,
        y_train
    )

    predictions = model.predict(
        Xte
    )

    return accuracy_score(
        y_test,
        predictions
    )


results = {}

for name, (Xtr, Xte) in feature_sets.items():
    results[name] = run_logistic_regression(
        Xtr,
        Xte
    )


results_df = pd.DataFrame({
    "Feature Set":
        list(results.keys()),

    "Accuracy":
        list(results.values())
})
```

```
results_df["Accuracy (%)"] = (
    100 *
    results_df["Accuracy"]
)


results_df = (
    results_df
    .sort_values(
        "Accuracy",
        ascending=False
    )
    .reset_index(
        drop=True
    )
)


print(
    results_df.to_string(
        index=False,
        formatters={
            "Accuracy":
                "{:.4f}".format,

            "Accuracy (%)":
                "{:.2f}".format
        }
    )
)


results_df.to_csv(
    "mnist_skyrmion_results_2ns.csv",
    index=False
)


# Feature-duplication controls
control_sets = {

    "x+x": (
        physical_features(
            train_states,
            [0, 0]
        ),
        physical_features(
            test_states,
            [0, 0]
        )
    ),

    "x+x+x": (
        physical_features(
            train_states,
            [0, 0, 0]
        ),
```

```
        physical_features(
            test_states,
            [0, 0, 0]
        )
    ),
}


for name, (Xtr, Xte) in control_sets.items():

    accuracy = run_logistic_regression(
        Xtr,
        Xte
    )

    print(
        f"{name}: "
        f"{100 * accuracy:.2f}%"
    )
```

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